

Amenable groups with nearly exponential sofic profile, and quantum channels that need nearly linear memory

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Abstract

How far from finite can an amenable group be? Cornulier’s sofic profile measures the least number of points on which a finite piece of a group can be modelled by permutations at accuracy $1/r$. In earlier work we showed, by a counting criterion that can never give more than $\exp(O(r))$, that Houghton’s group has a finite piece with profile $\exp(\Omega(r^{0.129}))$; to our knowledge it was the first amenable group shown to have superpolynomial profile. We construct a finitely presented elementary amenable group with a finite piece whose profile is $\exp(r^{1+o(1)})$, with a lower bound certified by 25 explicit relations: at least $\exp(cr/(\log r)^{12.75})$ and at most $\exp(O(r \log r))$. The idea is to place copies of S_3 not on points but on configurations: Houghton’s group moves the points of three rays, finitely supported affine maps act on their binary configurations, and one copy of S_3 sits on each configuration. Then N points carry 2^N copies, while any two copies are brought together by relations that touch only three points at a time. We reach this group through a ladder of transport mechanisms, including an elementary amenable group at exponent $1/3$ and a subgroup of Thompson’s group V whose models, if they exist, need $\exp(cr/\log^2 r)$ points. Our group embeds in Brin’s higher-dimensional Thompson group $3V$, so $3V$ has a chunk with finite profile $\exp(r^{1+o(1)})$. Finally, the group becomes an explicit quantum channel on $\mathbb{C}^{873}$ whose repeated use is expensive. A device that serves n uses, releasing each output before the next input arrives, needs memory about n/B when it may spend purity B , and exact devices attain this trade-off up to polylogarithmic factors; with logarithmic purity it needs memory $n^{1-o(1)}$, and at the optimal exchange rate its memory is $n^{1/2+o(1)}$, attained with purity of the same order. The engine is a one-round theorem that reads an approximate representation of the group directly off the memory state of any such device. As a consequence, a local lattice process in ν dimensions whose bath starts maximally mixed needs time about $\delta^{-1/\nu}$ to produce this channel to accuracy δ . Autonomous dynamics can still produce such channels: a bounded time-independent Hamiltonian, not local, coupled to a tracial bath, produces in the long run a channel with the same memory-purity trade-off.

Contents

1	Introduction	2
2	Counting approximately commuting copies of S_3	6
3	Two steps up the ladder	7
3.1	An amenable group at exponent $1/3$	7
3.2	Thompson’s group V	8
3.3	From orbits to configurations	12
4	Configuration lamps	12
4.1	A finite presentation	13
4.2	An embedding in a Brin–Thompson group	15

5	Houghton inputs	16
6	A certificate of 25 relations and local fillings	18
6.1	Local fillings	19
7	Pair areas and the profile	21
7.1	Affine hashing and the upper bound	22
7.2	The profile	24
7.3	The grid host	26
8	Quantum channels that need nearly linear memory	27
8.1	Setting	27
8.2	The weighted one-round theorem	29
8.3	The channel of the configuration lamps	30
8.4	A purity budget at the rank rate	32
8.5	An upper bound from sofic models	33
8.6	Streaming at the rank rate	34
8.7	Proof of Theorem 8.5	37
9	Proof of the weighted one-round theorem	38
9.1	Weighted norms	38
9.2	Extraction	39
9.3	Charged area	40
9.4	Exact S_3 rounding in place	41
9.5	A weighted sector bound	42
9.6	Completion	44
10	Local physics	45
11	Open problems	54
	Use of AI	55

1 Introduction

A group is sofic when its finite pieces can be modelled by permutations of finite sets, up to a small error. How large must those models be? For a finite piece E of a group, a *chunk*, Cornuier’s *sofic profile* $\sigma_E(r)$ is the least number of points on which E can be modelled with multiplication correct on all but a fraction $1/r$ of the points and distinct elements separated almost everywhere [18]. Amenable groups are sofic, so their profiles are finite. Cornuier asked whether a sofic group can have profile that is unbounded and not linear in r [18, Problem 3.18]. In [19] we showed that Houghton’s group H_3 , which is finitely presented [10], with an explicit presentation due to Johnson [34], and elementary amenable, has a chunk whose profile is at least $\exp(\Omega(r^c))$, $c = 1/(1 + \log 107) \approx 0.129$. The proof rests on a counting criterion (Theorem 2.1): if finitely many relations force many displaced copies of S_3 to commute cheaply, every model must be large, because approximately commuting copies of S_3 carry entropy. That criterion can never give more than $\exp(O(r))$.

This paper shows that the criterion nearly reaches its ceiling in an amenable group, and that the group then becomes a quantum channel whose repeated use needs nearly linear memory.

Theorem A. There is a finitely presented elementary amenable group Γ , with five generators, and a chunk E of Γ such that, for all $r \geq 2$,

$$\exp\left(\frac{cr}{(\log r)^{12.75}}\right) \leq \sigma_E(r) \leq \exp(Cr \log r).$$

So $\sigma_E(r) = \exp(r^{1+o(1)})$. The lower bound is certified by an explicit set of 25 relations of Γ . Moreover, for every $\Delta \in (0, 1]$ there are $c_\Delta, e_\Delta > 0$ such that every assignment of unitaries in $U(D)$ to the five generators under which the 25 relations have defect at most $e \leq e_\Delta$ in the normalized Hilbert–Schmidt norm, and ab stays at distance at least Δ from I , has $\log D \geq c_\Delta e^{-2}/(\log(1/e))^{25.49}$.

Theorem 7.5 gives the precise statement, and Proposition 4.2 the finite presentation, which follows from Cornuier’s criterion for permutational wreath products [17, Theorem 1.1] once the affine group acting on the configurations is shown to be finitely presented. The lower bound is within a polylogarithmic factor in the exponent of the most the criterion can give, and the upper bound shows that the truth is not much larger. The upper bound comes from models that hash configurations by random affine maps; they are far smaller than the models that Følner sets give [18, Proposition 3.14], since the Følner function of Γ is doubly exponential (Proposition 7.7). The group Γ also embeds in Brin’s higher-dimensional Thompson group $3V$ (Proposition 4.3), so $3V$, a finitely presented simple group whose soficity is open, has a chunk whose profile is finite and $\exp(r^{1+o(1)})$.

Configuration lamps. In Houghton’s group the copies of S_3 sit on points of a ray; carrying a copy a distance k takes a word of length k , and commuting two copies at distance k costs at least of order k^2 relations [19, Proposition 3.4]. A family of N copies spreads over distance N , and its commutators are expensive. We place the copies instead on the finitely supported binary configurations of the three rays of H_3 . Houghton’s group moves the points, finitely supported affine maps act on the configurations (a translation flips one point, a transvection adds one point into another), and one copy of S_3 sits on each configuration. The 2^N configurations supported on the first N points of a ray are reached by words of length $O(N^2)$, and two of them differ by a configuration that at most N transvections clear down to a single point. Each step touches at most three points, so its cost is governed by the Dehn function of H_3 at scale N , which is polynomial by [19, Theorem 3.3]. The total cost of commuting two of the 2^N copies is therefore polynomial in N : polylogarithmic in the number of copies. This is what the criterion needs.

A ladder. The construction completes a sequence of transport mechanisms, each of which the criterion turns into a profile bound.

transport	group	$\log \sigma_E(r) \gtrsim$	profile finite
along a ray	H_3 [19]	$r^{0.129}$	yes
by one element	dyadic $\mathcal{G}$ (§3.1)	$r^{1/3}$	yes
exponential orbits	$\mathcal{L} \leq V$ (§3.2)	$r/\log^2 r$	if F is amenable
configurations	Γ (§§4–7)	$r/(\log r)^{12.75}$	yes

Transport by the powers of one element gives exponent $1/(p+1)$ when the far commutators have area of order k^p ; along a ray of H_3 the area is at least quadratic [19, Proposition 3.4], so that transport stops at $1/3$, and $\mathcal{G}$ attains $1/3$ with seventeen relations. Thompson’s group F transports cheaply along exponentially growing orbits, which gives nearly exponential profile in a finitely presented subgroup $\mathcal{L}$ of Thompson’s group V , provided its models exist at all: they do if F is amenable, and we show that $\mathcal{L}$ is hyperlinear exactly when F is, so that unitary models of this chunk exist whenever F is hyperlinear (Proposition 3.4). The configuration lamps keep the exponential transport and drop the condition. For a variant on the configurations of a grid (Remark 7.8) a sketched argument gives a stronger lower bound, $r/(\log r)^{7.38}$, with the same upper bound.

Quantum memory. Kretschmann and Werner showed that every causal quantum process is a concatenation of memory channels [37], and Rybár and Ziman asked when a single memory device, which is not reset between uses, applies the same channel to each of its inputs [47]. In [20] we asked how much memory such a device needs to apply the same quantum channel n times, when each output must be released before the next input arrives and every retained qubit counts. The device may export qubits and import fresh ones; the *exchange* J counts the qubits exchanged, and the *purity*, measured as the number of qubits minus the entropy of the initial memory and of the imports, is a resource. The error is the half trace distance to the ideal experiment, against adaptive observers that keep quantum references. For a gadget channel in the tracial closure, $\frac{1}{2} \log r_*$ qubits per use, with r_* the Choi rank, is the optimal exchange rate [20, Propositions 3.1 and 9.1]. A channel built from a finite set of relations inherits the approximation profile of the group: small memory would give small approximate representations. From H_3 we obtained an explicit channel needing memory $(n/\log n)^{0.069}$. The configuration lamps give nearly linear memory, and a trade-off between memory and purity that is sharp up to polylogarithmic factors.

Theorem B. The certificate of Theorem A defines an explicit quantum channel Φ on $\mathbb{C}^{873}$, in the closure of channels with finite tracial factorizations. For causal services of n uses of Φ with error at most $1/16$ and large n :

- (a) with purity at most B , where $\log n \leq B \leq c_0 n$, the least memory lies between $cn/(B(\log n)^{25.49})$, at any exchange, and $K(n/B) \log n$, attained by exact services that refresh their memory every round, with exchange $(n-1)Q$; so the least memory at purity B , times B , is $n^{1+o(1)}$, and purity $O(\log n)$ forces memory $n^{1-o(1)}$;
- (b) at the optimal exchange rate, that is, with exchange $J \leq \frac{n}{2} \log r_*$ over n uses, the least memory lies between $c\sqrt{n}/(\log n)^{12.75}$, whatever the purity, and $K\sqrt{n \log n}$; services near the least memory spend purity $n^{1/2 \pm o(1)}$, at least that much by (a) and at most $3Q + 6$, so at that rate memory and purity are both about $\sqrt{n}$.

Theorem 8.5 gives the precise statement. Its engine is a one-round theorem for every gadget channel whose relations contain those of S_3 and a name for ab (Theorem 8.2): a single round of any device reveals an approximate representation of the group on the device's memory, with a relator defect of the order of the square root of the leakage and entropy production of that round. In [20, Theorem 5.5 and Appendix B] this was proved for one gadget, built from Houghton's group. The upper bounds come from sofic models, which give devices directly (Proposition 8.9): at most points of a model the pure bath state reproduces the channel exactly, so a memory that is maximally mixed on those points serves the channel with no error at all. So the memory of Φ is caught between two readings of the profile of Γ at scale n , and for Γ both are $n^{1+o(1)}$. At the optimal exchange rate a device can spend at most $3Q + 6$ bits of purity (Proposition 8.7), which gives the lower bound in part (b). The upper bound needs a device that keeps a sofic model in memory for all n rounds instead of refreshing it: a streaming theorem (Theorem 8.10) uses the good points of a model as an exact dilation whose bath grows slightly in every round, so that the bad points cost purity rather than accumulated error.

Physics. A local lattice system whose bath starts in its maximally mixed state, run for time T , produces channels that are cheap for fixed T : they need memory only polynomial in T and $\log n$ (Proposition 10.1). So a local process that produces Φ to accuracy δ takes time at least about $\delta^{-1/\nu}$ in lattice dimension ν (Corollary 10.2). Autonomous dynamics with a bath in its tracial state face a different constraint: by the mean ergodic theorem, their long-time limits are compressions of conditional expectations and hence self-adjoint channels [35,

Remark 10.3], [25, Remark 5.2], and they satisfy a further inequality between coherence and return (Propositions 10.3 and 10.4). Every channel $\Psi\Psi^*$ with Ψ in the tracial closure is such a limit (Proposition 10.6), but for gadget channels this route gives nothing expensive: composing a gadget channel with its adjoint erases the multiplication of the group, and $\Phi\Phi^*$ has an exact finite factorization on at most $3r_* - 2$ qubits (Proposition 10.8). Averaging the doubling of Φ with the identity does give an expensive limit: a bounded time-independent Hamiltonian coupled to a tracial bath produces a channel on M_{1746} exactly at the times $\frac{\pi}{2}k$, and in the limit of long times, that trades memory and purity at the same rate as Φ (Proposition 10.9). The Hamiltonian is not local; whether local autonomous dynamics can produce an expensive channel is open.

Dependencies on the companion papers. Table 1 lists every result of [19] and [20] used here, with the numbering of their version 1.1.

result	source	used in
counting criterion	[19, Thm. 8.1]	Thm. 2.1, every lower bound
transport by the powers of one element	[19, Cor. 8.2]	Section 2
Johnson’s presentation of H_3 and its conventions	[19, Section 2]	Section 5
far-commutator area and polynomial Dehn bound for H_3	[19, Thms. 3.2, 3.3]	Prop. 5.3
quadratic lower bound along a ray	[19, Prop. 3.4]	Sections 1, 2, 3
bounded or at least $2r$ (sharpening Cornulier’s Fact 3.5); embeddings in finite groups	[19, Lemma 2.1]	Section 3, after Thm. 7.5
superpolynomial profile of H_3	[19, Thm. 1.1]	Section 3.2
projection products	[19, Lemma 4.1]	Lemma 9.4
causal services, exchange, purity and error	[20, Def. 2.3, Section 3]	Section 8
gadget channels: Choi rank r_* , flat probe, entropy $\log r_*$	[20, Prop. 9.1]	Section 8.1
the rank rate is the optimal exchange rate of gadget channels in $\mathcal{K}_d$	[20, Props. 3.1, 9.1]	Thm. B(b)
sequential extraction	[20, Thm. 5.2]	Thm. 8.1
purity budget $14Q + 3$ at the rank rate (sharpened here)	[20, Thm. 5.8]	Prop. 8.7
entropy variance	[20, Lemma 5.9]	Lemma 8.8
rolling-reservoir compiler	[20, Appendix A, Lemmas A.1, A.4]	Thm. 8.10
channel profiles are group profiles; hyperlinear groups give channels in $\mathcal{K}_d$	[20, Thm. 9.2]	Section 8
one-round theorem for the Houghton gadget (generalised here)	[20, Thm. 5.5, Appendix B]	Thm. 8.2, Section 9
weighted norms; charged transport and weighted extraction for the Houghton gadget (generalised here)	[20, Appendix B, Lemmas B.2, B.4]	Lemmas 9.1, 9.2
weighted S_3 rounding	[20, Lemma B.3]	Lemma 9.3
weighted sector bound (proved here by sequential measurement)	[20, Thm. B.1]	Thm. 9.6

Table 1: Results of the companion papers used in this paper (numbering of their version 1.1).

Organization. Section 2 recalls the counting criterion. Section 3 treats the dyadic group and the subgroup of Thompson’s group V . Sections 4–7 construct the configuration lamps,

show that they are finitely presented and embed in $3V$, prove the local fillings and the pair-area theorem, and prove Theorem A. Section 8 proves Theorem B, with the one-round theorem proved in Section 9. Section 10 treats local physics, and Section 11 lists open problems. The certificate of 25 relations is in the accompanying files, together with programs that check it; logarithms are to base two unless written $\ln$.

2 Counting approximately commuting copies of S_3

All lower bounds in this paper come from one principle, proved in [19]: if finitely many relations force many displaced copies of S_3 to commute, every model of a fixed finite piece must be large. We recall it in the form we use, with a, b in place of the letters u, v of [19]. The scheme, many cheaply transported copies of a finite group, relations whose cost is controlled by area, and a dimension bound, follows Slofstra [48, Sections 2–3]; the criterion needs only pairwise approximate commutation of displaced copies of S_3 .

For a finite set S and a finite set R of words in $S^{\pm 1}$, $\text{Area}_R(w)$ denotes the least number of conjugates of words of $R^{\pm 1}$ whose product is w in the free group on S , and ∞ if there is none. A *chunk* of a group is a finite subset $E \ni 1$ with the partial multiplication $xy = z$ whenever $x, y, z \in E$ and $xy = z$ in the group. Its sofic profile $\sigma_E(r)$ is the least n for which some $f : E \rightarrow \text{Sym}_n$ with $f(1) = \text{id}$ has $d(f(x)f(y), f(xy)) \leq 1/r$ whenever xy is defined in E and $d(f(x), f(y)) \geq 1 - 1/r$ for distinct $x, y \in E$, where d is the normalized Hamming distance [18]. Logarithms are to base two.

Theorem 2.1 ([19, Theorem 8.1]). *Let G be a group, $S \subset G$ a finite set containing a, b , and R a finite set of words in $S^{\pm 1}$, each representing 1 in G , with $a^2, b^2, (ab)^3 \in R$ and $ab \neq 1$ in G . Let l be the largest length of a word in R , and let E be the chunk consisting of 1, $S^{\pm 1}$, the prefixes of the words of R and of ab , and the inverses of these elements. Let $w_1, \dots, w_N$ be words in $S^{\pm 1}$ and $\Lambda \geq 1$ such that*

$$\text{Area}_R([w_i x w_i^{-1}, w_j y w_j^{-1}]) \leq \Lambda \quad (i \neq j, x, y \in \{a, b\}).$$

Then $\log \sigma_E(r) \geq N/16$ for every $r \geq 10^8 N \Lambda (2l - 1)$. Moreover, for $0 < \Delta \leq 1$ and $K_\Delta = \sqrt{80000(1 + \log(1/\Delta))}/\Delta$, every assignment ϕ of unitaries in $U(D)$ to S under which every word of R lies within ϵ of I in the normalized Hilbert–Schmidt norm and $\|\phi(ab) - I\|_2 \geq \Delta$ satisfies $\log D \geq N\Delta^2/12$ whenever $K_\Delta \sqrt{N} \Lambda \epsilon \leq 1$.

The mechanism is short to describe. Conjugating one approximate copy of S_3 by the words w_i produces N copies whose three-cycles are displaced from the identity and which commute up to Λ times the relator error. Each copy is rounded to an exact representation of S_3 , and a state on the model space made invariant under all of them must carry about one bit of entropy per copy. The chunk E does not depend on N or on the words w_i ; only the areas matter.

Two consequences frame everything below. First, the theorem never gives more than $\log \sigma_E(r)$ of order r , nor $\log D$ of order e^{-2} : its hypotheses force $N \leq r/10^8$ and $N \leq (K_\Delta \epsilon)^{-2}$. Second, the result depends on how cheaply the group can carry copies of S_3 to many places. When the copies are transported by the powers of one element, with area of order k^p at distance k , the theorem gives exponent $1/(p+1)$:

Corollary 2.2 ([19, Corollary 8.2]). *Let G, S, R, l and E be as in Theorem 2.1, let $t \in S$, and suppose that $\text{Area}_R([x, t^k y t^{-k}]) \leq Ck^p$ for $x, y \in \{a, b\}$ and $k \geq 1$, with $C \geq 1$ and $p \geq 0$. Then*

$$\log \sigma_E(r) \geq \frac{1}{16} \left\lfloor \left(\frac{r}{10^8 C (2l - 1)} \right)^{1/(p+1)} \right\rfloor \quad (r > 1),$$

and every unitary assignment as in Theorem 2.1 has $\log D \geq \frac{\Delta^2}{12} ((CK_\Delta \epsilon)^{-2/(2p+1)} - 1)$.

For Houghton's group H_3 and its copies of S_3 along a ray, the area grows at least quadratically [19, Proposition 3.4], so transport along a ray can never pass the exponent $1/3$. To do better, a group must reach many sites cheaply. Finally, by Cornulier's dichotomy [18, Fact 3.5] the profile of a chunk is bounded exactly when the chunk embeds in a finite group, and otherwise it is at least r ; this lower bound is sharpened to $2r$ in [19, Lemma 2.1]. So a chunk with unbounded profile certifies that its group is not locally embeddable into finite groups (LEF) [18, Fact 3.8].

3 Two steps up the ladder

Before the main construction we climb two steps that show what the criterion needs. The first keeps transport by the powers of one element and makes it cheap: in an elementary amenable group the far commutators have at most quadratic area, which gives the exponent $1/3$. Transport by the powers of one element gives exponent $1/(p+1)$ when the far commutators have area of order k^p , and along a ray of H_3 the area is at least quadratic [19, Proposition 3.4], so that transport stops at $1/3$. The second changes the kind of transport: Thompson's group F has orbits that grow exponentially, so words of length $O(\log N)$ reach N sites, and its quadratic Dehn function fills all pairwise commutators cheaply. This comes within a factor $\log^2 r$ in the exponent of the ceiling of Theorem 2.1, but in a group that contains F , whose soficity is unknown. Section 4 keeps the exponential transport and removes the condition.

3.1 An amenable group at exponent $1/3$

In this subsection and the next, functions are composed from right to left, and a group acting on a set X acts on $\bigoplus_{x \in X} S_3$ by carrying the copy of S_3 at x to the copy at $g(x)$ by the identity of S_3 .

Let $X = \mathbb{Z}[\frac{1}{2}]$, and let t and u be the permutations of X given by $t(x) = x + 1$ for $x \in \mathbb{Z}$, $t(x) = x$ for $x \notin \mathbb{Z}$, and $u(x) = 2x$. Let $\mathcal{B} = \langle t, u \rangle$ and $\mathcal{G} = (\bigoplus_{x \in X} S_3) \rtimes \mathcal{B}$, and let a, b be generating involutions of the copy of S_3 at 0. Put $s = utu^{-1}$ and $v = tst^{-1}$. Then s adds 2 to the even integers and v adds 2 to the odd integers, each fixing every other point, so $t^2 = sv$ and $[s, v] = 1$. Moreover u and v fix 0, and s fixes $1 = t(0)$. Since $\mathcal{B}$ acts transitively on X , the group $\mathcal{G}$ is generated by a, b, t and u . So the following seventeen words in a, b, t, u, s, v , which make up the set $R_{\mathcal{G}}$, are relations of $\mathcal{G}$:

$$a^2, b^2, (ab)^3, s^{-1}utu^{-1}, v^{-1}tst^{-1}, t^2v^{-1}s^{-1}, [s, v], [u, x], [v, x], [s, txt^{-1}], [x, tyt^{-1}],$$

where $x, y \in \{a, b\}$.

Theorem 3.1. *The group $\mathcal{G}$ is elementary amenable and not solvable, and every commutator $[x, t^k y t^{-k}]$ with $x, y \in \{a, b\}$ and $k \geq 1$ has area at most $5k^2$ with respect to $R_{\mathcal{G}}$. Consequently the chunk $E_{\mathcal{G}}$ that Theorem 2.1 associates with $R_{\mathcal{G}}$ has finite sofic profile with*

$$\log \sigma_{E_{\mathcal{G}}}(r) \geq \frac{1}{16} \left\lfloor \left(\frac{r}{75 \cdot 10^8} \right)^{1/3} \right\rfloor \quad (r > 1),$$

every unitary assignment ϕ as in Theorem 2.1 has $\log D \geq \frac{\Delta^2}{12} ((5K_{\Delta}e)^{-2/5} - 1)$, and $\mathcal{G}$ is not LEF.

Proof. Amenability. For $j \in \mathbb{Z}$ let $t_j = u^j t u^{-j}$; it adds 2^j to the points of $2^j \mathbb{Z}$ and fixes the others. The group $\mathcal{K}$ generated by all t_j is normalized by u and contains t , so it is normal in $\mathcal{B}$ and $\mathcal{B}/\mathcal{K}$ is cyclic. For $n \geq 1$ the elements t_j with $|j| \leq n$ move only points of $2^{-n} \mathbb{Z}$, and they commute with the translation $x \mapsto x + 2^n$ of $2^{-n} \mathbb{Z}$, which has 4^n orbits. A permutation

of $2^{-n}\mathbb{Z}$ that commutes with this translation maps each orbit onto an orbit by a translation, so these t_j generate a subgroup of $\mathbb{Z}^{4^n} \rtimes \text{Sym}_{4^n}$, which is virtually abelian. Hence $\mathcal{K}$ is a directed union of virtually abelian groups, $\mathcal{B}$ is elementary amenable, and so is $\mathcal{G}$, because $\bigoplus_X S_3$ is locally finite.

Not solvable. For $n \geq 1$ the elements $t_0, \dots, t_{n-1}$ preserve $\mathbb{Z}$ and commute with $x \mapsto x + 2^n$, so they act on $\mathbb{Z}/2^n$, where they permute the classes modulo 2^i for every $i \leq n$. So they generate a subgroup Γ_n of the iterated wreath product W_n of n copies of $\mathbb{Z}/2$, which has order 2^{2^n-1} . On the even classes, identified with $\mathbb{Z}/2^{n-1}$ by halving, $t_1, \dots, t_{n-1}$ act as $t_0, \dots, t_{n-2}$ act on $\mathbb{Z}/2^{n-1}$, and they fix the odd classes; conjugation by t_0 carries this action to the odd classes, and t_0 exchanges the two halves. So $|\Gamma_n| \geq 2|\Gamma_{n-1}|^2$, and $\Gamma_n = W_n$ by induction from $\Gamma_1 = W_1 = \mathbb{Z}/2$. The derived subgroup of W_n lies in $W_{n-1} \times W_{n-1}$ and projects onto each factor, since the commutator of $(g, 1)$ with the exchange is (g^{-1}, g) ; by induction the $(n-1)$ st derived subgroup of W_n is nontrivial. Since Γ_n is a quotient of the subgroup $\langle t_0, \dots, t_{n-1} \rangle$ of $\mathcal{B}$, the derived length of $\mathcal{B}$ is at least n for every n , so neither $\mathcal{B}$ nor $\mathcal{G}$ is solvable.

Area. Let $A_{\mathcal{G}}(k)$ be the largest area of $[x, t^k y t^{-k}]$ over $x, y \in \{a, b\}$, so that $A_{\mathcal{G}}(1) = 1$. Each of the following moves costs one cell: replacing t^2 by sv ; exchanging adjacent letters $s^{\pm 1}$ and $v^{\pm 1}$; replacing s by utu^{-1} ; moving a letter $x^{\pm 1}$, $x \in \{a, b\}$, across $u^{\pm 1}$ or $v^{\pm 1}$; and moving txt^{-1} or its inverse across $s^{\pm 1}$. The commutator contains four powers $t^{\pm k}$.

For $k = 2m$, write each $t^{\pm 2m}$ as $(s^m v^m)^{\pm 1}$, with m replacements and $m(m-1)/2$ exchanges each, $2m^2 + 2m$ cells in all. Moving $y^{\pm 1}$ across $v^{\pm m}$ ($2m$ cells) leaves $[x, s^m y s^{-m}]$. Replacing its $4m$ letters $s^{\pm 1}$ gives $[x, ut^m u^{-1} y ut^{-m} u^{-1}]$, and moving $y^{\pm 1}$ and $x^{\pm 1}$ across u (four cells) gives $u[x, t^m y t^{-m}]u^{-1}$. Hence $A_{\mathcal{G}}(2m) \leq A_{\mathcal{G}}(m) + 2m^2 + 8m + 4$.

For $k = 2m + 1$, put $z = tyt^{-1}$, so that the commutator is freely $[x, t^{2m} z t^{-2m}]$. Write each $t^{\pm 2m}$ as $(v^m s^m)^{\pm 1}$, now with $m(m+1)/2$ exchanges each, $2m^2 + 6m$ cells in all. Moving $z^{\pm 1}$ across $s^{\pm m}$ ($2m$ cells) leaves $[x, v^m z v^{-m}]$, moving $x^{\pm 1}$ across $v^{\pm m}$ ($2m$ cells) leaves $v^m[x, z]v^{-m}$, and $[x, z]$ is a relation. Hence $A_{\mathcal{G}}(2m + 1) \leq 2m^2 + 10m + 1$.

By induction $A_{\mathcal{G}}(k) \leq 5k^2$: $A_{\mathcal{G}}(2) \leq 15$, $A_{\mathcal{G}}(m) + 2m^2 + 8m + 4 \leq 7m^2 + 8m + 4 \leq 20m^2$ for $m \geq 1$, and $2m^2 + 10m + 1 \leq 5(2m + 1)^2$.

Profile. The words of $R_{\mathcal{G}}$ have length at most 8, so Corollary 2.2 with $C = 5$, $p = 2$ and $l = 8$ gives the bounds. Since $\mathcal{G}$ is amenable, it is sofic and every chunk has finite profile. The profile of $E_{\mathcal{G}}$ is unbounded, so $E_{\mathcal{G}}$ does not embed in a finite group ([18, Fact 3.5]; see also [19, Lemma 2.1]), and $\mathcal{G}$ is not LEF [18, Fact 3.8]. $\square$

The fillings in the proof have been generated and checked by free reduction for $k \leq 64$; the largest ratio of cell count to k^2 among them is $15/4$, at $k = 2$. In H_3 the doubling endomorphism of [19, Section 3], which differs by a conjugation from the doubling map of Burillo, Cleary, Martino and Röver [11, Section 7], multiplies a reweighted area by 107 at each step, which gives the exponent $\log 107$; in $\mathcal{G}$ the dilation u relates distance $2m$ to distance m exactly, at a cost of $O(m^2)$ cells for the two commuting shifts. The orbit of 0 under $\mathcal{B}$ grows exponentially, as the orbits of Thompson's group F do, but we do not know whether $\mathcal{G}$ supports the transport of many copies used for F below (Section 11).

3.2 Thompson's group V

Röver proved that every Houghton group embeds in Thompson's group V [46, Proposition 2.6], and Genevois extended the embedding to generalised Houghton groups [22, Corollary 5.11]. Since H_3 is amenable, V therefore has a chunk whose profile is finite at every r and super-polynomial by [19, Theorem 1.1]. A different subgroup of V gives a much larger bound.

Let $\mathbb{D}$ be the set of dyadic rationals in $(0, 1)$, and let Thompson's group F act on $[0, 1]$ by its piecewise-linear homeomorphisms. Let $\mathcal{L} = (\bigoplus_{x \in \mathbb{D}} S_3) \rtimes F$.

Lemma 3.2 (Gap coding). *$\mathcal{L}$ embeds in V .*

Proof. Let c be the substitution $0 \mapsto 0, 1 \mapsto 11$ on finite binary words. An infinite binary word either is a concatenation of blocks 0 and 11, or it has a first occurrence of 10 at a block boundary. So Cantor space is the disjoint union of the set $\mathcal{C}$ of block sequences and the cylinders $Z_w = c(w)10\{0,1\}^{\mathbb{N}}$, one for each finite word w . Index Z_w by the dyadic $x_w = 0.w1 \in \mathbb{D}$ in binary; this is a bijection. Let $f \in F$ have a tree-pair diagram with leaves $w_1, \dots, w_n$ and $w'_1, \dots, w'_n$, so that $f(0.w_k z) = 0.w'_k z$. Define $\hat{f}$ by the prefix replacements $c(w_k) \mapsto c(w'_k)$ and $c(v)10 \mapsto c(v')10$, where the interior vertices v of the first tree are matched in order with the interior vertices v' of the second. Both families of prefixes are complete prefix codes, because splitting a leaf w replaces $c(w)$ by $c(w0)$, $c(w)10$ and $c(w1)$; expanding a matched pair of leaves adds a matched pair of gaps and changes nothing else, so $\hat{f}$ depends only on f . It maps Z_w onto $Z_{w'}$ with $x_{w'} = f(x_w)$: below a leaf this is the formula for f , and interior vertices correspond to the breakpoints between leaves. On $\mathcal{C}$ it acts as f acts on binary sequences, so $f \mapsto \hat{f}$ is an injective homomorphism $F \rightarrow V$. Inside each Z_w let S_3 permute the three cylinders $c(w)10y\{0,1\}^{\mathbb{N}}$, $y \in \{0, 10, 11\}$. Copies with different w have disjoint supports, and $\hat{f}$ carries the copy at x to the copy at $f(x)$. An element of the group they generate that acts trivially on $\mathcal{C}$, so its F -part is trivial, and then each of its S_3 coordinates is trivial. $\square$

Let a, b generate the copy of S_3 at $\frac{1}{2}$, let $t \in F$ be

$$t(x) = \begin{cases} 2x, & 0 \leq x \leq \frac{1}{4}, \\ x + \frac{1}{4}, & \frac{1}{4} \leq x \leq \frac{1}{2}, \\ \frac{x+1}{2}, & \frac{1}{2} \leq x \leq 1, \end{cases}$$

so that $t(\frac{1}{2}) = \frac{3}{4}$, and for $m \geq 1$ let $\mathbb{D}_m$ be the set of the $2^m - 1$ points of $\mathbb{D}$ with denominator at most 2^m .

The group $\mathcal{L}$ is finitely presented, by Cornulier's criterion for permutational wreath products [17, Theorem 1.1] applied to the action of F on $\mathbb{D}$, whose hypotheses he verifies in [17, Example 3.4]. Indeed, Thompson's group F is finitely presented, and it acts transitively on increasing pairs of points of $\mathbb{D}$ [13, Lemma 4.2], so it has finitely many orbits on $\mathbb{D} \times \mathbb{D}$; and the stabilizer $F_{1/2}$ of $\frac{1}{2}$ is the product of the elements supported in $[0, \frac{1}{2}]$ and in $[\frac{1}{2}, 1]$, two copies of F , so it is finitely generated. The bounds below do not use this.

Proposition 3.3. *The group $\mathcal{L}$ contains F and contains no Houghton group H_m , so the bounds below do not come from the Houghton subgroups of V . There are a finite set R of relations of $\mathcal{L}$, containing $a^2, b^2, (ab)^3$, and a constant C with the following property: for every $m \geq 1$ there are words w_p , $p \in \mathbb{D}_m$, with $w_p(\frac{1}{2}) = p$ and*

$$\text{Area}_R([w_p x w_p^{-1}, w_q y w_q^{-1}]) \leq C m^2 \quad (p \neq q \in \mathbb{D}_m, x, y \in \{a, b\}).$$

Consequently a chunk E of $\mathcal{L}$, and hence of V , has $\log \sigma_E(r) \geq c r / \log^2 r$ for all large r , and for $0 < \Delta \leq 1$ every unitary assignment as in Theorem 2.1 has $\log D \geq c_\Delta e^{-2} / \log^4(1/e)$ for all small e , where $c > 0$ is a constant and $c_\Delta > 0$ depends only on Δ .

Proof. Transport. A standard dyadic interval is one of the form $[k2^{-j}, (k+1)2^{-j}]$. If two partitions of $[0, 1]$ into standard dyadic intervals have the same number n of intervals, the map sending the i th interval of the first affinely onto the i th interval of the second lies in F and has a tree-pair diagram with $n - 1$ carets, and every standard dyadic interval can be split into any given number of standard dyadic intervals. For $p < q$ in $\mathbb{D}_m$, the binary digits of p split $[0, p]$ and $[p, 1]$ into at most m standard dyadic intervals each, and $[p, q]$, cut at its point c with the shortest binary expansion, splits into at most m standard dyadic intervals on each side of c . Hence there are $w_p \in F$ with $w_p(\frac{1}{2}) = p$ and fewer than $2m$ carets, and

$g_{pq} \in F$ with $g_{pq}(\frac{1}{2}) = p$, $g_{pq}(\frac{3}{4}) = q$ and fewer than $4m$ carets. The word length in F is comparable to the number of carets of the reduced tree-pair diagram [12], so these elements are represented by words of length $O(m)$ in the generating set of the finite presentation of F used in R below; the Dehn functions of two finite presentations of F are equivalent, so Guba's quadratic bound holds for it. We write w_p also for such a word. An element fixing $\frac{1}{2}$ maps $[0, \frac{1}{2}]$ to itself, so the left and right subtrees of its reduced diagram are diagrams for its two restrictions, which are elements of F after affine rescaling, with fewer carets in total. Hence $F_{1/2} \cong F \times F$ is undistorted in F : its elements of length n in F are words of length $O(n)$ in a fixed finite generating set $h_1, \dots, h_q$ of $F_{1/2}$.

Relations. Let R consist of a finite presentation of F on a generating set that includes t and the h_j , the words $a^2, b^2, (ab)^3$, the commutators $[x, h_j]$, and the commutators $[x, tyt^{-1}]$, for $x, y \in \{a, b\}$. They hold in $\mathcal{L}$: each h_j fixes $\frac{1}{2}$, and the copies at $\frac{1}{2}$ and at $t(\frac{1}{2}) = \frac{3}{4}$ commute.

Area. Let $p < q$ in $\mathbb{D}_m$ and $g = g_{pq}$. The elements $g^{-1}w_p$ and $t^{-1}g^{-1}w_q$ fix $\frac{1}{2}$ and have length $O(m)$, so they are represented by words ω, ω' of length $O(m)$ in the h_j . The words $w_p^{-1}g\omega$ and $w_q^{-1}gt\omega'$ are null in F and have length $O(m)$, so Guba's quadratic Dehn function [24] fills each with $O(m^2)$ cells. Replacing the four occurrences of $w_p^{\pm 1}$ in $[w_p x w_p^{-1}, w_q y w_q^{-1}]$ by $(g\omega)^{\pm 1}$, and the four of $w_q^{\pm 1}$ by $(gt\omega')^{\pm 1}$, therefore costs $O(m^2)$ cells. Moving $x^{\pm 1}$ across ω and $y^{\pm 1}$ across ω' with the relations $[x, h_j]$ costs $O(m)$ cells and leaves $g[x, tyt^{-1}]g^{-1}$, a conjugate of a relation. For $p > q$ the commutator is the inverse of the one with the two copies exchanged. So every area is $O(m^2)$.

Profile. Apply Theorem 2.1 with $N = 2^m - 1$ and $\Lambda = Cm^2$, where $C \geq 1$, and let l be the largest length of a word of R . Given r , let m be the largest integer with $2^m m^2 \leq r/(10^8 C(2l - 1))$; then 2^m is at least of order $r/\log^2 r$, and $\log \sigma_E(r) \geq (2^m - 1)/16$. Given e , let m be the largest integer with $2^{m/2} m^2 \leq 1/(CK_\Delta e)$; then 2^m is at least of order $(CK_\Delta)^{-2} e^{-2}/\log^4(1/e)$, and $\log D \geq (2^m - 1)\Delta^2/12$.

No Houghton subgroup. A torsion element of $\mathcal{L}$ maps to a torsion element of the torsion-free group F , so it lies in $\bigoplus_{\mathbb{D}} S_3$, which has exponent six. Every H_m contains a five-cycle. Finally, $\mathcal{L}$ embeds in V by Lemma 3.2. $\square$

The subgroup $\mathcal{L}$ contains F , whose soficity is open, so we do not know whether this chunk has permutation models at all; if it does, they are very large. In particular, either V is not sofic, or its models of this chunk need $2^{c r/\log^2 r}$ points, close to the largest bound that Theorem 2.1 can give. If F is amenable, so is $\mathcal{L}$, and then this chunk has finite profile. If F is hyperlinear, so is $\mathcal{L}$ (Proposition 3.4), and then unitary models of the chunk exist at every defect. Slofstra's bounds are of the same kind: they bound models that may not exist, since his group is not known to be hyperlinear. Any unitary models of it need $\log D \gtrsim e^{-c}$ for every $c < \frac{1}{2}$ [48, Theorem 1.2 and the remark after it], and the proof of [48, Corollary 1.3] shows that any permutation models of a suitable chunk need $\log \sigma \gtrsim r^c$ for every $c < \frac{1}{4}$; these profiles may be infinite. The group $\mathcal{L}$ gives e^{-2} and r up to logarithms. Since $\mathbb{D}$ is the coset space $F/F_{1/2}$, the group $\mathcal{L} = (\bigoplus_{F/F_{1/2}} S_3) \rtimes F$ has the shape $(\bigoplus_{G/H} A) \rtimes G$ of the generalised wreath products that Kun and Thom prove non-sofic [39, Theorem A], with $A = \mathbb{Z}/2$, when $H < G$ is infranormal and not normal (the elements g with $gHg^{-1} \subseteq H$ generate G) and both H and G have property (T); here the input is instead cheap transport of non-abelian copies of S_3 .

Soficity and hyperlinearity pass to wreath products whose acting group is sofic [28], and to permutational wreath products whose acting group is sofic (respectively hyperlinear) and whose action is sofic [21, Theorems 3.6 and 3.8], [1]. Since F acts on $\mathbb{D}$ transitively and faithfully, a sofic action of F on $\mathbb{D}$ would already make F sofic [21, Theorem 2.12], so these results give nothing here; the next proposition assumes only that F is hyperlinear.

Proposition 3.4. *The group $\mathcal{L}$ is hyperlinear if and only if F is.*

Proof. Subgroups of hyperlinear groups are hyperlinear, so suppose that F is hyperlinear. A countable group Γ is hyperlinear if and only if its von Neumann algebra $L(\Gamma)$ has a trace-preserving embedding into an ultrapower $\mathcal{R}^\omega$ of the hyperfinite II_1 factor [14], and then every separable von Neumann subalgebra of $L(\Gamma)^\omega$ embeds in $\mathcal{R}^\omega$ as well, by a diagonal argument. It therefore suffices to embed $L(\mathcal{L})$ into $L(F)^\omega$ preserving the trace, for a free ultrafilter ω on $\mathbb{N}$. We write τ for the canonical traces and F_J for the subgroup of elements supported in an interval J . If J, J' are disjoint, $L(F_J)$ and $L(F_{J'})$ commute and $\tau(xy) = \tau(x)\tau(y)$ for $x \in L(F_J)$ and $y \in L(F_{J'})$, since F_J and $F_{J'}$ generate their direct product. If $f, f' \in F$ agree on J , then $\lambda(f)x\lambda(f)^{-1} = \lambda(f')x\lambda(f')^{-1}$ for $x \in L(F_J)$, because $f'^{-1}f$ fixes J pointwise and therefore commutes with F_J .

A localized copy of S_3 . Choose $\sigma \in F$ supported in $[\frac{1}{2}, \frac{3}{4}]$, with $\sigma(x) < x$ on $(\frac{1}{2}, \frac{3}{4})$ and $\sigma(x) = \frac{1}{2} + \frac{1}{2}(x - \frac{1}{2})$ near $\frac{1}{2}$, and a dyadic $x_0 \in (\frac{1}{2}, \frac{3}{4})$. The intervals $J_k = \sigma^k((\sigma(x_0), x_0))$, $k \in \mathbb{Z}$, are disjoint and accumulate at $\frac{1}{2}$ as $k \rightarrow \infty$. For a nonempty open interval J with dyadic endpoints, $F_J \cong F$ has infinite conjugacy classes, since a nontrivial element has conjugates with infinitely many different supports, so $L(F_J)$ is a II_1 factor. In $L(F_{J_0})$ choose a unital copy of M_6 and, in its relative commutant, which is again a II_1 factor, a self-adjoint Z_0 with uniform distribution on $[0, 1]$. Let $v_0 : S_3 \rightarrow U(M_6)$ be the regular representation, so that $\tau(v_0(g)) = 0$ for $g \neq 1$, and put $v_k = \lambda(\sigma)^k v_0 \lambda(\sigma)^{-k}$ and $Z_k = \lambda(\sigma)^k Z_0 \lambda(\sigma)^{-k}$, which lie in $L(F_{J_k})$. By the factorization of traces, the Z_k are independent and uniform. For a finite $K \subset \mathbb{Z}$ and $k \in K$, let p_k^K be the spectral projection of the event that Z_k is the largest of the Z_j , $j \in K$. Ties have trace zero, so these projections are orthogonal and sum to 1, and they commute with every $v_j(g)$. Hence

$$V_K(g) = \sum_{k \in K} p_k^K v_k(g)$$

is a representation of S_3 . Since $v_k(g)$ lies in a copy of M_6 whose relative commutant contains Z_k , and the other Z_j lie in algebras with factorized trace, $\tau(p_k^K v_k(g)) = \tau(p_k^K) \tau(v_0(g))$; so $\tau(V_K(g)) = 0$ for $g \neq 1$. If K is an interval of n integers and $|q| \leq n$, the two selected indices agree unless the largest Z_j over $K \cup (K + q)$ has its index outside $K \cap (K + q)$, an event of trace $2|q|/(n + |q|)$, and on its complement $V_K(g) = V_{K+q}(g)$. Hence $\|V_{K+q}(g) - V_K(g)\|_2^2 \leq 8|q|/(n + |q|)$.

Invariance. Put $V_n = V_{\{n, \dots, 2n-1\}}$, and let $h \in F_{1/2}$. Near $\frac{1}{2}$ on the right, $h(x) = \frac{1}{2} + 2^m(x - \frac{1}{2})$ for an integer m , so h agrees with σ^{-m} on J_k for all large k . Conjugation by $\lambda(h)$ therefore carries v_k and Z_k to v_{k-m} and Z_{k-m} for large k , and V_n to $V_{\{n, \dots, 2n-1\}-m}$ for large n , so $\|\lambda(h)V_n(g)\lambda(h)^{-1} - V_n(g)\|_2 \rightarrow 0$. Thus $V(g) = (V_n(g))_\omega$ is a representation of S_3 in $L(F)^\omega$ with $\tau(V(g)) = 0$ for $g \neq 1$, and it commutes with $\lambda(F_{1/2})$.

The embedding. For $x \in \mathbb{D}$ choose $f_x \in F$ with $f_x(\frac{1}{2}) = x$ and put $V_x = \lambda(f_x)V\lambda(f_x)^{-1}$. By the invariance this does not depend on the choice of f_x , and $\lambda(f)V_x\lambda(f)^{-1} = V_{f(x)}$ for $f \in F$. The representatives $V_n(g)$ lie in $L(F_{I_n})$, where $I_n = \bigcup_{n \leq k < 2n} J_k$ shrinks to $\frac{1}{2}$ from the right, so those of V_x lie in $L(F_{f_x(I_n)})$, and for $x \neq y$ the intervals $f_x(I_n)$ and $f_y(I_n)$ are disjoint for large n . So copies at distinct points commute, and the V_x together with λ define a homomorphism $\pi : \mathcal{L} \rightarrow U(L(F)^\omega)$. Let $\gamma = (\prod_{x \in X} \ell_x) f$, with X finite, ℓ_x in the copy of S_3 at x and $f \in F$. If $f = 1$ and some $\ell_x \neq 1$, the factors lie in commuting algebras with factorized trace for large n , so $\tau(\pi(\gamma)) = \prod_x \tau(V(\ell_x)) = 0$. If $f \neq 1$, choose a point moved by f outside X . For large n it lies outside the closure of every $f_x(I_n)$, so f is not in the group K_n generated by the $F_{f_x(I_n)}$, and $\tau(a\lambda(f)) = 0$ for every $a \in L(K_n)$, in particular for the representatives of $\prod_x \pi(\ell_x)$. Hence $\tau(\pi(\gamma)) = 0$ whenever $\gamma \neq 1$, and π extends to a trace-preserving embedding of $L(\mathcal{L})$ into $L(F)^\omega$. $\square$

3.3 From orbits to configurations

The two steps isolate what is needed: exponentially many sites reachable by short words, pairwise commutators of polylogarithmic area, and an amenable ambient group. In $\mathcal{G}$ the orbit of 0 grows exponentially, but we do not know how to commute two copies at points of opposite parity cheaply. In $\mathcal{L}$ everything works except amenability, which is the open question of whether F is amenable. The configuration lamps of the next section obtain all three at once by a change of viewpoint: the sites are finite binary configurations of the points of Houghton's three rays rather than points, the transport is by affine maps that touch a bounded number of coordinates at a time, and the ambient group is locally finite-by- $\mathbb{Z}^2$.

4 Configuration lamps

Theorem 2.1 turns M copies of S_3 whose pairwise commutators have area at most Λ into $\log \sigma_E(r) \geq M/16$ at r of order $M\Lambda$, so it approaches its ceiling $\exp(O(r))$ only when Λ is tiny compared with M . The group of this section places the copies of S_3 not on the points of Houghton's three rays but on their finitely supported binary configurations: Houghton's group moves the points, finitely supported affine maps move the configurations, and one copy of S_3 sits on each configuration. Then N points carry 2^N configurations, and hence 2^N copies of S_3 , while every local move we use involves at most three points. Two of the 2^N copies are compared by about N^2 local moves, each filled with the Dehn function of H_3 at scale N , so the area is polylogarithmic in the number of copies.

Throughout, functions are composed from right to left, groups act on the left, $[x, y] = xyx^{-1}y^{-1}$, and log is to base two.

Houghton's group. Let $Y = \{1, 2, 3\} \times \{1, 2, \dots\}$; the point (i, j) lies on ray i at height j . Houghton's group $H = H_3$ consists of the permutations g of Y that are eventually translations on each ray: there are integers $m_1(g), m_2(g), m_3(g)$ with $g(i, j) = (i, j + m_i(g))$ for all large j . The map $g \mapsto (-m_2(g), -m_3(g))$ is a surjective homomorphism $H \rightarrow \mathbb{Z}^2$ whose kernel is the group $\text{FSym}(Y)$ of finitary permutations [32]. Let p be the permutation with $p(1, j) = (1, j + 1)$, $p(2, 1) = (1, 1)$ and $p(2, j) = (2, j - 1)$ for $j \geq 2$, fixing ray 3, and let q be defined in the same way with rays 2 and 3 exchanged. They generate H (Section 5), and their images in $\mathbb{Z}^2$ are $(1, 0)$ and $(0, 1)$.

The group. Let $V = \mathbb{F}_2^{(Y)}$ be the space of finitely supported functions $Y \rightarrow \mathbb{F}_2$, the configurations, with basis $(e_y)_{y \in Y}$. The group H acts on V by permutation matrices, $g(e_y) = e_{g(y)}$, and we identify H with its image in $\text{GL}(V)$. Let $\text{GL}_{\text{fin}}(Y)$ be the group of linear automorphisms of V that fix all but finitely many basis vectors. It is the directed union of the finite groups $\text{GL}(\mathbb{F}_2^J)$, $J \subset Y$ finite, each extended by the identity on the other basis vectors; it is normalized by H ; and $H \cap \text{GL}_{\text{fin}}(Y) = \text{FSym}(Y)$. Let $L = \text{GL}_{\text{fin}}(Y)H \leq \text{GL}(V)$. Write τ_u for the translation $v \mapsto v + u$ of V , let $A = V \rtimes L$ be the group of affine maps $v \mapsto g(v) + u$ with $g \in L$ and $u \in V$, and let

$$\Gamma = \left(\bigoplus_{v \in V} S_3 \right) \rtimes A,$$

where A permutes the summands through its action on V : writing λ_v for the copy at v of $\lambda \in S_3$, we have $g\lambda_v g^{-1} = \lambda_{g(v)}$ for $g \in A$. Equivalently, Γ is the group of permutations of $V \times S_3$ generated by the maps $(v, s) \mapsto (g(v), s)$ for $g \in A$ and $(v, s) \mapsto (v, \lambda(v)s)$ for finitely supported $\lambda : V \rightarrow S_3$; this action is faithful. Groups obtained from a permutation action by adjoining the finitary linear or symmetric groups of the permuted set, and lamps over it, appear as enrichments [4, 1] and as halo products [23].

For distinct points $y, y' \in Y$ let $E_{y \leftarrow y'}$ be the transvection that sends $e_{y'}$ to $e_{y'} + e_y$ and fixes the other basis vectors. For $g \in H$ and a linear map M of V ,

$$g\tau_{e_y}g^{-1} = \tau_{e_{g(y)}}, \quad gE_{y \leftarrow y'}g^{-1} = E_{g(y) \leftarrow g(y')}, \quad M\tau_uM^{-1} = \tau_{M(u)}. \quad (1)$$

Linear maps fix 0, so they commute with the copy of S_3 at 0. On the first ray we abbreviate: for $x \geq 1$ we write x for the point $(1, x)$, e_x for $e_{(1, x)}$ and τ_x for τ_{e_x} .

Let a and b be the transpositions $(1\ 2)$ and $(2\ 3)$ in the copy of S_3 at 0, so that ab has order three, and put

$$t = \tau_1, \quad d = E_{2 \leftarrow 1}, \quad c = dab;$$

thus t adds the point 1 to a configuration, and d adds the point 2 to every configuration that contains the point 1.

Lemma 4.1. (a) *The element c has order six, $c^3 = d$ and $c^4 = ab$; hence $b = ac^4$.*

(b) *Γ is generated by the five elements p, q, t, c, a .*

(c) *Let $\pi : \Gamma \rightarrow \mathbb{Z}^2$ be the composite of $\Gamma \rightarrow A \rightarrow L \rightarrow L/\mathrm{GL}_{\mathrm{fin}}(Y) \cong H/\mathrm{FSym}(Y) \cong \mathbb{Z}^2$. Its kernel $(\bigoplus_V S_3) \rtimes (V \rtimes \mathrm{GL}_{\mathrm{fin}}(Y))$ is locally finite. Hence Γ is locally finite-by- $\mathbb{Z}^2$, and in particular elementary amenable.*

Proof. (a) The linear map d fixes 0, so it commutes with ab ; since d has order two and ab order three, c has order six, $c^3 = d^3(ab)^3 = d$ and $c^4 = d^4(ab)^4 = ab$.

(b) Let Γ' be the subgroup generated by the five elements. It contains H , because $H = \langle p, q \rangle$. Since $\mathrm{FSym}(Y)$ acts 2-transitively on Y , (1) shows that Γ' contains every τ_{e_y} , hence V , and, since $d = c^3$, every transvection $E_{y \leftarrow y'}$. Transvections generate every $\mathrm{GL}(\mathbb{F}_2^J)$, so $\mathrm{GL}_{\mathrm{fin}}(Y) \leq \Gamma'$, and $A \leq \Gamma'$. Finally $b = ac^4 \in \Gamma'$, and the copy of S_3 at $u \in V$ is $\tau_u \langle a, b \rangle \tau_u^{-1}$.

(c) Since $H \cap \mathrm{GL}_{\mathrm{fin}}(Y) = \mathrm{FSym}(Y)$, we have $L/\mathrm{GL}_{\mathrm{fin}}(Y) \cong H/\mathrm{FSym}(Y)$, and the kernel of π is as stated. It is an extension of $\bigoplus_V S_3$ by $V \rtimes \mathrm{GL}_{\mathrm{fin}}(Y)$, the directed union of the finite groups $\mathbb{F}_2^J \rtimes \mathrm{GL}(\mathbb{F}_2^J)$; both are locally finite. An extension of a locally finite group K by a locally finite group is locally finite: a finitely generated subgroup Λ has finite image modulo K , so $\Lambda \cap K$ has finite index in Λ , is finitely generated, and is finite. Locally finite groups are directed unions of finite groups, and elementary amenable groups are closed under directed unions and extensions [16]. $\square$

4.1 A finite presentation

The certificate of Section 6 is a finite set of relations and does not present Γ : the group it presents has abelianization $\mathbb{Z}^2 \times \mathbb{Z}/2 \times \mathbb{Z}/4$, in which c has order four (a computation with the exponent sums; for instance $a^2, b^2 = (ac^4)^2$ and $(ab)^3$ give $2a = 8c = 12c = 0$), while in Γ the element $c = dab$ is a product of commutators, since $d = [E_{2 \leftarrow 3}, E_{3 \leftarrow 1}]$ and ab is a three-cycle. The group itself is finitely presented.

Proposition 4.2 (Finite presentation). *The groups L , A and Γ are finitely presented.*

Proof. Stabilizers. For a finite set $D \subset Y$ the pointwise stabilizer $H_{(D)}$ of D in H is isomorphic to H : renumbering the points of each ray outside D in increasing order turns $Y \setminus D$ into three rays, and a permutation fixing D is eventually a translation on the old rays exactly when it is on the new ones. So $H_{(D)}$ is finitely generated. Since H contains $\mathrm{FSym}(Y)$, it acts transitively on ordered k -tuples of distinct points for every k .

A presentation lemma. Let a finitely presented group H act on a set $\mathcal{X}$ with finitely many orbits and finitely generated stabilizers, and let $F(\mathcal{X})$ be the free group on $\mathcal{X}$. Then $F(\mathcal{X}) \rtimes H$,

with H permuting the free generators, is finitely presented: adjoin to a presentation of H one representative x_i of each orbit and the relations $[x_i, k] = 1$ for k in a finite generating set of its stabilizer; then hx_ih^{-1} depends only on $h \cdot x_i$, and the two groups are isomorphic (both are the iterated amalgam of H with the groups $\text{Stab}(x_i) \times \langle x_i \rangle$ over the stabilizers $\text{Stab}(x_i)$). Imposing, in addition, a set of relations that is a finite union of H -orbits keeps the group finitely presented, since one representative of each orbit suffices.

A presentation of A . For distinct $i, j \in Y$ let x_{ij} be the transvection $e_j \mapsto e_j + e_i$, let $t_i = \tau_{e_i}$, and let $\varsigma_{ij} \in \text{FSym}(Y) \leq H$ be the transposition of i and j . Let $\tilde{A}$ be the quotient of $F(\mathcal{X}) \rtimes H$, where $\mathcal{X} = \{x_{ij}\} \sqcup \{t_i\}$ and H relabels indices, by the relations

$$\begin{aligned} x_{ij}^2 &= 1, & [x_{ij}, x_{kl}] &= 1 \ (i \neq l, j \neq k), & [x_{ij}, x_{jk}] &= x_{ik} \ (i, j, k \text{ distinct}), \\ \varsigma_{ij} &= x_{ij}x_{ji}x_{ij}, & t_i^2 &= 1, & [t_i, t_j] &= 1, \\ x_{ij}t_jx_{ij}^{-1} &= t_it_j, & x_{ij}t_kx_{ij}^{-1} &= t_k \ (k \neq j). \end{aligned}$$

They hold in A , so there is a surjection $\tilde{A} \rightarrow A$. Each relation involves at most four points, so by transitivity the relations form finitely many H -orbits; the stabilizers of x_{ij} and t_i are $H_{(\{i,j\})}$ and $H_{(\{i\})}$. Since H is finitely presented [10], for instance by Johnson's presentation (4) [34], $\tilde{A}$ is finitely presented, and it remains to show that $\tilde{A} \rightarrow A$ is injective. The linear relations are Steinberg's [43]; we give the short direct argument.

Fix a finite set $D \subset Y$, number its points $1, \dots, m$ (a numbering unrelated to heights), and let $U_D \leq \tilde{A}$ be generated by the x_{ij} with $i < j$ in D . By the first three relations, $x_{1m}, \dots, x_{m-1,m}$ are commuting involutions, and they generate a subgroup normalized by the x_{ij} with $i < j < m$; by induction $|U_D| \leq 2^{m-1}2^{(m-1)(m-2)/2} = 2^{m(m-1)/2}$, the order of the upper unitriangular group $\text{UT}_m(\mathbb{F}_2)$, onto which U_D maps. So $U_D \cong \text{UT}_m(\mathbb{F}_2)$. Let $W_D = \text{Sym}(D) \leq H$ and $\varsigma_i = \varsigma_{i,i+1}$. By the relation $\varsigma_{ij} = x_{ij}x_{ji}x_{ij}$ and relabeling, the subgroup G_D generated by all x_{ij} with $i, j \in D$ contains W_D and is generated by U_D and the ς_i . We claim $G_D = U_D W_D U_D$; since this set contains 1, it suffices that it is closed under right multiplication by U_D and by each ς_i . Every $b \in U_D$ is $x_{i,i+1}^\epsilon b_0$ with b_0 in the subgroup generated by the other positive roots, the kernel of the entry $(i, i+1)$, which ς_i normalizes. So it suffices to treat $w x_{i,i+1} \varsigma_i$ with $w \in W_D$; put $k = w(i)$ and $l = w(i+1)$. If $k < l$, then $w x_{i,i+1} \varsigma_i = x_{kl} w \varsigma_i \in U_D W_D$. If $k > l$, then $w \varsigma_i = \varsigma_{kl} w$ and $x_{kl} \varsigma_{kl} = x_{lk} x_{kl}$, so $w x_{i,i+1} \varsigma_i = x_{kl} \varsigma_{kl} w = x_{lk} w x_{i,i+1} \in U_D W_D U_D$. The map $G_D \rightarrow \text{GL}(\mathbb{F}_2^D)$ is onto, since transvections generate $\text{GL}(\mathbb{F}_2^D)$. If $b_1 w b_2$ maps to the identity of $\text{GL}(\mathbb{F}_2^D)$, the permutation matrix of w is upper unitriangular, so $w = 1$ in H , and then $b_1 b_2 = 1$ in U_D . Hence $G_D \cong \text{GL}(\mathbb{F}_2^D)$. Every element of the subgroup $\tilde{J}$ generated by all x_{ij} lies in some G_D , so $\tilde{J} \cong \text{GL}_{\text{fin}}(Y)$. By relabeling, every element of the subgroup generated by $\tilde{J}$ and H has the form jh ; if it maps to 1, then $h \in \text{GL}_{\text{fin}}(Y) \cap H = \text{FSym}(Y)$, so h is a product of transpositions, each equal in $\tilde{A}$ to a word in the x_{ij} , and $jh \in \tilde{J}$ maps to 1, hence is 1. Finally the t_i generate a normal elementary abelian subgroup, which maps isomorphically onto V (it is a quotient of $\mathbb{F}_2^{(Y)}$, and $t_i \mapsto e_i$), and the linear part acts on it as on V , so every element of $\tilde{A}$ is $t\ell$ with t in that subgroup; if its image is the identity affine map, evaluating at 0 gives $t = 1$, and then $\ell = 1$. So $\tilde{A} \cong A$. Omitting the t_i gives L .

The lamps. The group A acts on V doubly transitively: a translation carries v to 0, and a finitary linear map carries any nonzero vector to e_1 . The stabilizer of 0 is L , which is generated by p, q and d , because the H -conjugates of d are all the transvections $E_{y \leftarrow y'}$. So A is finitely presented, its point stabilizers are finitely generated, and it has two orbits on $V \times V$. By Cornulier's criterion for permutational wreath products [17, Theorem 1.1], $\Gamma = S_3 \wr_V A$ is finitely presented. Explicitly, if $\mathcal{R}_A$ presents A on generators including p, q, d, t , then adding the generators a, b and the relations $a^2, b^2, (ab)^3, [a, \ell]$ and $[b, \ell]$ for $\ell \in \{p, q, d\}$, and $[u, tvt^{-1}]$ for $u, v \in \{a, b\}$, presents Γ : the lamp at $v \in V$ is well defined as a conjugate of

$\langle a, b \rangle$ because L centralizes it, and double transitivity carries the commutation of the lamps at 0 and e_1 to every pair. $\square$

4.2 An embedding in a Brin–Thompson group

Let $\mathfrak{C} = \{0, 1\}^{\mathbb{N}}$ be Cantor space. For $n \geq 1$, Brin’s group nV [7] consists of the homeomorphisms h of $\mathfrak{C}^n$ given by a finite table of prefix replacements: there are two partitions of $\mathfrak{C}^n$ into the same number of rectangles $u_1\mathfrak{C} \times \cdots \times u_n\mathfrak{C}$, with $u_1, \dots, u_n$ finite binary words, matched in pairs, and h maps each rectangle of the first partition onto its partner by replacing the n prefixes and keeping the n suffixes. So $1V$ is Thompson’s group V , and nV embeds in $n'V$ for $n \leq n'$ by acting trivially on the additional coordinates. The groups nV are finitely presented, by Brin for $2V$ [8] and by Hennig and Matucci for every n [29], and simple [9], and, as for V , it is not known whether they are sofic. (The letter V in these names is not the space of configurations.)

Proposition 4.3. Γ embeds in $3V$, and hence in nV for every $n \geq 3$. Consequently $3V$ has a chunk E_{3V} with

$$\exp\left(\frac{cr}{(\log r)^{12.75}}\right) \leq \sigma_{E_{3V}}(r) \leq \exp(Cr \log r) \quad (r \geq 2).$$

Proof. Coding configurations. For a finite set $S = \{j_1 < \cdots < j_m\}$ of positive integers, with $j_0 = 0$, let

$$e(S) = \left(\prod_{k=1}^m 10^{j_k - j_{k-1} - 1} 1 \right) 0,$$

so that $e(\emptyset) = 0$, $e(\{1\}) = 110$ and $e(\{2\}) = 1010$. Read a word from the left: at a record boundary the letter 0 ends the word and the letter 1 opens a record $10^k 1$, which ends at its next letter 1. So the words $e(S)$ form a prefix code. A finite word whose reading has not ended can be extended to one that begins with some $e(S)$, by appending 0 at a record boundary or 10 inside a record. Hence the cylinders $e(S)\mathfrak{C}$ are disjoint and their union is dense in $\mathfrak{C}$. For $v \in V$ and $i = 1, 2, 3$ let $S_i(v) = \{j : v(i, j) = 1\}$, let

$$Z_v = e(S_1(v))\mathfrak{C} \times e(S_2(v))\mathfrak{C} \times e(S_3(v))\mathfrak{C},$$

and let $\iota_v : \mathfrak{C}^3 \rightarrow Z_v$ prefix the three codewords. The rectangles Z_v are disjoint, and their union is dense in $\mathfrak{C}^3$.

Prepending a site. Define $s_0, s_1 : \mathfrak{C} \rightarrow \mathfrak{C}$ by $s_0(0\xi) = 0\xi$, $s_0(1\xi) = 10\xi$ and $s_1(\xi) = 11\xi$. Their images $0\mathfrak{C} \cup 10\mathfrak{C}$ and $11\mathfrak{C}$ partition $\mathfrak{C}$, and, with $S + 1 = \{j + 1 : j \in S\}$,

$$s_0(e(S)\xi) = e(S + 1)\xi, \quad s_1(e(S)\xi) = e(\{1\} \cup (S + 1))\xi : \quad (2)$$

an empty site in front lengthens the first gap by one, and an occupied site in front adds the record 11.

The affine group. For $b \in \{0, 1\}$ put

$$\begin{aligned} \hat{P}(x, s_b(y), z) &= (s_b(x), y, z), & \hat{Q}(x, y, s_b(z)) &= (s_b(x), y, z), \\ \hat{T}(s_b(x), y, z) &= (s_{1-b}(x), y, z), \end{aligned}$$

and let $\hat{D}$ fix $s_0(\mathfrak{C}) \times \mathfrak{C}^2$ pointwise and send $(11x', y, z)$ to $(11x', y, z)$, where $\hat{T}(x, y, z) = (x', y, z)$. These maps lie in $3V$. With ϵ the empty word, $\hat{P}$ replaces the prefixes $(\epsilon, 11, \epsilon)$, $(0, 0, \epsilon)$, $(1, 0, \epsilon)$, $(0, 10, \epsilon)$ and $(1, 10, \epsilon)$ by $(11, \epsilon, \epsilon)$, $(0, 0, \epsilon)$, $(10, 0, \epsilon)$, $(0, 1, \epsilon)$ and $(10, 1, \epsilon)$; $\hat{Q}$ does the same with the roles of the second and third coordinates exchanged; $\hat{T}$ exchanges

the first-coordinate prefixes $0 \leftrightarrow 110$ and $10 \leftrightarrow 111$; and $\hat{D}$ exchanges $110 \leftrightarrow 11110$ and $1110 \leftrightarrow 11111$ and keeps 0 and 10 . The permutation p moves the first point of ray 2 to the first point of ray 1 and shifts the other points of these rays, q does the same with ray 3, t changes the value of a configuration at the point 1, and d adds the value at 1 to the value at 2. So (2) gives, for every $v \in V$,

$$\hat{P}\iota_v = \iota_{p(v)}, \quad \hat{Q}\iota_v = \iota_{q(v)}, \quad \hat{T}\iota_v = \iota_{t(v)}, \quad \hat{D}\iota_v = \iota_{d(v)}.$$

By the proof of Lemma 4.1(b), p, q, t, d generate A . If a word in them represents $g \in A$, the same word in $\hat{P}, \hat{Q}, \hat{T}, \hat{D}$ maps $\iota_v(\xi)$ to $\iota_{g(v)}(\xi)$ for all v and ξ ; if $g = 1$, it is the identity on the dense set $\bigcup_v Z_v$, hence the identity. So $p, q, t, d \mapsto \hat{P}, \hat{Q}, \hat{T}, \hat{D}$ defines a homomorphism $\rho : A \rightarrow 3V$ with

$$\rho(g)\iota_v = \iota_{g(v)} \quad (g \in A, v \in V), \quad (3)$$

and ρ is injective: if $g \neq 1$, then $g(v) \neq v$ for some v , and $\rho(g)$ maps Z_v onto the disjoint rectangle $Z_{g(v)}$.

The lamps. Let $\beta_1 = 0$, $\beta_2 = 10$ and $\beta_3 = 11$, and let S_3 act faithfully on $\mathfrak{C}^3$ by the prefix replacements $\theta(\lambda)(\beta_i \xi, y, z) = (\beta_{\lambda(i)} \xi, y, z)$. For $\lambda \in S_3$ and $v \in V$ let $\rho(\lambda_v)$ be $\iota_v \theta(\lambda) \iota_v^{-1}$ on Z_v and the identity off Z_v ; it lies in $3V$, since the complement of a rectangle is a finite union of rectangles. Copies at different configurations have disjoint supports and commute, each copy is faithful, and by (3) $\rho(g)\rho(\lambda_v)\rho(g)^{-1} = \rho(\lambda_{g(v)})$ for $g \in A$. These are the defining relations of Γ as a semidirect product, so ρ extends to a homomorphism $\rho : \Gamma \rightarrow 3V$. Suppose that $\rho(\gamma) = 1$, where γ has lamp part $\lambda : V \rightarrow S_3$ and affine part g . The lamps preserve every Z_v , so $\rho(g)$ does too, and $g(v) = v$ for all v ; so $g = 1$. Then $\rho(\gamma)$ restricts on Z_v to $\iota_v \theta(\lambda(v)) \iota_v^{-1}$, so $\lambda(v) = 1$ for all v . Hence ρ is injective. Explicitly, $\rho(a)$ exchanges the rectangles $00\mathfrak{C} \times 0\mathfrak{C} \times 0\mathfrak{C}$ and $010\mathfrak{C} \times 0\mathfrak{C} \times 0\mathfrak{C}$, and $\rho(c) = \hat{D}\rho(ab)$, where $\rho(ab)$ permutes the rectangles with prefixes $(00, 0, 0)$, $(010, 0, 0)$ and $(011, 0, 0)$ cyclically in this order and $\hat{D}$ is the identity on Z_0 .

The chunk. The sofic profile of a chunk depends only on its identity element and its partial multiplication (Section 2), and an injective homomorphism preserves both. So the chunk $E_{3V} = \rho(E)$, for the chunk E of Theorem 7.5, has $\sigma_{E_{3V}} = \sigma_E$, and parts (a) and (b) of that theorem give the bounds. $\square$

The script `verification/embedding3v_check.py` composes the prefix tables exactly and checks the identities of the proof and that the images of p, q, t, c, a satisfy the 25 relations of Table 2.

By Section 3.2, Thompson's group V has chunks with finite superpolynomial profile, from its Houghton subgroups, and a chunk with profile at least $\exp(cr/\log^2 r)$, which is finite if F is amenable. Proposition 4.3 gives finite, nearly exponential profile in $3V$ with no assumption. We do not know whether Γ embeds in V or in $2V$ (Section 11).

5 Houghton inputs

We need three facts about H : short words for finite permutations, generators of point stabilizers with linear word length, and a bound on the Dehn function. All transport below takes place among the first points of ray 1, so we only need words for permutations supported there. For $n \geq 1$ let $\Sigma_n \leq H$ be the group of permutations of Y supported in $\{(1, 1), \dots, (1, n)\}$, identified with $\text{Sym}(n)$; we write $(x \ y)$ for the transposition of the points x and y of the first ray.

Conventions. Put $\alpha = [p, q]$; a direct check shows $\alpha = (1\ 2)$. In [19] permutations act on the right, and Johnson's presentation of H [34] (see also [40, Theorem 2.8]) is written there in generators a, b, α that are the inverses, as maps, of our p, q, α . The map $g \mapsto g^{-1}$ is an isomorphism from the permutations of Y with the product of [19] to the permutations of Y under composition, and it sends a, b, α to p, q, α . So a word in a, b, α represents 1 in the conventions of [19] if and only if the same word in p, q, α represents 1 here. Hence the relators of [19, Section 2], read in our letters, give the presentation

$$P_J = \langle p, q, \alpha \mid \alpha^2, (\alpha p \alpha p^{-1})^3, [\alpha, p^2 \alpha p^{-2}], [p, q] \alpha^{-1}, p \alpha p^{-1} q \alpha^{-1} q^{-1} \rangle \quad (4)$$

of H ; in particular $H = \langle p, q \rangle$. Renaming the letters is an isomorphism of free groups that carries the relators of [19] to those of P_J , so every area computed in [19] for Johnson's presentation holds verbatim for P_J .

Eliminating α by the relator $[p, q] \alpha^{-1}$, a Tietze transformation, gives the presentation

$$P_H = \langle p, q \mid s^2, (s p s p^{-1})^3, [s, p^2 s p^{-2}], p s p^{-1} q s^{-1} q^{-1} \rangle, \quad s = [p, q], \quad (5)$$

of H . Let δ_H be its Dehn function: $\delta_H(n)$ is the largest area, with respect to the four relators of P_H , of a word of length at most n in $p^{\pm 1}, q^{\pm 1}$ that represents 1 in H . It is nondecreasing.

Lemma 5.1 (Finite permutations). *Put $\sigma_k = p^k \alpha p^{-k}$ for $k \geq 0$.*

- (a) $\sigma_k = (k+1\ k+2)$, and for $1 \leq x < y$ the word $\omega_{x,y} = p^{x-1}(\alpha p)^{y-x-1}(\alpha p^{-1})^{y-x-1} \alpha p^{1-x}$ represents $(x\ y)$. With $\alpha = p q p^{-1} q^{-1}$ it has length at most $10y - 8x - 8 < 10y$ in p, q .
- (b) Every element of Σ_n that moves $m \geq 2$ points is represented by a word in p, q of length less than $10(m-1)n$.
- (c) Let $x_1, \dots, x_l \in \{1, \dots, n\}$ be distinct. Put $\pi_0 = 1$ and, for $m = 1, \dots, l$, $w_m = \pi_{m-1}^{-1}(x_m)$ and $\pi_m = \pi_{m-1}(m\ w_m)$, where $(m\ m) = 1$. Then $\pi[x_1, \dots, x_l] := \pi_l$ sends m to x_m for $m \leq l$, it is a product of at most l transpositions of points in $\{1, \dots, n\}$, and the product of the corresponding words ω represents it with length less than $10ln$.

Proof. (a) Since p^k maps $(1, 1), (1, 2)$ to $(1, k+1), (1, k+2)$, σ_k is the stated transposition. Hence $(x\ y) = \sigma_{x-1} \sigma_x \cdots \sigma_{y-3} \sigma_{y-2} \sigma_{y-3} \cdots \sigma_{x-1}$, by induction on $y-x$ from $(x\ y) = \sigma_{x-1}(x+1\ y) \sigma_{x-1}$. In the product the powers of p telescope: $\sigma_k \sigma_{k+1} = p^k \alpha p \alpha p^{-k-1}$ and $\sigma_{k+1} \sigma_k = p^{k+1} \alpha p^{-1} \alpha p^{-k}$, which gives $\omega_{x,y}$ after free reduction. Its length is $2(x-1) + 10(y-x-1) + 4$.

(b) A permutation moving m points is a product of at most $m-1$ transpositions of those points, each of length less than $10n$ by (a).

(c) Since $\pi_{m-1}(l) = x_l \neq x_m$ for $l < m$, we have $w_m \geq m$, so $(m\ w_m)$ fixes $1, \dots, m-1$ and $\pi_m(l) = x_l$ for $l \leq m$. Each factor has length less than $10n$ by (a). $\square$

That a point stabilizer of a Houghton group is isomorphic to the group is known [17, Example 3.6]; part (a) of the next lemma gives such isomorphisms explicitly on generators.

Lemma 5.2 (Point stabilizers). *For $k \geq 1$ let $H^{(k)}$ be the pointwise stabilizer in H of $(1, 1), \dots, (1, k)$, put $\zeta_k = \sigma_{k-1} \cdots \sigma_1 \sigma_0$ and $K_k = (\zeta_k p, \zeta_k q)$.*

- (a) *There is an isomorphism $H \rightarrow H^{(k)}$ that sends p, q, α to $\zeta_k p, \zeta_k q, \sigma_k$ and the transposition $(x\ y)$ to $(x+k\ y+k)$. In particular $\zeta_k p$ and $\zeta_k q$ generate $H^{(k)}$.*
- (b) *Every element of $\Sigma_n \cap H^{(k)}$ that moves $m \geq 2$ points is represented by a word of length less than $10(m-1)n$ in $\zeta_k p$ and $\zeta_k q$.*

In p, q the words $\zeta_1 p = [p, q] p$ and $\zeta_2 p = p[p, q] p^{-1} [p, q] p$ have length at most 5 and 11, and likewise for q .

Proof. (a) Let $\theta_k : Y \rightarrow Y \setminus \{(1, 1), \dots, (1, k)\}$ send $(1, j)$ to $(1, j + k)$ and fix rays 2 and 3. For $g \in H$ let $g^{[k]}$ act as $\theta_k g \theta_k^{-1}$ on the image of θ_k and fix $(1, 1), \dots, (1, k)$. Then $g^{[k]}$ is eventually a translation on each ray, and $g \mapsto g^{[k]}$ is an isomorphism $H \rightarrow H^{(k)}$, since an element of $H^{(k)}$ restricts to a permutation of the image of θ_k . Clearly $(x y)^{[k]} = (x + k \ y + k)$, so $\alpha^{[k]} = \sigma_k$. The element ζ_k is the cycle $(1, 1) \mapsto (1, k + 1) \mapsto (1, k) \mapsto \dots \mapsto (1, 2) \mapsto (1, 1)$. Hence $\zeta_k p$ fixes $(1, j)$ for $j \leq k$, maps $(1, j)$ to $(1, j + 1)$ for $j > k$, maps $(2, 1)$ to $(1, k + 1)$, and agrees with p elsewhere; that is, $\zeta_k p = p^{[k]}$. Likewise $\zeta_k q = q^{[k]}$.

(b) An element $g \in \Sigma_n \cap H^{(k)}$ moving m points is $g'^{[k]}$ for some $g' \in \Sigma_{n-k}$ moving m points. Write g' as in Lemma 5.1(b) and substitute $\zeta_k p, \zeta_k q$ for p, q . $\square$

Proposition 5.3 (Dehn function; (a) from Lee [40, Theorem D] and (b) from [19, Theorems 3.2 and 3.3], by a Tietze transformation). *Let δ_J be the Dehn function of P_J . Then $\delta_H \leq \delta_J$, and:*

- (a) *there is an absolute constant K with $\delta_H(n) \leq e^{Kn}$ for $n \geq 1$;*
- (b) *there is an absolute constant C with $\delta_H(n) \leq Cn^{p_H}$ for $n \geq 1$, where $p_H = \log 107 + 4 \approx 10.74$.*

Proof. Let w be a word in p, q of length at most n representing 1. In the free group on p, q, α it is a product of at most $\delta_J(n)$ conjugates of relators of P_J and their inverses. Apply the homomorphism to the free group on p, q that fixes p, q and sends α to $[p, q]$. It fixes w , sends $[p, q]\alpha^{-1}$ to the trivial word, and sends each of the other four relators to a relator of P_H . Hence $\text{Area}_{P_H}(w) \leq \delta_J(n)$.

(a) Lee proved that H_3 satisfies an exponential isoperimetric inequality [40, Theorem D]. The Dehn functions δ, δ' of two finite presentations of the same group are equivalent: $\delta(n) \leq C'\delta'(C'n) + C'n + C'$ for a constant C' [6]. This bound preserves the class of functions bounded by e^{Kn} .

(b) By [19, Theorem 3.3], $\delta_J(n) \leq 400n^4(1 + A^*(4n + 2))$, where the far-commutator area satisfies $A^*(M) \leq 135M^{\log 107}$ [19, Theorem 3.2]; so $\delta_J(n) = O(n^{\log 107 + 4})$. By the discussion of conventions above this holds for P_J in our letters, and $\delta_H \leq \delta_J$. $\square$

Part (a) uses only the published exponential bound, and part (b) our polynomial one. Either enters the lower bounds below only through δ_H at scale N , the number of points that carry the configurations.

6 A certificate of 25 relations and local fillings

All fillings in Γ are built from one finite set R of words in the five generators of Lemma 4.1, each representing 1 in Γ . The set R does not present Γ (Section 4.1); areas are always taken with respect to R .

Abbreviations. Put

$$s = [p, q], \quad B = psp^{-1}, \quad t_m = p^{m-1} t p^{1-m} \quad (m = 1, 2, 3), \quad d = c^3, \quad b = ac^4.$$

By Lemma 5.1, s and B are the transpositions $(1\ 2)$ and $(2\ 3)$, and t_m represents τ_m , since $p^{m-1}(1, 1) = (1, m)$. By Lemma 5.2, with $\zeta_1 = s$ and $\zeta_2 = Bs$, the pairs

$$K_1 = (sp, sq), \quad K_2 = (Bsp, Bsq)$$

generate the pointwise stabilizers H_1 of the point 1 and H_2 of the points 1, 2. The elements t and d have the *prototypes* $P_t = \{1\}$ and $P_d = \{1, 2\}$, the points on which they depend: by (1), an element of H that fixes P_f pointwise commutes with f .

	relations	number	length
H	$s^2, (sps p^{-1})^3, [s, p^2 sp^{-2}], psp^{-1}qs^{-1}q^{-1}$	4	64
Q	t^2	1	2
S	$[t, k]$ for $k \in K_1$; $[d, k]$ for $k \in K_2$	4	56
Z	$[g, x]$ for $g \in \{p, q\}, x \in \{a, b\}$; $[d, a]$	5	40
C	$[t_1, t_2]$	1	8
A	$dt_1 d^{-1}(t_1 t_2)^{-1}, [d, t_2], [d, t_3]$	3	39
L	$a^2, b^2, (ab)^3$	3	30
P	$[x, tyt^{-1}]$ for $x, y \in \{a, b\}$	4	64
	total	25	303

Table 2: The certificate R : eight families of relations in the five generators p, q, t, c, a , written with the abbreviations above. Each relation is taken to be the freely reduced word obtained by expanding the abbreviations; lengths are those of these words, and the longest has length 24.

Lemma 6.1. *Every word of Table 2 represents 1 in Γ .*

Proof. H: the words of (5) hold in H . Q: V has exponent two. S: the elements of H_1 fix the point 1, so they commute with $t = \tau_1$; those of H_2 fix the points 1 and 2, so they commute with $d = E_{2 \leftarrow 1}$ (Lemma 4.1(a) and (1)). Z: p, q and d are linear, so they commute with a and with $b = ac^4 = a \cdot ab$, which lie in the copy of S_3 at 0. C: translations commute. A: $dt_m d^{-1} = \tau_{d(e_m)}$, and $d(e_1) = e_1 + e_2$, while d fixes e_2 and e_3 . L: these are relations of S_3 . P: tyt^{-1} lies in the copy of S_3 at e_1 , which commutes with the copy at 0. $\square$

The 25 freely reduced words are listed in the file `verification/relators1d.txt`. The script `verification/relators1d_check.py` rebuilds each word from the abbreviations and checks it, independently of Lemma 6.1, in the faithful action of Γ on $V \times S_3$. For a word w of length l it checks that the exponent sums of p and q vanish; that w fixes the empty configuration and every configuration $\{y\}$ with y of height at most $l + 2$ or far out on a ray; and that it fixes (v, s) for every $s \in S_3$ at every configuration v at which one of its lamp letters acts. These checks are exhaustive: while w is applied, a point of height greater than $l + 2$ never reaches the points 1 and 2 and is moved only by translations along its ray, so the checks determine the affine part of w , and then the lamps of w sit at the configurations just listed. The script also checks random configurations, that the abbreviations act as stated, and, as a negative control, that deleting any single letter from a sample of eleven relations produces a word that acts nontrivially.

6.1 Local fillings

Let $n \geq 3$. We use three elementary facts about areas. The area is invariant under free equality, inversion and conjugation, and subadditive under products. If N represents 1, then $\mu N \nu = (\mu N \mu^{-1}) \mu \nu$ for all words μ, ν ; so replacing a subword U by U' changes the area by at most $\text{Area}_R(UU'^{-1})$. And $[w, z_1 z_2] = [w, z_1] \cdot z_1 [w, z_2] z_1^{-1}$ and $[w, z^{-1}] = z^{-1} [w, z]^{-1} z$, so $\text{Area}_R([w, z])$ is at most the sum of the areas $\text{Area}_R([w, z_l])$ over the letters z_l of any factorization $z = z_1^{\pm 1} \cdots z_m^{\pm 1}$.

Canonical transporters. For points $i \neq j$ in $\{1, \dots, n\}$ put

$$U_i = \pi[i], \quad \tau_i = U_i t U_i^{-1}, \quad U_{ij} = \pi[j, i], \quad X_{ij} = U_{ij} d U_{ij}^{-1},$$

with the words of Lemma 5.1(c). We use τ_i both for this word and for the translation by e_i that it represents, since $U_i(1) = i$; and X_{ij} represents $E_{i \leftarrow j}$, since U_{ij} maps 1 to j and 2 to i .

By Lemma 5.1, $|U_i| < 10n$ and $|U_{ij}| < 20n$, and $U_i (U_{ij})$ is a product of at most one (two) transpositions.

The next lemma says that it does not matter which short word transports a prototype: two such words differ by an element of a stabilizer, which commutes with the transported element by the family S. We allow the transporting word to end with a power of p , because the relations of Table 2 use $t_2 = ptp^{-1}$ and $t_3 = p^2tp^{-2}$.

Lemma 6.2 (Change of transporter). *Let $n \geq 3$ and $f \in \{t, d\}$. Let U and W_0 be words in $p^{\pm 1}, q^{\pm 1}$ of length less than $30n$ that represent elements of Σ_n , such that $W_0^{-1}U$ moves at most 10 points, and let $W = W_0p^j$, where $j = 0$ if $f = d$ and $j \in \{0, 1, 2\}$ if $f = t$. If W and U agree on P_f , then*

$$\text{Area}_R(WfW^{-1} \cdot Uf^{-1}U^{-1}) \leq \mathcal{E}(n) := 2\delta_H(1500n) + 130n.$$

Proof. Let $k = |P_f|$ and let g be the element represented by $W^{-1}U$; it fixes P_f pointwise, so $g \in H^{(k)}$. If $j = 0$, then $g \in \Sigma_n$ moves at most 10 points, and by Lemma 5.2(b) it is represented by a word Z with fewer than $90n$ letters from $K_k^{\pm 1}$. If $j \geq 1$, then $f = t$, $k = 1$, and $g = (sp)^{-j}g'$ with $g' = (sp)^jp^{-j} \cdot W_0^{-1}U$. Here $(sp)p^{-1} = s$ and $(sp)^2p^{-2} = sB$ lie in Σ_3 , so $g' \in \Sigma_n$ moves at most 13 points; and g' fixes the point 1, since g and sp do. By Lemma 5.2(b), g' is represented by a word Z' with fewer than $120n$ letters from $K_1^{\pm 1}$, and we put $Z = (sp)^{-j}Z'$. In both cases Z has fewer than $122n$ letters from $K_k^{\pm 1}$, each of length at most 11 in p, q .

Let $N = W^{-1}UZ^{-1}$. It represents 1 in H and has length less than $30n + 2 + 30n + 1342n \leq 1500n$, so by the family H its area is at most $\delta_H(1500n)$. By the family S, $\text{Area}_R([f, Z]) < 122n$. Since $U = WNZ$ freely,

$$WfW^{-1} \cdot Uf^{-1}U^{-1} = W(fNf^{-1})[f, Z]N^{-1}W^{-1},$$

whose area is at most $2\delta_H(1500n) + 122n \leq \mathcal{E}(n)$. □

Lemma 6.3 (Local fillings). *Let $n \geq 3$ and*

$$L(n) = 10\delta_H(1500n) + 700n.$$

Let $i, j, k \in \{1, \dots, n\}$ be points with $i \neq j$, and $x, y \in \{a, b\}$, where b stands for the word ac^4 . The following words represent 1, and each has area at most $L(n)$ with respect to R :

- (a) τ_i^2 , of area 1, and $[\tau_i, \tau_j]$;
- (b) $X_{ij}\tau_k X_{ij}^{-1}\tau_k^{-1}$ if $k \neq j$, and $X_{ij}\tau_j X_{ij}^{-1}\tau_i^{-1}\tau_j^{-1}$;
- (c) $[X_{ij}, x]$, of area less than $40n + 1$;
- (d) $[x, \tau_i y \tau_i^{-1}]$, of area less than $40n + 1$.

Proof. All the words represent 1 by (1). Call replacing a subword UfU^{-1} by WfW^{-1} as in Lemma 6.2 a *change of transporter*; it costs at most $\mathcal{E}(n)$, and $5\mathcal{E}(n) + 1 \leq L(n)$. Below, W_0 is always a product of at most three transpositions, of length less than $30n$, and $U \in \{U_i, U_j, U_k, U_{ij}\}$ is a product of at most two, so $W_0^{-1}U$ moves at most 10 points.

(a) $\tau_i^2 = U_i t^2 U_i^{-1}$ freely, a conjugate of the relation Q. For $[\tau_i, \tau_j]$ let $W_0 = \pi[i, j]$. Replace both occurrences of $\tau_i^{\pm 1}$ by $(W_0 t W_0^{-1})^{\pm 1}$, and both occurrences of $\tau_j^{\pm 1}$ by $(W_0 t_2 W_0^{-1})^{\pm 1} = (W_0 p t p^{-1} W_0^{-1})^{\pm 1}$. These are four changes of transporter, since W_0 and U_i both map 1 to i , and $W_0 p$ and U_j both map 1 to j . The result $W_0[t_1, t_2]W_0^{-1}$ is a conjugate of the relation C, so the area is at most $4\mathcal{E}(n) + 1$.

(b) Let $W_0 = \pi[j, i, k]$, or $\pi[j, i]$ if $k \in \{i, j\}$, so that W_0 maps 1 to j , 2 to i , and 3 to k when $k \notin \{i, j\}$; let $m \in \{1, 2, 3\}$ be the point with $W_0(m) = k$, so that $m = 1$ exactly when $k = j$. Replace $X_{ij}^{\pm 1}$ by $(W_0 d W_0^{-1})^{\pm 1}$, a change of transporter because W_0 and U_{ij} agree on $P_d = \{1, 2\}$. Replace $\tau_k^{\pm 1}$ by $(W_0 t_m W_0^{-1})^{\pm 1} = (W_0 p^{m-1} t p^{1-m} W_0^{-1})^{\pm 1}$, and $\tau_i^{\pm 1}$ by $(W_0 t_2 W_0^{-1})^{\pm 1}$, again changes of transporter. If $k \neq j$, four of them turn the word into $W_0(dt_m d^{-1} t_m^{-1}) W_0^{-1}$ with $m \in \{2, 3\}$, a conjugate of a relation A. If $k = j$, five of them turn it into $W_0(dt_1 d^{-1} t_2^{-1} t_1^{-1}) W_0^{-1}$, again a conjugate of a relation A. So the area is at most $5\mathcal{E}(n) + 1$.

(c) Moving x across a letter of U_{ij} costs one relation Z, and moving it across $d = c^3$ costs one relation $[d, a]$, since $[c^3, ac^4] = [c^3, a]$ freely. Hence $\text{Area}_R([X_{ij}, x]) \leq 2|U_{ij}| + 1 < 40n + 1$.

(d) Freely the word is $U_i[U_i^{-1} x U_i, t(U_i^{-1} y U_i) t^{-1}] U_i^{-1}$. Replacing the two occurrences of $U_i^{-1} x^{\pm 1} U_i$ and the two of $U_i^{-1} y^{\pm 1} U_i$ by $x^{\pm 1}$ and $y^{\pm 1}$ costs at most $4|U_i|$ relations Z and leaves $U_i[x, tyt^{-1}] U_i^{-1}$, a conjugate of a relation P; so the area is at most $4|U_i| + 1 < 40n + 1$. $\square$

7 Pair areas and the profile

We now bring together two of the 2^N copies of S_3 supported on the first N points of ray 1. Their transport words are products of the τ_i ; the commutator of two copies reduces to the copy at a single point after clearing the difference of the two configurations by at most N transvections. Clearing naively would let the translation word grow; we keep it normalized after every step, so that it never has more than N factors, and each step costs $O(N)$ local fillings.

Theorem 7.1 (Pair areas). *Let $N \geq 1$ and $n = \max(3, N)$. For a configuration u supported in the first N points of ray 1 let $T_u = \tau_{i_1} \cdots \tau_{i_m}$, where $i_1 < \cdots < i_m$ are the points of the support of u . Then T_u represents τ_u , so $T_u a T_u^{-1}$ and $T_u b T_u^{-1}$ generate the copy of S_3 at u ; $|T_u| \leq 21Nn$; and for distinct such u, w and $x, y \in \{a, b\}$, with b the word ac^4 ,*

$$\text{Area}_R([T_u x T_u^{-1}, T_w y T_w^{-1}]) \leq \Lambda_N := 20N^2 L(n) \leq C_0 N^2 (\delta_H(C_0 N) + N)$$

with $C_0 = 4.5 \cdot 10^4$. Consequently there are absolute constants K' and C' with

$$\Lambda_N \leq e^{K'N} \quad \text{and} \quad \Lambda_N \leq C' N^{2+p_H} \quad (N \geq 1), \quad p_H = \log 107 + 4.$$

Proof. Translations commute, so T_u represents τ_u , and $|T_u| \leq N(2 \cdot 10n + 1) \leq 21Nn$. Throughout, $L = L(n)$; by Lemma 6.3 the local identities (a)–(d) have area at most L .

Step 1: one translation word. Freely, $[T_u x T_u^{-1}, T_w y T_w^{-1}] = T_u [x, Z y Z^{-1}] T_u^{-1}$ with $Z = T_u^{-1} T_w$. Replace each of the at most N factors τ_i^{-1} of T_u^{-1} by τ_i (one relation Q each), sort the resulting product of at most $2N$ factors τ_i into increasing order by at most $\binom{2N}{2}$ swaps of adjacent distinct factors (Lemma 6.3(a)), and cancel the at most N adjacent equal pairs (one relation Q each). The result is T_{u_0} with $u_0 = u + w \neq 0$, and $\text{Area}_R(Z T_{u_0}^{-1}) \leq 2N + N(2N - 1)L \leq 2N^2 L$. Since $Z^{\pm 1}$ occurs four times in $[x, Z y Z^{-1}]$,

$$\text{Area}_R([T_u x T_u^{-1}, T_w y T_w^{-1}]) \leq 8N^2 L + \text{Area}_R([x, T_{u_0} y T_{u_0}^{-1}]).$$

Step 2: normalized clearing. Let $s \leq N$ be the size of the support of u_0 , choose a point j in it, and list the other points of the support as $i_1, \dots, i_{s-1}$. Put $G_l = X_{i_l j} \cdots X_{i_1 j}$ and $u_l = u_0 + e_{i_1} + \cdots + e_{i_l}$, whose support is that of u_0 with $i_1, \dots, i_l$ removed; it still contains j , and $u_{s-1} = e_j$. We claim

$$\text{Area}_R(G_l T_{u_0} G_l^{-1} T_{u_l}^{-1}) \leq 2lsL \quad (0 \leq l \leq s-1).$$

For $l = 0$ there is nothing to prove. Given the claim for l , write $X = X_{i_{l+1}j}$ and $i = i_{l+1}$. Freely $G_{l+1}T_{u_0}G_{l+1}^{-1} = X(G_lT_{u_0}G_l^{-1})X^{-1}$, which differs from $XT_{u_l}X^{-1}$ by a conjugate of a word of area at most $2lsL$. Next, $XT_{u_l}X^{-1}$ is freely the product of the words $X\tau_{i'}X^{-1}$ over the support of u_l , in order. By Lemma 6.3(b), at a cost of at most sL , each of these becomes $\tau_{i'}$, except that $X\tau_jX^{-1}$ becomes $\tau_j\tau_i$. The new factor τ_i has a partner τ_i in T_{u_l} , because i lies in the support of u_l ; at most $s - 1$ swaps (Lemma 6.3(a)) make the two adjacent, and one relation Q cancels them. What remains is $T_{u_{l+1}}$. This step costs at most $sL + (s - 1)L + 1 \leq 2sL$, which proves the claim. The translation word is renormalized after every transvection and never has more than $s \leq N$ factors.

Step 3: a single point. Let $G = G_{s-1}$ and let $M = GT_{u_0}G^{-1}\tau_j^{-1}$, which has area at most $2s(s - 1)L \leq 2N^2L$ by Step 2. Freely $T_{u_0} = G^{-1}M\tau_jG$. Substituting this into the four occurrences of $T_{u_0}^{\pm 1}$ in $[x, T_{u_0}yT_{u_0}^{-1}]$ and deleting the four occurrences of $M^{\pm 1}$ costs at most $8N^2L$. By Lemma 6.3(c), $\text{Area}_R([G, z]) \leq (s - 1)(40n + 1)$ for $z \in \{a, b\}$, and the same bound holds for $[G, z^{-1}]$ and $[z, G^{-1}]$. Moving $y^{\pm 1}$ across G and G^{-1} , and then $x^{\pm 1}$, leaves $G^{-1}[x, \tau_jy\tau_j^{-1}]G$, at a cost of at most $4(s - 1)(40n + 1)$; and $[x, \tau_jy\tau_j^{-1}]$ has area at most $40n + 1$ by Lemma 6.3(d). Altogether, since $L \geq 700n$,

$$\text{Area}_R([x, T_{u_0}yT_{u_0}^{-1}]) \leq 8N^2L + 4N(40n + 1) + 40n + 1 \leq 9N^2L,$$

and with Step 1 the area is at most $17N^2L \leq \Lambda_N$.

Growth. Since $n \leq 3N$, $L(n) \leq 10\delta_H(4500N) + 2100N$, which gives the first bound on Λ_N . With Proposition 5.3(a) it is at most $e^{K'N}$ for a suitable K' . With Proposition 5.3(b) it is at most $C_0N^2(C(C_0N)^{p_H} + N) \leq C'N^{2+p_H}$, since $p_H > 1$. $\square$

The factor N^2 counts local moves: at most N transvections, each followed by at most N commutations. The Dehn function of H enters only through the local fillings, at the scale N of the transport.

7.1 Affine hashing and the upper bound

The Følner sets of Γ are doubly exponential (Proposition 7.7), but its chunks have far smaller models. A Følner set gives models of every chunk, by Cornulier's comparison of the sofic profile with the Følner function [18, Proposition 3.14], but it records the lamp at every configuration of a large finite part of V . A model need only record the lamps that are lit, and it can store them through a random affine hash of the configurations, which the affine group moves covariantly. Random $\mathbb{F}_2$ -linear maps form a universal hash family of Carter and Wegman [15], and a random translation does not change collision probabilities.

Let W be a finite-dimensional vector space over $\mathbb{F}_2$, and let $K_W = S_3^W \rtimes \text{AGL}(W)$, where S_3^W is the group of functions $W \rightarrow S_3$ under pointwise multiplication, and $g \in \text{AGL}(W)$ acts by $(g\lambda)(w) = \lambda(g^{-1}w)$; thus $(\lambda, g)(\mu, h) = (\lambda \cdot g\mu, gh)$. The support of λ is the set of w with $\lambda(w) \neq 1$. Fix $k \geq 1$ and a total order on W , let $\text{Aff}(W, \mathbb{F}_2^k)$ be the set of affine maps $W \rightarrow \mathbb{F}_2^k$, and let

$$\Omega = \text{Aff}(W, \mathbb{F}_2^k) \times S_3^{\mathbb{F}_2^k}, \quad |\Omega| = 2^{k(\dim W + 1)} 6^{2^k}.$$

For $\eta \in \text{Aff}(W, \mathbb{F}_2^k)$ and $\lambda \in S_3^W$ let $\lambda^\eta : \mathbb{F}_2^k \rightarrow S_3$ send z to the product, in the order of W , of the $\lambda(w)$ with $\eta(w) = z$. For $\gamma = (\lambda, g) \in K_W$ define a permutation Θ_γ of Ω by

$$\Theta_\gamma(\eta, u) = (\eta \circ g^{-1}, \lambda^{\eta \circ g^{-1}} \cdot u);$$

its inverse sends (η', u') to $(\eta' \circ g, (\lambda^{\eta'})^{-1}u')$, and Θ_1 is the identity.

Lemma 7.2 (Affine hashing). *Let $b \geq 1$, let $\gamma_1 = (\lambda, g)$ and $\gamma_2 = (\mu, h)$ be elements of K_W whose lamp parts λ, μ have supports of size at most b , and let ω be uniformly distributed on Ω . Then*

(i) $\Theta_{\gamma_1} \Theta_{\gamma_2} \omega = \Theta_{\gamma_1 \gamma_2} \omega$ except with probability at most $\binom{2b}{2} 2^{-k}$;

(ii) if $\gamma_1 \neq \gamma_2$, then $\Theta_{\gamma_1} \omega \neq \Theta_{\gamma_2} \omega$ except with probability at most $\binom{2b}{2} 2^{-k}$.

Proof. Write $\omega = (\eta, u)$ and $\eta = \eta_0 + z_0$ with η_0 linear; then η_0 and z_0 are independent and uniform. For $w \neq w'$ in W , $\eta(w) - \eta(w') = \eta_0(w - w')$ is uniform on $\mathbb{F}_2^k$, because $\eta_0 \mapsto \eta_0(w - w')$ is a surjective linear map; so $\eta(w) = \eta(w')$ with probability exactly 2^{-k} . Precomposition with a fixed affine bijection of W permutes $\text{Aff}(W, \mathbb{F}_2^k)$, so $\eta \circ f$ is again uniform for every $f \in \text{AGL}(W)$. Hence, for a set $S \subset W$ of size at most $2b$, $\eta \circ f$ is injective on S except with probability at most $\binom{2b}{2} 2^{-k}$. If η' is injective on a set S containing the support of λ , then $\lambda^{\eta'}(\eta'(w)) = \lambda(w)$ for $w \in S$, and $\lambda^{\eta'} = 1$ off $\eta'(S)$.

(i) Put $\eta' = \eta \circ (gh)^{-1}$. Then $\Theta_{\gamma_1} \Theta_{\gamma_2}(\eta, u) = (\eta', \lambda^{\eta'} \mu^{\eta \circ h^{-1}} u)$ and $\Theta_{\gamma_1 \gamma_2}(\eta, u) = (\eta', (\lambda \cdot g\mu)^{\eta'} u)$. Let S be the union of the support of λ and the image under g of the support of μ , and suppose that η' is injective on S . Since $\eta \circ h^{-1}(v) = \eta'(gv)$, the map $\eta \circ h^{-1}$ is injective on the support of μ , and $\mu^{\eta \circ h^{-1}}(\eta'(w)) = \mu(g^{-1}w)$ for $w \in S$. The support of $\lambda \cdot g\mu$ lies in S . So all three functions are trivial off $\eta'(S)$, and at $\eta'(w)$, $w \in S$, both sides equal $\lambda(w)\mu(g^{-1}w)$.

(ii) If $g \neq h$, choose w_0 with $h^{-1}g(w_0) \neq w_0$. If $\eta \circ g^{-1} = \eta \circ h^{-1}$, then $\eta = \eta \circ h^{-1}g$, so $\eta(h^{-1}g(w_0)) = \eta(w_0)$, which has probability 2^{-k} . If $g = h$, then $\lambda \neq \mu$; put $\eta' = \eta \circ g^{-1}$ and let S be the union of the supports. If η' is injective on S , then $\lambda^{\eta'}$ and $\mu^{\eta'}$ take the different values $\lambda(w)$ and $\mu(w)$ at $\eta'(w)$ for some $w \in S$. $\square$

To use the lemma in Γ we need finite subgroups that contain the relevant cocycles. For $z \in \mathbb{Z}^2$ put $g_z = p^{z_1} q^{z_2}$, so that $\pi(g_z) = z$, and for $m \geq 1$ let $B_m = \{-m, \dots, m\}^2$. For $R \geq 1$ let $Y_R \subset Y$ be the set of points of height at most R , let $W_R = \mathbb{F}_2^{Y_R} \subset V$, and let $K_R \leq \ker \pi$ consist of the elements whose lamps lie at configurations in W_R and whose affine part lies in $W_R \rtimes \text{GL}(W_R)$, with $\text{GL}(W_R)$ extended by the identity on the other basis vectors. Restricting to W_R identifies K_R with K_{W_R} .

Lemma 7.3 (Cocycles). *Let $m \geq 1$, $R = 2m + 3$, $z \in B_m$ and $s \in \{p, q, t, c, a\}^{\pm 1}$. Then $g_{z+\pi(s)}^{-1} s g_z$ and $g_z s g_{z+\pi(s)}^{-1}$ lie in K_R , and their lamps sit at no more than one configuration.*

Proof. Each of $p^{\pm 1}, q^{\pm 1}$ changes heights by at most one and acts on the points of height at least 2 as a translation along the rays, and these translations commute. Hence for $(y_1, y_2) \in \mathbb{Z}^2$ the commutator $[p^{y_1}, q^{y_2}]$ fixes every point of height greater than $|y_1| + |y_2| + 1$: while it is applied, such a point stays at height at least 2. For $s = p^{\pm 1}$ we have $\pi(s) = (\pm 1, 0)$, $g_{z+\pi(s)}^{-1} s g_z = 1$ and $g_z s g_{z+\pi(s)}^{-1} = p^{z_1} [q^{z_2}, p^{\pm 1}] p^{-z_1}$, which is supported in heights at most $|z_2| + 2 + |z_1| \leq 2m + 2$. For $s = q^{\pm 1}$ we have $\pi(s) = (0, \pm 1)$, $g_z s g_{z+\pi(s)}^{-1} = 1$ and $g_{z+\pi(s)}^{-1} s g_z = q^{-z_2 \mp 1} [p^{-z_1}, q^{\pm 1}] q^{z_2 \pm 1}$, supported in heights at most $|z_1| + 2 + |z_2| + 1 \leq 2m + 3$. These are permutations of Y_R , hence elements of $\text{GL}(W_R)$. For $s \in \{t, c, a\}^{\pm 1}$ we have $\pi(s) = 0$, and $g_z^{\pm 1}$ moves the points 1 and 2 to height at most $2m + 2$; so the conjugates of t and d by $g_z^{\pm 1}$ are a translation and a transvection of W_R , while those of a and ab are a and ab , at the configuration 0. $\square$

Proposition 7.4 (Hash models). *Let E' be a chunk of Γ each of whose elements is a product of at most $\ell \geq 1$ of the generators p, q, t, c, a and their inverses. Then*

$$\log \sigma_{E'}(r) \leq 500 \ell^2 r \log(2 + r) \quad (r > 1).$$

Proof. Let $m = \lceil 8\ell r \rceil$, $R = 2m + 3$, $D = |Y_R| = 6m + 9$ and $k = \lceil \log(4\ell^2 r) \rceil$, and let Ω be the set above for $W = W_R$. For $g \in \Gamma$ and $z \in \mathbb{Z}^2$ put $\psi(g, z) = g_{z+\pi(g)}^{-1} g g_z \in \ker \pi$; then $\psi(gh, z) = \psi(g, z + \pi(h)) \psi(h, z)$. If g is a product of at most ℓ generators and inverses and $z \in B_{m-\ell}$, then $\psi(g, z)$ is the product of the $\psi(s, z')$ for the letters s of the word, at points $z' \in B_m$, so by Lemma 7.3 it lies in K_R with lamps at no more than ℓ configurations.

Let $X = B_m \times \Omega$. For $g \in E'$ let F_g be a permutation of X that extends the injective partial map $(z, \omega) \mapsto (z + \pi(g), \Theta_{\psi(g,z)}\omega)$, defined for $z \in B_{m-\ell}$, with $F_1 = \text{id}$. Let $g, h, gh \in E'$ and $z \in B_{m-2\ell}$. Then z and $z + \pi(h)$ lie in $B_{m-\ell}$, so $F_g F_h(z, \omega) = (z + \pi(gh), \Theta_{\psi(g, z + \pi(h))} \Theta_{\psi(h, z)} \omega)$ and $F_{gh}(z, \omega) = (z + \pi(gh), \Theta_{\psi(g, z + \pi(h))} \psi(h, z) \omega)$; by Lemma 7.2(i) these agree for all but a fraction $\binom{2\ell}{2} 2^{-k}$ of the ω . Since $|B_{m-2\ell}| \geq (1 - 8\ell/(2m+1))|B_m|$,

$$d(F_g F_h, F_{gh}) \leq \varepsilon := \frac{8\ell}{2m+1} + \binom{2\ell}{2} 2^{-k} < \frac{1}{2r} + \frac{1}{2r} = \frac{1}{r}.$$

If $g \neq h$ in E' and $z \in B_{m-\ell}$, then either $\pi(g) \neq \pi(h)$ and $F_g(z, \omega) \neq F_h(z, \omega)$, or $\psi(g, z) \neq \psi(h, z)$ and Lemma 7.2(ii) applies; so $d(F_g, F_h) \geq 1 - \varepsilon$. Hence $\sigma_{E'}(r) \leq |X|$.

Finally $\log |X| = 2 \log(2m+1) + k(D+1) + 2^k \log 6$. Here $2^k < 8\ell^2 r$, $k < \log(8\ell^2 r) \leq 5\ell \log(2+r)$ and $D+1 \leq 64\ell r$, so $k(D+1) \leq 320\ell^2 r \log(2+r)$; also $2^k \log 6 \leq 21\ell^2 r \leq 14\ell^2 r \log(2+r)$ and $2 \log(2m+1) \leq 10\ell^2 r \log(2+r)$. The sum is less than $500\ell^2 r \log(2+r)$. $\square$

7.2 The profile

Theorem 7.5. *Let $E \subset \Gamma$ be the chunk consisting of 1, the five generators p, q, t, c, a and their inverses, the elements represented by the prefixes of the 25 freely reduced words of Table 2, and the inverses of these elements. Let $p_H = \log 107 + 4$ and*

$$\kappa = 2 + p_H = 6 + \log 107 < 12.75.$$

There are absolute constants $c, C > 0$ such that for every $r \geq 2$:

- (a) $\log \sigma_E(r) \geq cr/(\log r)^\kappa$;
- (b) $\log \sigma_E(r) \leq Cr \log r$; more generally, every chunk E' of Γ satisfies $\log \sigma_{E'}(r) \leq C_{E'} r \log r$.

In particular $\sigma_E(r) = \exp(r^{1+o(1)})$. Moreover:

- (c) *For every $0 < \Delta \leq 1$ there are $c_\Delta, e_\Delta > 0$ such that the following holds for $0 < e \leq e_\Delta$. Let ϕ assign unitaries in $U(D)$ to the five generators so that every word of Table 2 is mapped within e of I in the normalized Hilbert–Schmidt norm, and put $\phi(b) = \phi(a)\phi(c)^4$. If $\|\phi(a)\phi(b) - I\|_2 \geq \Delta$, then*

$$\log D \geq c_\Delta \frac{e^{-2}}{(\log(1/e))^{2\kappa}}, \quad 2\kappa < 25.49.$$

- (d) Γ is finitely presented and elementary amenable.

Proof. Setting up the criterion. Theorem 2.1 asks for a and b among the generators, while in R the letter b abbreviates ac^4 . So let S^+ consist of the five generators and $b = ac^4 \in \Gamma$, and let R^+ consist of R and the three words $b^{-1}ac^4$, b^2 and $(ab)^3$ in the letters of S^+ . These words represent $1, a^2, b^2, (ab)^3 \in R^+$, $ab \neq 1$, and the longest word of R^+ has length $l = 24$. The chunk E^+ that Theorem 2.1 associates with S^+ and R^+ is contained in E : besides the elements of E , it contains $b^{\pm 1}$, the prefixes $b^{-1}, c^{-4}, c^{-3}, c^{-2}, c^{-1}$ of $b^{-1}ac^4$, and the prefixes $b, ab = c^4, aba = c^4a, abab = c^2$ and $ababa = c^2a$ of b^2 , $(ab)^3$ and ab . These lie in E , because ac^4, a^2c^4, a^2c^4a and $a^2c^4a^2c^4a$ are prefixes of b^2 and $(ab)^3$ in Table 2, c^2 and c^3 are prefixes of the relations $[d, k]$ of the family $\mathbf{S}$, $a^2 = 1$, and $c^6 = 1$. Restricting a model of E to E^+ gives a model of E^+ , so $\sigma_E \geq \sigma_{E^+}$.

For $N \geq 1$ take the $M = 2^N$ words T_u of Theorem 7.1, one for each configuration u supported in the first N points of ray 1. A commutator $[T_u x T_u^{-1}, T_w y T_w^{-1}]$ with $x, y \in \{a, b\}$

contains the letter $b^{\pm 1}$ at most four times, and replacing each occurrence by $(ac^4)^{\pm 1}$ costs one relation $b^{-1}ac^4$. So its area with respect to R^+ is at most $\Lambda_N^+ = \Lambda_N + 4$, and Theorem 2.1 gives

$$\log \sigma_E(r) \geq 2^N/16 \quad \text{whenever} \quad r \geq 4.7 \cdot 10^9 \cdot 2^N \Lambda_N^+. \quad (6)$$

Since $a \neq 1$, a $(1 - 1/r)$ -expansive map needs at least two points, so $\log \sigma_E(r) \geq 1$; for r below any fixed bound r_0 claim (a) therefore holds once c is small enough, and we may assume r large.

(a) By Theorem 7.1, $\Lambda_N^+ \leq C_1 N^\kappa$ with C_1 absolute. Let $r' = r/(4.7 \cdot 10^9 C_1)$ and $N = \lfloor \log r' - \kappa \log \log r' \rfloor$, which is at least 1 for large r . Then $2^N \Lambda_N^+ \leq C_1 r' (\log r')^{-\kappa} (\log r')^\kappa = r/(4.7 \cdot 10^9)$, so (6) applies, and $\log \sigma_E(r) \geq 2^N/16 \geq r'/(32(\log r')^\kappa)$. Since $\log r' \leq \log r$, this is (a).

(b) Every element of E is a generator, the inverse of one, or represented by a prefix of a word of length at most 24 or by its inverse. Proposition 7.4 with $\ell = 24$ gives $\log \sigma_E(r) \leq 500 \cdot 24^2 r \log(2 + r)$, and $\log(2 + r) \leq 2 \log r$ for $r \geq 2$. The same proposition applies to every chunk.

(c) Extend ϕ to S^+ by $\phi(b) = \phi(a)\phi(c)^4$. Then $b^{-1}ac^4$ is mapped to I exactly, and b^2 and $(ab)^3$ are mapped to the images of the words b^2 and $(ab)^3$ of Table 2; so every word of R^+ is mapped within e of I . With $\Lambda_N^+ \leq C_1 N^\kappa$ as in (a), the condition $K_\Delta \sqrt{M} \Lambda_N^+ e \leq 1$ of Theorem 2.1 holds when $2^N N^{2\kappa} \leq Y := (C_1 K_\Delta e)^{-2}$, which is the case for $N = \lfloor \log Y - 2\kappa \log \log Y \rfloor$; this N is at least 1 for e small in terms of Δ . Then $2^N \geq Y/(2(\log Y)^{2\kappa})$, and since $C_1 K_\Delta \geq 1$ we have $\log Y \leq 2 \log(1/e)$. Hence

$$\log D \geq \frac{\Delta^2}{12} 2^N \geq \frac{\Delta^2}{24 \cdot 4^\kappa C_1^2 K_\Delta^2} \cdot \frac{e^{-2}}{(\log(1/e))^{2\kappa}}.$$

(d) Proposition 4.2 gives finite presentation and Lemma 4.1 elementary amenability. $\square$

The group Γ is not LEF. This is a standard consequence of $H_3 \leq \Gamma$: the group H_3 is finitely presented and not residually finite, since the finitary alternating group of an infinite set is simple, a finitely presented LEF group is residually finite [49], and subgroups of LEF groups are LEF. The chunk E also shows it: by Theorem 7.5(a) its profile is unbounded, so it does not embed in a finite group [18, Fact 3.5] (see also [19, Lemma 2.1]).

Remark 7.6. Part (a) uses the polynomial Dehn bound of Proposition 5.3(b). With Lee's exponential bound alone, Theorem 7.1 gives $\Lambda_N^+ \leq 2^{BN}$ for a constant B that depends on the constant in Lee's inequality, and $N = \lfloor \log(r/(4.7 \cdot 10^9))/(1 + B) \rfloor$ in (6) gives only $\log \sigma_E(r) \geq cr^{1/(1+B)}$: a superpolynomial profile, with an exponent we cannot make explicit. Transport at scale N is too expensive for an exponential Dehn bound. The grid host of Remark 7.8 transports at scale $\sqrt{N}$ and would reach, by the argument sketched there, $\log \sigma \geq cr e^{-C\sqrt{\log r}}$ with Lee's bound alone.

The hash models are far smaller than any Følner set. For $r \geq 1$ let $\text{Føl}(r)$ be the least size of a finite nonempty set $F \subset \Gamma$ with $|Fs \setminus F| \leq |F|/r$ for $s \in \{p, q, t, c, a\}$.

Proposition 7.7. *For $r \geq 12$,*

$$2^{2^{\lfloor r/12 \rfloor - 1}} \leq \text{Føl}(r) \leq 2^{2^{3r+14}}.$$

So the Følner function of Γ is doubly exponential, while by Theorem 7.5(b) the sofic profile of every chunk is $\exp(O(r \log r))$.

Proof. Lower bound. Let F be such a set and $l = \lfloor r/12 \rfloor \geq 1$. For $v \in \{0, 1\}^l$ the word $T_v = t^{v_1} p t^{v_2} p \cdots p t^{v_l} p^{1-l}$ is freely equal to $\prod_{j=1}^l p^{j-1} t^{v_j} p^{1-j}$, so it represents the translation

by $\sum_j v_j e_j$, and it has length less than $3l$. The elements $a_v = T_v a T_v^{-1}$ are the copies of a at 2^l distinct configurations, so they are commuting involutions and form a basis of an elementary abelian group Q of order 2^M , $M = 2^l$. Each a_v is a product of fewer than $6l$ generators and inverses; since $|Fxy \setminus F| \leq |Fx \setminus F| + |Fy \setminus F|$ and $|Fx^{-1} \setminus F| = |Fx \setminus F|$, we get $|Fa_v \setminus F| \leq 6l|F|/r \leq |F|/2$, and so $\sum_v |Fa_v \setminus F| \leq M|F|/2$.

On the other hand, for every finite $S \subset Q$,

$$\sum_v |Sa_v \setminus S| \geq |S|(M - \log |S|). \quad (7)$$

Indeed, the left side is $M|S| - 2e(S)$, where $e(S)$ is the number of pairs $\{x, xa_v\} \subset S$, and $e(S) \leq \frac{1}{2}|S|\log |S|$ by the edge-isoperimetric inequality of the cube [27]. We recall the proof, by induction on M : splitting S by the last coordinate into S_0, S_1 of sizes $s_0 \leq s_1$ gives $e(S) \leq e(S_0) + e(S_1) + s_0 \leq \frac{1}{2}(s_0 \log s_0 + s_1 \log s_1) + s_0$, which is at most $\frac{1}{2}s \log s$, $s = s_0 + s_1$, because the binary entropy function h satisfies $h(x) \geq 2x$ for $0 \leq x \leq \frac{1}{2}$. The right multiplications by the a_v preserve each coset xQ ; applying (7) to the sets $x^{-1}(F \cap xQ)$ and using $|F \cap xQ| \leq |F|$ gives $\sum_v |Fa_v \setminus F| \geq |F|(M - \log |F|)$. Hence $\log |F| \geq M/2 = 2^{l-1}$.

Upper bound. Let $m = \lceil (r-1)/2 \rceil$, $R = 2m+3$ and $F = \bigcup_{z \in B_m} K_R g_z$, a disjoint union because $\pi(g_z) = z$. By Lemma 7.3, $g_z s \in K_R g_{z+\pi(s)}$ for $z \in B_m$ and each generator s , so $|Fs \setminus F| \leq |K_R|(2m+1) = |F|/(2m+1) \leq |F|/r$. With $D = |Y_R| = 6m+9$ we have $|K_R| = 6^{2^D} \cdot 2^D \cdot |\text{GL}_D(\mathbb{F}_2)|$, so $\log |F| \leq 2^D \log 6 + D + D^2 + 2 \log(2m+1) \leq 2^{D+2} \leq 2^{3r+14}$. $\square$

7.3 The grid host

Remark 7.8. Placing the lamps on configurations of the cells of a grid, rather than of the points of Y , should give a stronger lower bound with the same upper bound, and a nearly linear exponent $r^{1-o(1)}$ even from Lee's exponential Dehn bound alone. We only sketch the proofs, and the conclusions of this remark are conditional on them. Let $I = Y \times Y$, let $T = H \times H$ act on I coordinatewise, and let

$$\Gamma_2 = \left(\bigoplus_{v \in V_2} S_3 \right) \rtimes (V_2 \rtimes (\text{GL}_{\text{fin}}(I) \rtimes T)), \quad V_2 = \mathbb{F}_2^{(I)}.$$

It is locally finite-by- $\mathbb{Z}^4$ and generated by seven elements (commutators $[E_{x \leftarrow y}, E_{y \leftarrow z}] = E_{x \leftarrow z}$ of transvections of the third kind give those within a row or a column): the generators p_1, q_1, p_2, q_2 of the two copies of H , the translation t by the cell $o = ((1,1), (1,1))$, the element $c = dab$, where d is the transvection that adds e_o to $e_{((1,2), (1,2))}$, and a . A certificate of 77 relations, of total length 5445 and maximum length 224, is listed in `verification/relators.txt` and checked by `verification/relators_check.py`. It has the families of Table 2, with two copies of the Houghton relators and their commutation, and with stabilizer and action relations for three kinds of transvection: within a row, within a column, and neither.

The gain is in the scale of transport. N cells fit in a square of side $\lceil \sqrt{N} \rceil$, so the transporters have length $O(\sqrt{N})$, and the local fillings cost $C(\delta_H(C\sqrt{N}) + N)$; the term N counts the swaps of letters from the two copies of H in a change of transporter. The clearing argument of Theorem 7.1 then bounds the pair areas by $CN^2(\delta_H(C\sqrt{N}) + N)$, which is at most $CN^{2+p_H/2}$ by Proposition 5.3(b) and at most $e^{C\sqrt{N}}$ by Proposition 5.3(a). As in the proof of Theorem 7.5, the chunk E_2 of this certificate should therefore have

$$\log \sigma_{E_2}(r) \geq c \frac{r}{(\log r)^{2+p_H/2}}, \quad 2 + \frac{p_H}{2} < 7.38,$$

and, using only Lee's bound, $\log \sigma_{E_2}(r) \geq c r e^{-C\sqrt{\log r}}$; unitary models need $\log D \geq c_\Delta e^{-2}/(\log(1/e))^{4+p_H}$. In the other direction, hashing the configurations directly, as in

Proposition 7.4, would need a configuration space of dimension of order r^2 . Hashing the cells first avoids this: a cell (y_1, y_2) gets the label $a_1(f_1^{-1}y_1) + a_2(f_2^{-1}y_2) \in \mathbb{F}_q$, where the f_j are positions in finite Følner charts of H of size $e^{O(r \log r)}$ (inverses of right Følner sets, since the charts move by left multiplication) and the a_j are random tables on $O(r)$ points; this label is covariant under the charts and separates the finitely many cells of a chunk with probability $1 - O(1/q)$. The configurations are then compressed linearly to $\mathbb{F}_2^{\mathbb{F}_q}$ and hashed affinely as before. With q of order $\ell^2 r$, for a chunk of words of length at most ℓ , and the other parameters of order r , this would give $\log \sigma_{E'}(r) \leq C_{E'} r \log(2 + r)$ for every chunk E' of Γ_2 , and the same for the d -dimensional grid host, built in the same way on the cells of Y^d with H^d acting, for every fixed d . So $\sigma_{E_2}(r)$ would be $\exp(r^{1+o(1)})$ as well, and no fixed-dimensional grid host would exceed $\exp(r^{1+o(1)})$. The proofs follow those above with the changes indicated, and we omit them.

8 Quantum channels that need nearly linear memory

A finite set of relations becomes a quantum channel of fixed dimension: each relation contributes a few small unitary blocks, and the channel evaluates the blocks in the regular representation of the group and traces out the group von Neumann algebra. In [20] we showed that the memory needed to apply such a channel repeatedly, releasing each output before the next input arrives, is bounded below by the size of approximate representations of the relations. This section sharpens that link, so that it loses a single square root, and applies it to the configuration lamps. The result is an explicit channel, in the closure of finite tracial channels, whose repeated use with logarithmic purity needs memory $n^{1+o(1)}$: at least $n/(\log n)^{O(1)}$ for n uses (Theorem 8.5), and at most $O(n \log n)$, because the sofic models of Γ also give devices (Proposition 8.9). More precisely, memory and purity trade at a fixed rate: a device allowed purity B needs memory about n/B , up to polylogarithmic factors, and exact devices attain this. At the optimal exchange rate the memory is $n^{1/2+o(1)}$, attained by devices that keep one sofic model for all rounds (Theorem 8.10).

8.1 Setting

We recall the framework of [20] and keep its notation.

Channels. Channels act on $M_d = M_d(\mathbb{C})$. The normalized Choi state of a channel Ψ is $\mathcal{J}(\Psi) = (\Psi \otimes \text{id})(|\Omega\rangle\langle\Omega|)$, with $|\Omega\rangle = d^{-1/2} \sum_i |ii\rangle$, and $d_\diamond(\Psi, \Psi') = \frac{1}{2} \|\Psi - \Psi'\|_\diamond$ is the diamond distance. We write τ_M for the normalized trace on M_M . A channel Ψ has a *finite tracial factorization* with bath dimension M if $\Psi(X) = (\text{id} \otimes \tau_M)[U(X \otimes I_M)U^*]$ for a unitary U on $\mathbb{C}^d \otimes \mathbb{C}^M$, and $\mathcal{K}_d$ denotes the closure of the set of channels on M_d with a finite tracial factorization of some bath dimension. It is also the closure of the convex hull of that set [44, Proposition 3.8(ii)], and it consists of unital channels. Channels with a finite tracial factorization are exactly the noisy operations of [31] from M_d to M_d , and $\mathcal{K}_d$ is their closure.

Causal services. A *causal service* of a channel Φ on M_d for n rounds consists of a memory register of Q qubits in a fixed initial state β_0 and, for each round, a fixed unitary on the input and the memory, after which the d -dimensional output is released; after each round but the last, a prescribed set of memory qubits is exported for good and replaced by the same number of imported qubits in a fixed state (possibly correlated within one step), independent of everything else. An observer chooses the inputs adaptively, possibly entangled with references of its own, and the *error* of the service is the largest half trace distance between the observer's final state in this experiment and in the ideal experiment with n independent uses of Φ . The

resources are the memory Q , the *exchange* J (the total number of exported qubits), and the *purity* $P + C$, where $P = Q - S(\beta_0)$ and C is the sum over the imports of their number of qubits minus their entropy. Every retained qubit counts; there is no free reset, measurement record or postselection.

Gadget channels. Let G be a group generated by a finite set S , and let R be a finite set of words in $S^{\pm 1}$, each of length at least two and each representing 1 in G . Write $r_i = \sigma_{i,1} \cdots \sigma_{i,\ell_i}$ ($1 \leq i \leq r_0$) for the words of R , and put $s = |S|$ and $\Lambda = \sum_i \ell_i$. The *symbols* are the $2s$ letters $x^{\pm 1}$, $x \in S$, and the proper prefixes $p_{i,l} = \sigma_{i,1} \cdots \sigma_{i,l}$, $2 \leq l \leq \ell_i - 1$, one for each pair (i, l) ; we also put $p_{i,1} = \sigma_{i,1}$ and let p_{i,ℓ_i} be the empty word. The *triangles* are the triples $(p_{i,l-1}, \sigma_{i,l}, p_{i,l})$, $2 \leq l \leq \ell_i$, and $(x, x^{-1}, \emptyset)$, $x \in S$. For unitaries $\lambda(w)$ assigned to the symbols, with $\lambda(\emptyset) = I$, let

$$U_\lambda = \text{diag}(I, (\lambda(w))_w) \oplus \bigoplus_{(x,y,z)} \frac{1}{\sqrt{2}} \begin{pmatrix} I & \lambda(y) \\ \lambda(x) & -\lambda(x)\lambda(y) \end{pmatrix},$$

where w runs over the symbols and (x, y, z) over the triangles. Each triangle block equals $\text{diag}(I, \lambda(x)) (H \otimes I) \text{diag}(I, \lambda(y))$ with H the Hadamard matrix, so U_λ is unitary for all unitary λ . There are $2s + \Lambda - 2r_0$ symbols and $\Lambda - r_0 + s$ triangles, so the system dimension is

$$d = 1 + (2s + \Lambda - 2r_0) + 2(\Lambda - r_0 + s) = 1 + 4s + 3\Lambda - 4r_0. \quad (8)$$

The *gadget channel* Φ_G on M_d evaluates each symbol w in the left regular representation, $\lambda(w) = \lambda([w])$ with $[w] \in G$ the element that w represents, and traces out the group von Neumann algebra $L(G)$ with its canonical trace τ : $\Phi_G(X) = (\text{id} \otimes \tau)[U_\lambda(X \otimes 1)U_\lambda^*]$. Since the relations hold, the lower right entry of the block of a triangle (x, y, z) is $-\lambda([z])$.

We write the gadget as $U_\lambda = \sum_{a \in \mathcal{T}} c_a E_a \otimes \lambda([w_a])$, where $\mathcal{T}$ is the set of *slots* (the nonzero entries), E_a is the matrix unit of slot a , $c_a \in \{\pm 1, \pm 2^{-1/2}\}$, and the word w_a is empty, a symbol, or the concatenation xy in the lower right corner of a triangle. The elements $[w_a]$ are the *labels*. With $A_g = \sum_{[w_a]=g} c_a E_a$, the matrices A_g have disjoint supports, $\Phi_G(X) = \sum_g A_g X A_g^*$ and $\mathcal{J}(\Phi_G) = \frac{1}{d} \sum_g |A_g\rangle\langle A_g|$, because $\tau(\lambda(g)\lambda(h)^*) = \delta_{gh}$. So the Choi rank r_* of Φ_G is the number of distinct labels. Every label is carried by an *anchor*: a diagonal slot with coefficient 1 whose row and column contain no other slot, namely the first diagonal entry for the label 1 and a symbol entry otherwise, since the lower right entry of a triangle carries the label of a prefix symbol or 1. Feeding one half of a maximally entangled state on the r_* anchor positions gives a pure probe whose output is flat of rank r_* , and $\log r_*$ is the largest entropy of a complementary output of Φ_G [20, Proposition 9.1].

One round. Let U be a unitary on $\mathbb{C}^d \otimes \mathbb{C}^M$ and ρ a state on $\mathbb{C}^M$, and put

$$\Psi(X) = \text{Tr}_M U(X \otimes \rho)U^*, \quad T(\rho) = \text{Tr}_d U(\tau_d \otimes \rho)U^*.$$

Relative to a channel Φ with Choi support projection Π_Φ , the *support leakage* of Ψ is $\ell = \text{Tr}[(I - \Pi_\Phi)\mathcal{J}(\Psi)]$. The *entropy production* of the round is $\Delta S = S(T(\rho)) - S(\rho)$, which is nonnegative because T is unital. A service produces such rounds: the next result, from [20], finds one in which both quantities are small.

Theorem 8.1 (Sequential extraction [20]). *Let a causal service of a channel Φ on M_d for n rounds have memory Q , error $\epsilon < 1/2$ and $P + C \leq B$, and put*

$$t_n = \frac{B + \epsilon + h_2(\epsilon) + \epsilon \log n}{n(1 - \epsilon)}.$$

Then some round t , with its unitary U_t and the memory state ρ_t entering it, defines a channel $\Psi_t(X) = \text{Tr}_M U_t(X \otimes \rho_t) U_t^*$ with bath dimension $M = 2^Q$, $d_\diamond(\Psi_t, \Phi) \leq \epsilon/(1 - \epsilon)$, and support leakage ℓ_t and entropy production ΔS_t satisfying $\ell_t + \Delta S_t \leq t_n$.

Since the bath has Q qubits, $S(\rho_t) \leq Q$. Every lower bound on memory below is a lower bound on the entropy of a memory state that realizes one good round.

8.2 The weighted one-round theorem

The following theorem is the engine of this section. It reads an approximate representation of the relations directly off the memory state of one round, weighted by that state, and turns many cheaply commuting copies of S_3 into entropy of the memory. Transport words $u_1, \dots, u_N$ over $S^{\pm 1}$ have length at most L and pairwise areas at most A if $|u_i| \leq L$ for all i and

$$\text{Area}_R([u_i x u_i^{-1}, u_k y u_k^{-1}]) \leq A \quad (i \neq k, x, y \in \{a, b\}),$$

where Area_R is the area in the free group on S , as in Section 2.

Theorem 8.2 (Weighted one-round theorem). *Let G , S and R be as above, and suppose that S contains a , b and j , that a^2 , b^2 , $(ab)^3$ and $j^{-1}ab$ lie in R , and that $j \neq 1$ in G . Put $m = \max\{6, \ell_1, \dots, \ell_{r_0}\}$. Let $u_1, \dots, u_N$ be transport words of length at most L and pairwise areas at most A . Let $\Psi(X) = \text{Tr}_M U(X \otimes \rho) U^*$ be one round with $d_\diamond(\Psi, \Phi_G) \leq 1/8$, support leakage ℓ relative to Φ_G and entropy production ΔS , and put $\nu = \ell + \Delta S$. If*

$$2^{25} m^2 \sqrt{d} \sqrt{N} (A + L + 1) \sqrt{\nu} \leq 1, \quad (9)$$

then $S(\rho) \geq N/16$.

Two features matter. First, the relator defect that the round reveals is of order $\sqrt{\nu}$, a single square root of the quantity that Theorem 8.1 controls, so a service with purity $O(\log n)$ is read at defect about $n^{-1/2}$. Second, there is no tracialization: the argument works with the actual memory state ρ , which need not be flat. Replacing the round by a nearby finite tracial mixture, as the general correspondence of [20] does, costs a second square root, so that correspondence reads the profile at defect about $n^{-1/4}$ instead of $n^{-1/2}$. The weighted route was carried out in [20] for a single gadget built from Houghton's group; Theorem 8.2 holds for every relation set that contains the relations of S_3 and a name for ab . It is proved in Section 9.

Remark 8.3 (The constant term). The term $+1$ in (9) cannot be dropped. Take $N = 1$, the empty transport word, so that $L = 0$, and $A = 0$, which is allowed since there are no pairs. The gadget channel has a Stinespring dilation with a pure environment of dimension r_* , so there is a round with $M = r_*$, ρ pure and $\Psi = \Phi_G$ exactly. Then $d_\diamond(\Psi, \Phi_G) = 0$ and the condition with $A + L$ in place of $A + L + 1$ holds, since its left side vanishes, but $S(\rho) = 0 < 1/16$. With the $+1$, the condition forces $\nu \leq 2^{-50} m^{-4} d^{-1}$, while this round has $\ell = 0$ and $\Delta S = S(T(\rho))$, the entropy of the complementary output at the maximally mixed input. That output is diagonal with entries $\|A_g\|_F^2/d \geq 1/d$, and there are at least two labels, since $ab \neq 1$; so $\Delta S \geq h_2(1/d)$, far above the threshold.

Applied to a family of transport words that grows with a parameter, Theorem 8.2 gives a lower bound on the entropy of one round in terms of ν alone.

Corollary 8.4 (Amplification). *Keep the assumptions of Theorem 8.2 on G , S , R and the round, and let $K \geq 1$ and $\kappa \geq 0$.*

- (a) Suppose that for every integer $k \geq 1$ there are 2^k transport words of length at most L_k and pairwise areas at most A_k , with $A_k + L_k + 1 \leq Kk^\kappa$. Put $C_R = 2^{50}m^4dK^2$. There is y_0 depending only on κ such that

$$S(\rho) \geq \frac{y}{32(\log y)^{2\kappa}} \quad \text{for every } y \geq y_0 \text{ with } C_R \nu y \leq 1.$$

- (b) Suppose instead that $A_k + L_k + 1 \leq 2^{K\sqrt{k}}$ for every $k \geq 1$, and put $C'_R = 2^{50}m^4d$. There is y_1 depending only on K such that

$$S(\rho) \geq \frac{y}{32} 2^{-2K\sqrt{\log y}} \quad \text{for every } y \geq y_1 \text{ with } C'_R \nu y \leq 1.$$

Proof. (a) Let $y_0 \geq 4$ be such that $k = \lfloor \log y - 2\kappa \log \log y \rfloor \geq 1$ for $y \geq y_0$. Then $k \leq \log y$ and $2^k \leq y(\log y)^{-2\kappa}$, so $2^k k^{2\kappa} \leq y$. With $N = 2^k$, the left side of (9) is at most $2^{25}m^2\sqrt{d}K(2^k k^{2\kappa})^{1/2}\sqrt{\nu} \leq (C_R \nu y)^{1/2} \leq 1$. So $S(\rho) \geq 2^k/16$, and $2^k > y(\log y)^{-2\kappa}/2$.

(b) Let y_1 be such that $k = \lfloor \log y - 2K\sqrt{\log y} \rfloor \geq 1$ for $y \geq y_1$. Then $K\sqrt{k} \leq K\sqrt{\log y}$, so $2^k(A_k + L_k + 1)^2 \leq 2^{k+2K\sqrt{\log y}} \leq y$, and (9) holds with $N = 2^k$ as in (a). Hence $S(\rho) \geq 2^k/16 > y 2^{-2K\sqrt{\log y}}/32$. $\square$

8.3 The channel of the configuration lamps

Theorem 8.2 needs a , b and a name for ab among the generators. Let R be the certificate of Table 2, in the five generators p, q, t, c, a , with b standing for the word ac^4 . Let $S_\Phi = \{p, q, t, c, a, b, j\}$, where the new letters represent $b = ac^4$ and $j = ab$ in Γ , and let R_Φ consist of the 25 words of R and the four words

$$b^{-1}ac^4, \quad b^2, \quad (ab)^3, \quad j^{-1}ab$$

in the letters of S_Φ . They represent 1 in Γ , $a^2 \in R$, and $j \neq 1$ because ab has order three (Lemma 4.1). The channel of the configuration lamps is the gadget channel $\Phi = \Phi_\Gamma$ of R_Φ .

Dimension and rank. Here $s = 7$, $r_0 = 29$, and the total length is $\Lambda = 303 + 6 + 2 + 6 + 3 = 320$, so (8) gives

$$d = 1 + 4 \cdot 7 + 3 \cdot 320 - 4 \cdot 29 = 873,$$

with 276 symbols and 298 triangles; the longest word has length 24, so $m = 24$. The Choi rank r_* is the number of distinct labels, at most $1 + 276 = 277$. It is smaller: each of the 26 words of R_Φ of length at least three has a longest proper prefix that represents the inverse of its last letter, so these 26 prefix symbols carry labels already carried by letters, and $a^2 = b^2 = t^2 = 1$ identifies the labels of a^{-1}, b^{-1}, t^{-1} with those of a, b, t . Hence $2 \leq r_* \leq 248$, where the lower bound holds because 1 and j are distinct labels.

Transport. For $N \geq 1$, the proof of Theorem 7.5 provides 2^N words T_u , one for each configuration u supported in the first N points of ray 1, with $|T_u| \leq 21N \max(3, N) \leq 63N^2$ (Theorem 7.1) and $\text{Area}_{R_\Phi}([T_u x T_u^{-1}, T_w y T_w^{-1}]) \leq \Lambda_N + 4 \leq C_1 N^\kappa$ for $u \neq w$ and $x, y \in \{a, b\}$, where

$$p_H = \log 107 + 4, \quad \kappa = 2 + p_H < 12.75,$$

and C_1 is absolute; the letter b is converted to ac^4 by the word $b^{-1}ac^4$, at most four times, and areas with respect to $R_\Phi \supseteq R$ are no larger than with respect to R . So the hypothesis of Corollary 8.4(a) holds with this κ and $K = C_1 + 64$.

Theorem 8.5 (Theorem B). *The channel Φ on M_{873} lies in $\mathcal{K}_{873}$, and its Choi rank satisfies $2 \leq r_* \leq 248$. Let $\kappa = 2 + p_H = 6 + \log 107$. There are constants $c, c_0, K > 0$ and n_0 such that the following hold; all services have error at most $1/16$.*

- (a) *Memory against purity. For $n \geq n_0$ and $\log n \leq B \leq c_0 n$, every causal service of Φ for n rounds with purity $P + C \leq B$ has, at any exchange,*

$$Q \geq c \frac{n}{B (\log n)^{2\kappa}}.$$

For every $n \geq 1$ and $0 < B \leq n$ there is an exact causal service, with error 0, purity $P + C < B$, exchange $J = (n - 1)Q$ and memory $Q \leq K(n/B) \log(2n/B)$. In particular, for every $A_0 \geq 0$ there is $c' > 0$ such that services with $P + C \leq A_0 \log n$ have $Q \geq c'n/(\log n)^{2\kappa+1}$ for all large n , where $2\kappa + 1 = 2p_H + 5 < 26.49$, and there are services with $P = C = 0$ and $Q \leq Kn \log n$.

- (b) *The rank rate. For $n \geq n_0$, every causal service of Φ for n rounds at the rank rate, $J \leq \frac{n}{2} \log r_*$, has, whatever its purity,*

$$Q \geq c \frac{\sqrt{n}}{(\log n)^\kappa}.$$

For every $n \geq 2$ there is a causal service at the rank rate with $C = 0$ and $P, Q \leq K\sqrt{n \log n}$.

So memory and purity trade at a fixed rate: for $n \geq n_0$ and $\log n \leq B \leq c_0 n$ the least memory of a service with purity at most B lies between $cn/(B(\log n)^{2\kappa})$ and $K(n/B) \log(2n/B)$, and the product of memory and purity is $n^{1+o(1)}$. At the rank rate the least memory is $n^{1/2+o(1)}$, and it is attained with purity of the same order: the rank rate sits where memory and purity are equal.

The exponent 2κ in (a) is the unitary exponent of Theorem 7.5(c): a service with purity B is read at relator defect about $(B/n)^{1/2}$. The lower bound in (b) needs no assumption on purity, because at the rank rate purity is paid for with memory (Proposition 8.7): there $P + C < 3Q + 6$, and the argument of (a) with B of order Q gives $Q^2 \gtrsim n$ up to logarithms. The exact services in (a) refresh the whole memory in every round, so their exchange is far above the rank rate; the services in (b) keep a sofic model in memory for all n rounds and pay for its bad points with purity (Theorem 8.10). We prove Theorem 8.5 in Section 8.7, after the purity budget and the upper bound.

Remark 8.6 (The grid host). The grid host Γ_2 of Remark 7.8 should give a stronger lower bound with the same upper bound; both rest on sketches. Its certificate has 7 generators and 77 relations of total length 5445 and maximum length 224. Adding b and j and the same four words gives $s = 9$, $r_0 = 81$ and $\Lambda = 5462$, so its gadget channel Φ_2 acts on M_d with

$$d = 1 + 4 \cdot 9 + 3 \cdot 5462 - 4 \cdot 81 = 16099,$$

has $m = 224$, and has Choi rank at most $1 + 18 + 5300 - 78 - 3 = 5238$ by the count above (78 of its 81 words have length at least three). With the transport of Remark 7.8, words of length $O(N^{3/2})$ and pair areas $O(N^2(\delta_H(C\sqrt{N}) + N))$, the hypothesis of Corollary 8.4(a) holds with $\kappa = 2 + p_H/2$, and part (b) of that corollary holds with Lee's exponential Dehn bound [40] alone. The proof below then gives, for Φ_2 : memory at least $cn/(\log n)^{p_H+5}$ at purity $O(\log n)$, where $p_H + 5 < 15.75$; memory at least $n2^{-C\sqrt{\log n}}$ at purity $O(\log n)$ using only Lee's bound; and memory at least $c\sqrt{n}/(\log n)^{2+p_H/2}$ at the rank rate, whatever the purity. These rest on the pair-area bound for Γ_2 , whose proof Remark 7.8 only sketches. By the two-stage hashing sketched in Remark 7.8, $\log \sigma(r) = O(r \log r)$ for Γ_2 as well, and so an upper bound of order $n \log n$. For Γ itself, Lee's bound alone gives pair areas $2^{O(N)}$, and the argument gives only memory n^c for some $c > 0$.

8.4 A purity budget at the rank rate

At the rank rate a device cannot buy memory with purity: the purity it can consume is linear in its memory. We sharpen the budget $14Q + 3$ of [20] to $3Q + 6$ by combining its rank count with a one-sided second-moment bound.

Proposition 8.7 (Purity budget). *Let Φ be a channel on M_d , and suppose that some pure state ψ_* of the input and a reference has output $\omega = (\Phi \otimes \text{id})(\psi_*)$ flat of rank r . Every causal service of Φ for n rounds with error $\epsilon \leq 1/16$ and exchange $J \leq \frac{n}{2} \log r$ satisfies*

$$P + C < 3Q + 6.$$

Every gadget channel has such a probe with $r = r_*$ (Section 8.1). The proof uses the variance of the surprisal of a state with small deficit, which is [20, Lemma 5.9]; we include the short argument.

Lemma 8.8 (Surprisal variance [20, Lemma 5.9]). *Let β be a state on $q \geq 1$ qubits with spectrum (β_i) and deficit $D = q - S(\beta)$. The surprisal $-\log \beta_i$, drawn with probability β_i , has variance at most $(q + 2/\ln 2) D$.*

Proof. Put $\varphi(x) = x \ln x - x + 1 \geq 0$. We claim $x(\ln x)^2 \leq (2 + \ln_+ x)\varphi(x)$ for $x \geq 0$. For $x = e^t$ with $t \geq 0$ the difference of the two sides is $f(t) = e^t(t - 2) + t + 2$, with $f(0) = f'(0) = 0$ and $f''(t) = te^t \geq 0$. For $t \leq 0$ it is $g(t) = 2(te^t - e^t + 1) - t^2e^t$, with $g(0) = 0$ and $g'(t) = -t^2e^t \leq 0$; and $x = 0$ is clear. Now let $Q_0 = 2^q$ and $x_i = Q_0\beta_i \leq Q_0$, the index running over all Q_0 eigenvalues. Then $\frac{1}{Q_0} \sum_i x_i = 1$ and $D \ln 2 = \sum_i \beta_i \ln(Q_0\beta_i) = \frac{1}{Q_0} \sum_i \varphi(x_i)$. The variance is at most

$$\sum_i \beta_i (\log(Q_0\beta_i))^2 = \frac{1}{Q_0(\ln 2)^2} \sum_i x_i (\ln x_i)^2 \leq \frac{2 + q \ln 2}{(\ln 2)^2} D \ln 2 = \left(q + \frac{2}{\ln 2}\right) D.$$

□

Proof of Proposition 8.7. Put $b = P + C$ and suppose, for a contradiction, that $b \geq 3Q + 6$.

Preloading. The imports are fixed states, independent of everything else, and the exported qubits are never used again. So the observer's final state does not change if every import is prepared at the start in a register of its own, and each replacement swaps it with the exported qubits, which then stay in the device. The device becomes a register of $L = Q + J$ qubits in the product state $\sigma_0 = \beta_0 \otimes \bigotimes_f \omega_f$ of the initial memory state and the imported states, and it applies a fixed unitary in each round.

The surprisal. Let (σ_i) be the spectrum of σ_0 and let $Z = L + \log \sigma_i$, with i drawn with probability σ_i . Then $\mathbb{E}Z = L - S(\sigma_0) = b$. Also Z is the sum of the independent variables $Z_f = q_f + \log \beta_i^{(f)}$ attached to the factors of σ_0 , where q_f is the number of qubits of the factor and $\beta^{(f)}$ its spectrum. Every factor has at most Q qubits, since an import replaces exported memory qubits, and the deficits $b_f = \mathbb{E}Z_f$ of the factors add up to b . By Lemma 8.8,

$$\text{Var } Z = \sum_f \text{Var } Z_f \leq \left(Q + \frac{2}{\ln 2}\right) b.$$

A tail bound from the rank. Let the observer feed the input halves of n copies of ψ_* and keep the references. Its final state is $\tau_O = \sum_i \sigma_i \tau_i$, where τ_i is its final state when the register starts in the eigenvector of σ_i . Each τ_i is a marginal of a pure state of the observer's system and the L -qubit register, so $\text{rank } \tau_i \leq 2^L$. Fix a real number k and let I_k be the set of indices with $\sigma_i \geq 2^{k-L}$, so that $|I_k| \leq 2^{L-k}$. Let E be the support projection of $\sum_{i \in I_k} \tau_i$. Then

$\text{rank } E \leq 2^{2L-k}$ and $\text{Tr}(E\tau_O) \geq \sum_{i \in I_k} \sigma_i = \Pr(Z \geq k)$. The ideal final state $\omega^{\otimes n}$ is flat of rank r^n , so $\text{Tr}(E\omega^{\otimes n}) \leq 2^{2L-k}r^{-n} \leq 2^{2Q-k}$, using $2J \leq n \log r$. Since the error is at most ϵ ,

$$\Pr(Z \geq k) \leq \epsilon + 2^{2Q-k} \quad (k \in \mathbb{R}). \quad (10)$$

The contradiction. Put $u = b - 2Q - 3$, so that $u \geq b/3 + 1 > 0$. By the one-sided Chebyshev inequality, $\Pr(Z \geq b - u) \geq u^2/(\text{Var } Z + u^2)$, while (10) with $k = b - u = 2Q + 3$ gives $\Pr(Z \geq b - u) \leq \frac{1}{16} + \frac{1}{8} = \frac{3}{16}$. Hence $13u^2 \leq 3 \text{Var } Z$, and with $u > b/3$,

$$\frac{13}{9}b^2 < 3\left(Q + \frac{2}{\ln 2}\right)b, \quad \text{that is} \quad b < \frac{27}{13}\left(Q + \frac{2}{\ln 2}\right) < 2.08Q + 5.993,$$

which contradicts $b \geq 3Q + 6$. $\square$

8.5 An upper bound from sofic models

Sofic models of the labels give devices. The point is that the triangle blocks can be made exactly unitary for arbitrary permutations, and that at most points of a model the pure bath vector reproduces the target channel exactly, not approximately. Recall from Section 2 that $\sigma_E(r)$ is the least M for which some $f : E \rightarrow \text{Sym}(\Omega)$, $|\Omega| = M$, with $f(1) = \text{id}$, is a $(1 - r^{-1})$ -expansive r^{-1} -morphism: $f(x)f(y)$ and $f(xy)$ differ on at most M/r points whenever $x, y, xy \in E$, and $f(x)$ and $f(y)$ agree on at most M/r points whenever $x \neq y$ in E .

Proposition 8.9 (Sofic models give devices). *Let G , S and R be as in Section 8.1, with t_Δ triangles and r_* labels, let E be a chunk of G containing all labels, and put $K_G = t_\Delta + \binom{r_*}{2}$. Every $(1 - r^{-1})$ -expansive r^{-1} -morphism $f : E \rightarrow \text{Sym}(\Omega)$, $|\Omega| = M$, gives a unitary U_f on $\mathbb{C}^d \otimes \mathbb{C}^\Omega$ whose tracial channel $\Phi_f(X) = (\text{id} \otimes \tau_M)[U_f(X \otimes I)U_f^*]$ satisfies*

$$\Phi_f = (1 - \vartheta)\Phi_G + \vartheta\Xi$$

for some channel Ξ and some $0 \leq \vartheta \leq K_G/r$. Consequently, for all $n \geq 1$ and $0 < \epsilon \leq 1$ with $\sigma_E(2K_G n/\epsilon) < \infty$, Φ_G has

- (a) a causal service with error at most ϵ , $P = C = 0$, $J = (n - 1)Q$ and $Q = \lceil \log \sigma_E(2K_G n/\epsilon) + \log(2n/\epsilon) \rceil$;
- (b) a causal service with error at most ϵ , $J = P = C = 0$ and $Q = \lceil n \log \sigma_E(2K_G n/\epsilon) + \log(2/\epsilon) \rceil$;

and, for all $n \geq 1$ and $0 < B \leq n$ with $\sigma_E(4K_G n/B) < \infty$,

- (c) an exact causal service, with error 0, purity $P + C < B$, $J = (n - 1)Q$ and $Q = \lceil \log \sigma_E(4K_G n/B) + \log(4n/B) \rceil$.

Proof. The unitary. For a symbol w let P_w be the permutation matrix of $f([w])$ on $\mathbb{C}^\Omega$, so that symbols with the same label get the same matrix, and let U_f be the gadget U_λ with $\lambda(w) = P_w$. Each triangle block is $\frac{1}{\sqrt{2}} \begin{pmatrix} I & P_y \\ P_x & -P_x P_y \end{pmatrix} = \text{diag}(I, P_x)(H \otimes I)\text{diag}(I, P_y)$, so U_f is unitary although no relation holds among the P_w . Writing π_a for the permutation of Ω in slot a , namely $f([w_a])$ for every slot except the lower right corner of a triangle (x, y, z) , where it is $f([x])f([y])$, we have $U_f = \sum_a c_a E_a \otimes P_{\pi_a}$.

Pure bath points. Since τ_M is the average of the vector states of the points of Ω , $\Phi_f = \frac{1}{M} \sum_{\omega \in \Omega} \Phi_\omega$ with $\Phi_\omega(X) = \text{Tr}_\Omega U_f(X \otimes |\omega\rangle\langle\omega|)U_f^*$. The isometry $|\psi\rangle \mapsto U_f(|\psi\rangle \otimes |\omega\rangle)$ is $\sum_a c_a E_a \otimes |\pi_a(\omega)\rangle$, so

$$\begin{aligned} \Phi_\omega(X) &= \sum_{a,b \in \mathcal{T}} c_a \overline{c_b} \mathbf{1}[\pi_a(\omega) = \pi_b(\omega)] E_a X E_b^*, \\ \Phi_G(X) &= \sum_{a,b \in \mathcal{T}} c_a \overline{c_b} \mathbf{1}[[w_a] = [w_b]] E_a X E_b^*, \end{aligned}$$

the second because $\tau(\lambda(g)\lambda(h)^*) = \delta_{gh}$. Call ω *good* if $f([x])f([y])(\omega) = f([z])(\omega)$ for every triangle (x, y, z) and $f(g)(\omega) \neq f(h)(\omega)$ for all distinct labels g, h . At a good point $\pi_a(\omega) = f([w_a])(\omega)$ for every slot, since $[w_a] = [z]$ at the lower right corner of a triangle (x, y, z) ; so $\pi_a(\omega) = \pi_b(\omega)$ exactly when $[w_a] = [w_b]$, and $\Phi_\omega = \Phi_G$.

Bad points. The labels of a triangle satisfy $[x][y] = [z]$ with all three in E , so $f([x])f([y])$ and $f([z])$ differ on at most M/r points; and distinct labels $g, h \in E$ have $f(g)(\omega) = f(h)(\omega)$ for at most M/r points. So the fraction ϑ of bad points is at most K_G/r , and $\Phi_f = (1 - \vartheta)\Phi_G + \vartheta\Xi$, where Ξ is the average of Φ_ω over the bad points (any channel if there are none).

Exchange. Let $r = 2K_G n/\epsilon$, let f have degree $M = \sigma_E(r)$, and let Q be as in (a), so that $M \leq 2^Q \epsilon/(2n)$. Split the Q -qubit memory as $(\mathbb{C}^\Omega \otimes \mathbb{C}^{k_0}) \oplus \mathbb{C}^{g_0}$ with $k_0 = \lfloor 2^Q/M \rfloor$ and $g_0 = 2^Q - k_0 M < M$. In every round apply $U_f \otimes I_{k_0}$ on the first summand and the identity on the second, and after every round but the last export all Q memory qubits and import Q fresh qubits in the maximally mixed state; the initial state is maximally mixed as well, so $P = C = 0$. Every round then meets a fresh flat memory, and the observer faces n independent uses of the channel $(1 - g_0/2^Q)\Phi_f + (g_0/2^Q)\text{id}$, which is within $\vartheta + g_0/2^Q \leq \epsilon/(2n) + \epsilon/(2n)$ of Φ_G in diamond distance, by convexity. A hybrid argument over the rounds bounds the error by ϵ .

Exact service. Let $0 < B \leq n$, put $\eta = B/(4n) \leq \frac{1}{4}$ and $r = K_G/\eta$, let f have degree $M = \sigma_E(r)$, and let Q be as in (c), so that $M \leq \eta 2^Q$. Let $\Omega_0 \subseteq \Omega$ be the set of good points, so that $|\Omega_0| \geq (1 - \eta)M$. Split the memory as $(\mathbb{C}^\Omega \otimes \mathbb{C}^{k_0}) \oplus \mathbb{C}^{g_0}$ as in the exchange case, with $k_0 = \lfloor 2^Q/M \rfloor$ and $g_0 < M \leq \eta 2^Q$, and let β be the maximally mixed state on $\text{span}\{|\omega\rangle : \omega \in \Omega_0\} \otimes \mathbb{C}^{k_0}$. Since $k_0 M = 2^Q - g_0$, its deficit is

$$Q - \log(|\Omega_0|k_0) = \log \frac{2^Q}{2^Q - g_0} + \log \frac{M}{|\Omega_0|} \leq 2 \log \frac{1}{1 - \eta} \leq \frac{8\eta}{3 \ln 2} < \frac{B}{n}.$$

Start the memory in β , apply $U_f \otimes I_{k_0}$ on the first summand in every round, and after every round but the last export all Q memory qubits and import Q fresh qubits in the state β . The n registers used have total deficit $P + C < B$. In every round the memory is a fresh copy of β , a uniform mixture of vector states $|\omega\rangle \otimes |j\rangle$ with ω good, and $\Phi_\omega = \Phi_G$ at good points; so every round applies exactly Φ_G , independently of the other rounds, and the error is 0.

No exchange. Let f be as in the exchange case and Q as in (b), so that $M^n \leq 2^Q \epsilon/2$. Split the memory as $(\mathbb{C}^{\Omega^n} \otimes \mathbb{C}^{k_0}) \oplus \mathbb{C}^{g_0}$ with $k_0 = \lfloor 2^Q/M^n \rfloor$ and $g_0 < M^n$, start it maximally mixed, and in round s apply U_f to the input and the s -th factor $\mathbb{C}^\Omega$ of the first summand, and the identity on the second. With probability $1 - g_0/2^Q \geq 1 - \epsilon/2$ the memory starts in the first summand, uniformly distributed, and then the n factors are independent flat baths, each used once: the observer faces n independent uses of Φ_f , which is within $n\vartheta \leq \epsilon/2$ of the ideal experiment. The remaining branch has weight at most $\epsilon/2$, so the error is at most ϵ . $\square$

In particular, if $\sigma_E(r)$ is finite for every r , as it is when G is sofic, then $\Phi_G \in \mathcal{K}_d$, since $d_\diamond(\Phi_f, \Phi_G) \leq K_G/r$ for models at every accuracy. This is a quantitative form, for sofic groups, of the fact that gadget channels of hyperlinear groups lie in $\mathcal{K}_d$ [20].

8.6 Streaming at the rank rate

The services of Proposition 8.9 refresh the whole memory in every round. At the rank rate the memory has to be kept, and a sofic model cannot simply be reused, since its bad points would then accumulate error. The following theorem keeps the model for all rounds and pays for its bad points in purity instead. It uses the good points of a model as an exact *rectangular* dilation, whose input bath is smaller than its output bath, and lets the part of the memory

in use grow slightly in every round. The mixing and export steps are those of the rolling reservoir of [20, Appendix A].

Theorem 8.10 (Streaming at the rank rate). *Let Φ be a channel on M_d with linearly independent Kraus operators $A_1, \dots, A_r$, $r \geq 2$, and let $1 \leq k \leq k'$. Suppose that $W = \sum_i A_i \otimes B_i$, with $B_i : \mathbb{C}^k \rightarrow \mathbb{C}^{k'}$, is an isometry from $\mathbb{C}^d \otimes \mathbb{C}^k$ to $\mathbb{C}^d \otimes \mathbb{C}^{k'}$ and that*

$$\text{Tr}(B_i^* B_j) = k \delta_{ij}, \quad \sum_i B_i^* B_i = r I_k. \quad (11)$$

Then for all $n \geq 2$ and $0 < \epsilon < 1$ there is a causal service of Φ for n rounds with error at most ϵ , $C = 0$, $J = \lfloor \frac{n-1}{2} \log r \rfloor$ and

$$P \leq n \log \frac{k'}{k} + 2, \quad Q \leq \log k + n \log \frac{k'}{k} + O_r \left(\log \frac{n}{\epsilon} + \log(2 + \log k) \right).$$

The first identity in (11) says that W with a flat input bath realizes Φ exactly: $\Phi(X) = \text{Tr}_{k'}[W(X \otimes I_k/k)W^*] = \sum_i A_i X A_i^*$. The second says that every Kraus history costs the same. For a channel with a flat probe of rank r it follows from $W^*W = I$ alone: apply $\text{Tr}(\rho_* \cdot)$ to the first factor of $\sum_{i,j} A_i^* A_j \otimes B_i^* B_j = I$, where ρ_* is the input marginal of the probe, for which $\text{Tr}(\rho_* A_i^* A_j) = \delta_{ij}/r$.

Proof. We use the frames of [20, Appendix A]. A *frame* is a linear map $F : \mathbb{C}^M \rightarrow \mathbb{C}^D \otimes \mathbb{C}^S$; its columns are the vectors, in the memory $\mathbb{C}^D$ and an inaccessible exterior $\mathbb{C}^S$, attached to the Kraus histories. The exterior holds the purifier of the initial memory, the exported qubits and the purifiers of the imported ones. Put $g(F) = \|F^*F\|$ and $\alpha(F) = \|\text{Tr}_{\mathbb{C}^D} F F^*\|$. Two facts are used.

- *Concentration* [20, Lemma A.1]. For Hermitian X on $\mathbb{C}^D$ and Haar-random $V \in U(D)$, the matrix $Z = F^*((V^* X V) \otimes I)F - \tau_D(X)F^*F$ satisfies $\Pr\{\|Z\| \geq s\} \leq 2M \exp[-Ds^2/(4\alpha(F)g(F)\|X\|^2)]$.
- *Export* [20, Lemma A.4]. If $\mathbb{C}^D = \mathbb{C}^{D/e} \otimes \mathbb{C}^e$, there is a unitary of $\mathbb{C}^D$ after which exporting the factor $\mathbb{C}^e$ and importing a maximally mixed e -dimensional register give a frame F' with $F'^*F' = F^*F$ and, for every $\eta > 0$, $\alpha(F') \leq (1 + \eta)\alpha(F)/e^2 + 4e^2 g(F) \ln(8Se^2)/(\eta D)$.

Schedule and dimensions. Put $h = \log r$, $E_p = 2^{\lceil h/2 \rceil}$, $J_t = \lfloor th/2 \rfloor$, $j_t = J_t - J_{t-1} \leq \lceil h/2 \rceil$ and $e_t = 2^{j_t}$, a divisor of E_p . Choose D_0 , a multiple of kE_p , with $D_0 \geq nkE_p$ and

$$D_0^2 \geq \frac{32 A_* r^4 n^2 \ln(8r^{n+2})}{\epsilon^2}, \quad \text{where } A_* = 12(1 + 8E_p^2 n^2 L_*), \quad L_* = \ln(8D_0 r^n E_p^2). \quad (12)$$

Put $q_0 = k'/k$. For $t \geq 1$ let $\tilde{D}_t = q_0 D_{t-1}$, a multiple of $k'E_p$, and let D_t be the least multiple of kE_p with $D_t \geq \tilde{D}_t$. Then $D_t \leq q_0 D_{t-1} + kE_p$, so $D_n \leq q_0^n (D_0 + nkE_p) \leq 2q_0^n D_0$. The memory has $Q = \lceil \log D_n \rceil$ qubits, of which the last $\lceil h/2 \rceil$ form a *port*. A space whose dimension D' is a multiple of E_p is realized as a subspace of the first $Q - \lceil h/2 \rceil$ qubits of dimension D'/E_p , tensored with the port; the *active space* of round t is such a space of dimension D_{t-1} , identified with $\mathbb{C}^k \otimes \mathbb{C}^{D_{t-1}/k}$.

The device. The memory starts maximally mixed on the active space of round 1, of dimension D_0 . In round t the device applies a unitary V_t of the active space and then $W \otimes I_{D_{t-1}/k}$, which maps the input and the active space isometrically onto the output and a space of dimension $\tilde{D}_t$; it releases the output, then applies a unitary V'_t of that space (it acts on the memory only, so it could equally act before the release), and, if $t < n$, exports j_t port qubits and imports j_t qubits in the maximally mixed state in their place. The resulting space

of dimension $\tilde{D}_t$ lies in the active space of round $t + 1$. These maps are completed arbitrarily to unitaries of the input and the whole memory; the completion never acts, since the memory never leaves the active spaces. So $C = 0$, $J = J_{n-1}$, and $P = Q - \log D_0 \leq n \log q_0 + 2$.

Frames. Purify the initial memory: its frame F_0 is one unit vector, with $\alpha(F_0) = 1/D_0$. Let F be the frame before round t , with one column F_y for each history $y \in [r]^{t-1}$. After the call the frame has the columns $(B_i \otimes I)(V_t \otimes I)F_y$, indexed by the histories (y, i) . Its Gram matrix has the blocks $F^*((V_t^*(B_i^*B_j \otimes I)V_t) \otimes I)F$, whose Haar means are $\tau(B_i^*B_j \otimes I)F^*F = \delta_{ij}F^*F$ by the first identity in (11). By the second identity and cyclicity of the partial trace over the memory, the exterior marginal after the call is exactly r times the one before, whatever V_t is; so the call multiplies α by r .

Choice of the unitaries. We choose $V_1, V'_1, V_2, \dots$ in turn so that the frame F_t after round t satisfies

$$\|F_t^*F_t - I\| \leq t\epsilon/n, \quad \alpha(F_t) \leq A_*/D_0. \quad (13)$$

This holds for $t = 0$. Suppose it holds for $t - 1$; then $g(F_{t-1}) \leq 2$. Since $B_i^*B_i \leq rI$, the Hermitian and anti-Hermitian parts of the $B_i^*B_j$ have norm at most r . Apply the concentration lemma to these $2r^2$ operators, tensored with $I_{D_{t-1}/k}$, with $D = D_{t-1} \geq D_0$, at most r^n columns and level $s = \epsilon/(2rn)$: by (12) the failure probabilities add up to at most $4r^{n+2} \exp[-D_0^2\epsilon^2/(32A_*r^4n^2)] \leq \frac{1}{2}$. So some V_t makes every block of the new Gram matrix deviate from its mean by at most $\epsilon/(rn)$, hence the $r \times r$ block matrix by at most ϵ/n , and the Gram matrix after the call is within $t\epsilon/n$ of I ; exports and imports preserve it. Choose V'_t by the export lemma with $e = e_t$, $\eta = 1/n$ and $D = \tilde{D}_t \geq D_0$. The exterior has dimension $S \leq D_0 2^{2J_{t-1}} \leq D_0 r^n$, so $\ln(8Se_t^2) \leq L_*$, and

$$\alpha(F_t) \leq \left(1 + \frac{1}{n}\right) \frac{r \alpha(F_{t-1})}{e_t^2} + \frac{8E_p^2 n L_*}{D_0}.$$

Put $c_t = 2^{2J_t} r^{-t}$, so that $c_0 = 1$, $\frac{1}{4} < c_t \leq 1$ and $c_t r/e_t^2 = c_{t-1}$. Multiplying by c_t and unrolling gives $c_t \alpha(F_t) \leq (1 + \frac{1}{n})^t (1 + 8E_p^2 n^2 L_*)/D_0$, hence $\alpha(F_t) \leq 4e(1 + 8E_p^2 n^2 L_*)/D_0 \leq A_*/D_0$. In round n there is no export, and only the Gram bound is needed.

Error. Purify the observer. In the ideal experiment, with the Stinespring isometry $\xi \mapsto \sum_i A_i \xi \otimes |i\rangle$ in each round, its final vector is $\sum_y \psi_y \otimes |y\rangle$; in the service it is $\sum_y \psi_y \otimes F_n |y\rangle$ with the same vectors ψ_y , because the device acts on the input and the memory through $W = \sum_i A_i \otimes B_i$ and unitaries of the memory, and the observer never touches the memory or the exterior. The polar part $T = F_n(F_n^*F_n)^{-1/2}$ is an isometry with $\|F_n - T\| = \|(F_n^*F_n)^{1/2} - I\| \leq \epsilon$. So the actual final vector is within ϵ of the ideal one mapped by $I \otimes T$, and after tracing out the memory and the exterior the observer's final states are within ϵ in trace distance.

Memory. Finally $Q \leq \log D_n + 1 \leq \log D_0 + n \log q_0 + 2$. The conditions on D_0 hold for $D_0 = kE_p[K_r(n/\epsilon)^4(1 + \ln k)]$ with K_r large enough in terms of r , and then $\log D_0 = \log k + O_r(\log(n/\epsilon) + \log(2 + \log k))$. $\square$

Corollary 8.11 (The lamp channel at the rank rate). *There is K such that for every $n \geq 2$ the channel Φ of Theorem 8.5 has a causal service for n rounds with error at most $1/16$, $C = 0$, $J = \lfloor \frac{n-1}{2} \log r_* \rfloor$ and $P, Q \leq K\sqrt{n \log n}$.*

Proof. Let E be the chunk of labels of Proposition 8.9, let $s \geq 2K_G$, and let $f : E \rightarrow \text{Sym}(\Omega)$ be a $(1 - s^{-1})$ -expansive s^{-1} -morphism of degree $M = \sigma_E(s)$. Let Ω_0 be its set of good points, so that $k = |\Omega_0| \geq (1 - K_G/s)M$, and for each label g let $B_g : \mathbb{C}^{\Omega_0} \rightarrow \mathbb{C}^\Omega$ send $|\omega\rangle$ to $|f(g)\omega\rangle$. At a good point ω every slot of U_f acts through the permutation of its label, so $U_f(\xi \otimes |\omega\rangle) = \sum_g A_g \xi \otimes |f(g)\omega\rangle$, and $W = \sum_g A_g \otimes B_g$ is the restriction of the unitary U_f to $\mathbb{C}^d \otimes \mathbb{C}^{\Omega_0}$, an isometry. Each $B_g^*B_g$ is I_k , and $\text{Tr}(B_g^*B_h) = 0$ for $g \neq h$ because distinct labels do not collide at good points; the A_g are linearly independent, having disjoint supports.

So Theorem 8.10 applies with $r = r_*$ and $k' = M$. Here $\log(M/k) \leq -\log(1 - K_G/s) \leq 2K_G/(s \ln 2)$, and $\log k \leq \log \sigma_E(s) \leq C_E s \log s$ by Theorem 7.5(b). With $\epsilon = 1/16$ the theorem gives $Q \leq C_E s \log s + 2K_G n/(s \ln 2) + O(\log n)$ and $P \leq 2K_G n/(s \ln 2) + 2$; take $s = \max(2K_G, \lceil \sqrt{n/\log n} \rceil)$. $\square$

8.7 Proof of Theorem 8.5

Closure and services. By Theorem 7.5(b) the chunk E of Γ that consists of 1 and the labels of the gadget of R_Φ has $\log \sigma_E(r) \leq C_E r \log r$. Proposition 8.9 therefore gives finite tracial channels within K_G/r of Φ for every r , so $\Phi \in \mathcal{K}_{873}$; this also follows from [20], since Γ is amenable (Lemma 4.1) and hence hyperlinear. Parts (c) and (a) of the proposition give the exact services and the zero-purity services in (a), since $\log \sigma_E(R) \leq C_E R \log R$ for $R \geq 2$ and $\log(4K_G n/B) \leq (1 + \log 2K_G) \log(2n/B)$, because $\log(2n/B) \geq 1$; for the zero-purity services use $\epsilon = 1/16$. Corollary 8.11 gives the services in (b). The bounds on r_* were proved in Section 8.3.

Part (a). Let a service have error $\epsilon \leq 1/16$ and $P + C \leq B$, where $\log n \leq B$. Theorem 8.1 gives a round with bath dimension 2^Q , $d_\diamond(\Psi_t, \Phi) \leq \epsilon/(1 - \epsilon) \leq 1/15$ and $\nu_t = \ell_t + \Delta S_t \leq t_n$. Since $\epsilon + h_2(\epsilon) < 1$, $\epsilon \log n \leq \log n$ and $1 \leq \log n \leq B$,

$$t_n \leq \frac{16}{15} \cdot \frac{B + 1 + \log n}{n} \leq \frac{16B}{5n} \quad (n \geq 2).$$

Let $C_R \geq 1$ be the constant of Corollary 8.4(a) for R_Φ and the transport of Section 8.3, and put $y_n = 5n/(16C_R B)$, so that $C_R \nu_t y_n \leq 1$ and $y_n \leq n$. Put $c_0 = 5/(16C_R y_0)$. If $B \leq c_0 n$, then $y_n \geq y_0$, and the corollary gives

$$Q \geq S(\rho_t) \geq \frac{y_n}{32(\log y_n)^{2\kappa}} \geq \frac{5}{512 C_R} \cdot \frac{n}{B(\log n)^{2\kappa}}.$$

For the second statement take $B = \max(1, A_0) \log n$, which lies between $\log n$ and $c_0 n$ for large n .

Part (b). Proposition 8.7, with the flat probe of rank r_* , gives $P + C \leq B := 3Q + 6$. Theorem 8.1 now gives a round with $\nu_t \leq t_n \leq \frac{16}{15}(3Q + 7 + \log n)/n$. Put $y_n = 15n/(16C_R(3Q + 7 + \log n)) \leq n$. If $y_n < y_0$, then $3Q + 7 + \log n > 15n/(16C_R y_0)$, so $Q \geq c'n$ for large n . Otherwise Corollary 8.4(a) gives $Q \geq y_n/(32(\log n)^{2\kappa})$, that is,

$$Q(3Q + 7 + \log n) \geq \frac{15}{512 C_R} \cdot \frac{n}{(\log n)^{2\kappa}}.$$

If $Q \leq \log n + 7$ the left side is at most $4(\log n + 7)^2$, which is smaller than the right side for large n . So $Q > \log n + 7$, the left side is at most $4Q^2$, and $Q \geq c\sqrt{n}/(\log n)^\kappa$. $\square$

Remark 8.12 (Memory between two readings of the profile). For every gadget channel of a sofic group that satisfies the hypotheses of Theorem 8.2, the least memory of services with error $1/16$ and purity $O(\log n)$ lies between two readings of the group at scale n : below, the entropy forced by as many cheaply commuting copies of S_3 as Corollary 8.4 fits at relator defect about $\sqrt{\log n/n}$; above, $\log \sigma_E(Cn) + O(\log n)$ by Proposition 8.9, already at zero purity and with exchange $(n - 1)Q$. For the configuration lamps both readings are $n^{1+o(1)}$. The upper reading is the sofic profile, not the Følner function: the latter is doubly exponential for Γ (Proposition 7.7), so models built from Følner sets [18, Proposition 3.14] would only give memory exponential in n , while the hash models behind Theorem 7.5(b) give $O(n \log n)$. For the grid host of Remark 8.6, whose proofs are only sketched, the two readings would be $n/(\log n)^{15.75}$ and $O(n \log n)$.

Remark 8.13 (The rank rate is the diagonal of the trade-off). At the rank rate a device with error at most $1/16$ can spend at most $3Q + 6$ bits of purity, so Theorem 8.5(a) with B of order Q forces $Q \gtrsim \sqrt{n}$, and the streaming services attain this with purity of the same order. In general, for every channel with a flat probe, padding a rank-rate service with error at most $1/16$ with $2Q + 6$ maximally mixed qubits turns it into a service whose purity is at most its memory; so the least memory at purity at most memory is at most three times the least memory at the rank rate, plus six. For Φ both are $n^{1/2+o(1)}$. The one-round method cannot give more than $\sqrt{n}$ at the rank rate, since condition (9) certifies at most about $1/\nu$ copies; Theorem 8.10 shows that for Φ nothing more is true.

9 Proof of the weighted one-round theorem

The proof has four steps. First, a round with small leakage and entropy production yields unitaries on the bath under which every relation holds up to $O(\sqrt{\nu})$, with errors weighted by the memory state (Lemma 9.1). Second, weighted errors add up over van Kampen diagrams, at a price for each face and each vertex that is paid once (Lemma 9.2). Third, each transported copy of S_3 is rounded to an exact representation on the same space, so the state and its entropy do not change (Lemma 9.3). Fourth, measuring the rounded three-cycles one after another on one half of a purification of the memory state produces about one bit per copy that the other half also knows (Theorem 9.6).

The section generalises the argument of [20, Appendix B], which proves the one-round theorem for the Houghton gadget [20, Theorem 5.5], to every gadget channel that satisfies the hypotheses of Theorem 8.2. The weighted norms are those of [20, Appendix B]; Lemmas 9.1 and 9.2 extend Lemmas B.4 and B.2 there from the Houghton gadget to every such gadget; Lemma 9.3 is Lemma B.3 there; and Theorem 9.6 is a form of Theorem B.1 there, proved by sequential measurement.

9.1 Weighted norms

Throughout, $\|\cdot\|_F$ is the unnormalized Hilbert–Schmidt norm on M_M , ρ is a state on $\mathbb{C}^M$ and $\xi = \sqrt{\rho}$, so $\|\xi\|_F = 1$. For operators W on $\mathbb{C}^M$ put

$$E_\rho(W) = \|(W - I)\xi\|_F, \quad \chi_\rho(W) = \|W\xi - \xi W\|_F,$$

the *displacement* and the *centrality defect* of W at ρ . We drop the subscript ρ . For unitaries u, v and any operator F :

- (W1) $E(uv) \leq E(u) + E(v)$ and $E(u^*) = E(u)$;
- (W2) $\chi(uv) \leq \chi(u) + \chi(v)$ and $\chi(u^*) = \chi(u)$;
- (W3) $\|Fv\xi\|_F \leq \|F\xi\|_F + \|F\|\chi(v)$;
- (W4) $|E(uWu^*) - E(W)| \leq 2\chi(u)$ for every W with $\|W\| \leq 1$;
- (W5) $\|\xi F\|_F = \|F^*\xi\|_F$, and $\|\xi F\|_F = \|F\xi\|_F$ if F is normal.

Indeed, $(uv - I)\xi = u(v - I)\xi + (u - I)\xi$ and $(u^* - I)\xi = -u^*(u - I)\xi$; $uv\xi - \xi uv = u(v\xi - \xi v) + (u\xi - \xi u)v$ and $u^*\xi - \xi u^* = -u^*(u\xi - \xi u)u^*$; $Fv\xi = F\xi v + F(v\xi - \xi v)$; $(uWu^* - I)\xi = u(W - I)u^*\xi$, to which (W3) applies with $\|W - I\| \leq 2$, in both directions; and $\|F^*\xi\|_F^2 = \text{Tr}(\xi F F^* \xi)$, which equals $\text{Tr}(\xi F^* F \xi)$ when F is normal. A word w in $S^{\pm 1}$ is evaluated by an assignment ϕ of unitaries to S through $\phi(x^{-1}) = \phi(x)^*$; by (W2), $\chi(\phi(w)) \leq |w| \max_{x \in S} \chi(\phi(x))$.

9.2 Extraction

Lemma 9.1 (Weighted extraction). *Let G , S and R be as in Section 8.1, with $a, b, j \in S$, $j^{-1}ab \in R$ and $j \neq 1$ in G , and let m be as in Theorem 8.2. Let $\Psi(X) = \text{Tr}_M U(X \otimes \rho)U^*$ be a round with $u = d_\diamond(\Psi, \Phi_G) \leq 1/8$, leakage ℓ relative to Φ_G and entropy production ΔS , and put $\nu = \ell + \Delta S$, $\eta = \sqrt{d\nu}$ and $e = 256m\eta$. There are unitaries $\phi(x)$, $x \in S$, on $\mathbb{C}^M$ such that*

- (i) $E(\phi(r)) \leq e$ for every $r \in R$;
- (ii) $\chi(\phi(x)) \leq e$ for every $x \in S$;
- (iii) $E(\phi(a)\phi(b)) \geq 1 - e$.

Proof. Normalization. Let (ζ_i) and (ζ'_i) be orthonormal eigenbases of ρ and $T(\rho)$ with decreasing eigenvalues λ_i and μ_i . Replacing U by $(I \otimes W)U$, where the unitary W maps ζ'_i to ζ_i , changes neither Ψ nor its leakage and conjugates $T(\rho)$ by W . So we may assume $T(\rho) = \sum_i \mu_i |\zeta_i\rangle\langle\zeta_i|$. Input and output bath are now the same space, and $\xi = \sum_i \sqrt{\lambda_i} |\zeta_i\rangle\langle\zeta_i|$ acts on both; this does not assert $T(\rho) = \rho$.

Entropy production controls commutators. Write $U_{ji} = (I \otimes \langle\zeta_j|)U(I \otimes |\zeta_i\rangle) \in M_d$ and $t_{ji} = \frac{1}{d} \|U_{ji}\|_F^2$. Since the columns and the rows of U are orthonormal, t is doubly stochastic, and the diagonal of $T(\rho)$ gives $\mu_j = \sum_i t_{ji} \lambda_i$. The distributions $p_{ji} = t_{ji} \lambda_i$ and $q_{ji} = t_{ji} \mu_j$ satisfy

$$D(p||q) = \sum_{j,i} t_{ji} \lambda_i \log \frac{\lambda_i}{\mu_j} = -S(\rho) + S(T(\rho)) = \Delta S.$$

By Jensen's inequality $-2 \ln \sum \sqrt{p_{ji} q_{ji}} \leq D(p||q) \ln 2$, and $-2 \ln x \geq 2(1-x)$, so $\sum_{j,i} t_{ji} (\sqrt{\lambda_i} - \sqrt{\mu_j})^2 \leq \Delta S \ln 2$. The matrix t is a convex combination of permutation matrices (Birkhoff), and both sequences decrease, so the rearrangement inequality gives $\sum_{j,i} t_{ji} \sqrt{\lambda_i \mu_j} \leq \sum_j \sqrt{\lambda_j \mu_j}$; expanding the squares and using double stochasticity, $\sum_j (\sqrt{\lambda_j} - \sqrt{\mu_j})^2 \leq \sum_{j,i} t_{ji} (\sqrt{\lambda_i} - \sqrt{\mu_j})^2$. The bath block (j, i) of $U(I \otimes \xi) - (I \otimes \xi)U$ is $(\sqrt{\lambda_i} - \sqrt{\lambda_j})U_{ji}$, and $(\sqrt{\lambda_i} - \sqrt{\lambda_j})^2 \leq 2(\sqrt{\lambda_i} - \sqrt{\mu_j})^2 + 2(\sqrt{\mu_j} - \sqrt{\lambda_j})^2$. Therefore, with $U = \sum_{o,o'} |o\rangle\langle o'| \otimes A_{oo'}$ in system blocks $A_{oo'} \in M_M$,

$$\sum_{o,o'} \| [A_{oo'}, \xi] \|_F^2 = \| [U, I \otimes \xi] \|_F^2 \leq 4d \Delta S \ln 2 =: \epsilon_S^2. \quad (14)$$

Leakage controls residuals. The Choi state $\mathcal{J}(\Psi)$ is the marginal on $\mathbb{C}^d \otimes \mathbb{C}^d$ of the unit vector $d^{-1/2} \sum_{o,o'} |oo'\rangle \otimes A_{oo'} \xi$ in $\mathbb{C}^d \otimes \mathbb{C}^d \otimes \mathcal{H}$, where $\mathcal{H}$ is M_M with the Hilbert–Schmidt inner product: indeed $\langle A_{o_2 o'_2} \xi, A_{o_1 o'_1} \xi \rangle = \text{Tr}(A_{o_1 o'_1} \rho A_{o_2 o'_2}^*) = \langle o_1 | \Psi(|o'_1\rangle\langle o'_2|) | o_2 \rangle$. The support of $\mathcal{J}(\Phi_G)$ is spanned by the orthogonal vectors $|A_g\rangle$, of squared norms $\mu_g = \|A_g\|_F^2$. Projecting onto it, put

$$X_g = \mu_g^{-1} \sum_{[w_a]=g} \bar{c}_a A_a, \quad R_{oo'} = A_{oo'} - \sum_g (A_g)_{oo'} X_g,$$

where A_a is the block at slot a . Then $\sum_{o,o'} \|R_{oo'} \xi\|_F^2 = d\ell =: \epsilon_\ell^2$. At a slot $A_a = c_a X_{[w_a]} + R_a$, and at every other position $A_{oo'} = R_{oo'}$. We have $\epsilon_\ell \leq \eta$, $\epsilon_S \leq 2\sqrt{\ln 2} \eta$, and, by Cauchy–Schwarz, $\epsilon_\ell + \epsilon_S \leq \sqrt{1 + 4 \ln 2} \eta < 2\eta$ and $2\epsilon_\ell^2 + 2\epsilon_S^2 \leq 8 \ln 2 \eta^2$.

Anchors. Every symbol w , and the empty word, occupies a diagonal position $z = z(w)$ whose row and column contain no other slot; there the coefficient is 1, so $B_w := A_{zz} = X_{[w]} + R_{zz}$. Column unitarity gives $\text{Tr} \rho(I - B_w^* B_w) = \sum_{o \neq z} \|R_{oz} \xi\|_F^2 \leq \epsilon_\ell^2$. Row unitarity gives $\text{Tr} \rho(I - B_w B_w^*) = \sum_{o' \neq z} \|\xi A_{zo'}\|_F^2 \leq \sum_{o' \neq z} (\|R_{zo'} \xi\|_F + \|[A_{zo'}, \xi]\|_F)^2 \leq 2\epsilon_\ell^2 + 2\epsilon_S^2$. Let $B_w = V_w |B_w| = |B_w^*| V_w$ with V_w unitary. Since $(1-x)^2 \leq 1-x^2$ on $[0, 1]$,

$$\|(V_w - B_w) \xi\|_F \leq \epsilon_\ell, \quad \|\xi(V_w - B_w)\|_F = \|\xi(I - |B_w^*|)\|_F \leq \sqrt{8 \ln 2} \eta,$$

and so $\chi(V_w) \leq \|(V_w - B_w)\xi\|_F + \|[B_w, \xi]\|_F + \|\xi(B_w - V_w)\|_F \leq \epsilon_\ell + \epsilon_S + \sqrt{8 \ln 2} \eta < 4.3\eta$, since $\epsilon_\ell + \epsilon_S \leq \sqrt{1 + 4 \ln 2} \eta$.

Triangle slots. In the block of a triangle (x, y, z) , compare each slot a with the anchor of the symbol $w'(a)$ that carries the same label: the empty word at the upper left, y at the upper right, x at the lower left and z at the lower right. Put $F_a = A_a - c_a V_{w'(a)}$. Then $F_a = R_a - c_a R_{z'z'} + c_a (B_{w'(a)} - V_{w'(a)})$ with $z' = z(w'(a))$, and since a and (z', z') are different positions, Cauchy-Schwarz gives

$$\begin{aligned} \|F_a \xi\|_F &\leq \sqrt{3/2} \epsilon_\ell + 2^{-1/2} \epsilon_\ell < 2\eta, & \|F_a\| &< 2, \\ \|\xi F_a\|_F &\leq \|F_a \xi\|_F + \|[A_a, \xi]\|_F + 2^{-1/2} \chi(V_{w'(a)}) < 7\eta. \end{aligned}$$

For two slots a_1, a_2 in the same row of the block, with $A_{a_i} = c_i V_i + F_i$,

$$A_{a_1}^* A_{a_2} - \bar{c}_1 c_2 V_1^* V_2 = A_{a_1}^* F_2 + c_2 F_1^* V_2,$$

whose right-weighted norm is at most $\|F_2 \xi\|_F + 2^{-1/2} (\|\xi F_1\|_F + \|F_1\| \chi(V_2)) < 2\eta + 2^{-1/2} \cdot 17\eta < 15\eta$ by (W3) and (W5). The two columns of the block are orthogonal columns of U , so the products for the two rows of the block add up to $-C_1^* C_2$, where C_1, C_2 are the parts of the two columns in the other rows. These are residuals, so $\|C_1^* C_2 \xi\|_F \leq \|C_2 \xi\|_F \leq \epsilon_\ell$. The coefficients are $\bar{c}_1 c_2 = \frac{1}{2}$ for the pair (upper left, upper right) and $-\frac{1}{2}$ for the pair (lower left, lower right), so

$$\frac{1}{2} \|(V_\emptyset^* V_y - V_x^* V_z) \xi\|_F < 15\eta + 15\eta + \eta.$$

Put $W_w = V_\emptyset^* V_w$ for every symbol w , and $W_\emptyset = I$. Since $W_x W_y - W_z = V_\emptyset^* V_x (V_\emptyset^* V_y - V_x^* V_z)$,

$$\|(W_x W_y - W_z) \xi\|_F \leq 64\eta \quad \text{for every triangle } (x, y, z), \quad \chi(W_w) < 10\eta. \quad (15)$$

Relations. Let $\phi(x) = W_x$ for $x \in S$. For a letter $x \in S$, the triangle $(x, x^{-1}, \emptyset)$ gives $\|(W_{x^{-1}} - \phi(x^{-1})) \xi\|_F = \|W_x^* (W_x W_{x^{-1}} - I) \xi\|_F \leq 64\eta$. Fix a word $r = \sigma_1 \cdots \sigma_\ell$ of R and let $D_l = \|(\phi(\sigma_1 \cdots \sigma_l) - W_{p_l}) \xi\|_F$, with p_l its prefix symbols, $p_1 = \sigma_1$ and $W_{p_\ell} = I$. Then $D_1 \leq 64\eta$, and

$$\phi(\sigma_1 \cdots \sigma_l) - W_{p_l} = (\phi(\sigma_1 \cdots \sigma_{l-1}) - W_{p_{l-1}}) \phi(\sigma_l) + W_{p_{l-1}} (\phi(\sigma_l) - W_{\sigma_l}) + (W_{p_{l-1}} W_{\sigma_l} - W_{p_l}).$$

By (W3), (15) and the triangle (p_{l-1}, σ_l, p_l) , $D_l \leq D_{l-1} + 2 \cdot 10\eta + 64\eta + 64\eta$. So $E(\phi(r)) = D_\ell \leq 148\ell\eta \leq e$, which is (i); and (ii) holds because $\chi(\phi(x)) < 10\eta \leq e$.

Displacement. Let $z = z(j)$ and let 0 be the position of the empty word. The labels there are $[j] \neq 1$ and 1, so $\langle z | \Phi_G(|z\rangle\langle 0|) | 0 \rangle = \tau(\lambda([j])) = 0$, while $\langle z | \Psi(|z\rangle\langle 0|) | 0 \rangle = \text{Tr}(B_j \rho B_\emptyset^*)$. Feed $(|zz\rangle + |00\rangle)/\sqrt{2}$ into $\Psi \otimes \text{id}$ and $\Phi_G \otimes \text{id}$. The difference Δ of the outputs is Hermitian and traceless with $\|\Delta\|_1 \leq 2u$, so each of its entries has modulus at most $\|\Delta\| \leq \|\Delta\|_1/2 \leq u$; the entry at $(zz, 00)$ is half the difference above, so $|\text{Tr}(\rho B_\emptyset^* B_j)| \leq 2u$. By the anchor bounds, $|\text{Tr}(\rho V_\emptyset^* V_j) - \text{Tr}(\rho B_\emptyset^* B_j)| \leq 2\epsilon_\ell$, and so

$$E(\phi(j))^2 = \|(V_j - V_\emptyset) \xi\|_F^2 = 2 - 2 \text{Re} \text{Tr}(\rho V_\emptyset^* V_j) \geq 2 - 4u - 4\epsilon_\ell \geq \frac{3}{2} - 4\eta.$$

By (i) for the word $j^{-1}ab$, $\|(\phi(a)\phi(b) - \phi(j)) \xi\|_F = E(\phi(j)^* \phi(a)\phi(b)) \leq e$, so $E(\phi(a)\phi(b)) \geq \sqrt{3/2 - 4\eta} - e \geq 1 - e$ if $\eta \leq 1/8$. If $\eta > 1/8$ then $e > 1$ and (iii) is trivial. $\square$

9.3 Charged area

Relator errors add up over a van Kampen diagram, but in the weighted norms every conjugation costs twice the centrality defect of the conjugator. Writing a null word naively as a product of conjugates of relations would charge each conjugating word once for every relation it encloses. Peeling the diagram one boundary face at a time charges instead each face a bounded amount and each vertex once.

Lemma 9.2 (Charged area). *Let S and R be finite, with every word of R of length at most m , and let ϕ assign unitaries to S with $E(\phi(r)) \leq e$ for $r \in R$ and $\chi(\phi(x)) \leq e$ for $x \in S$. Every word w of finite area satisfies*

$$E(\phi(w)) \leq (3m \text{Area}_R(w) + |w|)e \leq ((4m+1) \text{Area}_R(w) + 2|w|)e.$$

Proof. Since ϕ is a homomorphism on the free group, $E(\phi(w))$ depends only on the free reduction of w , which is not longer and has the same area; so let w be freely reduced. By van Kampen's lemma [42, Chapter V] there is a van Kampen diagram for w with at most $\text{Area}_R(w)$ faces: a finite, connected and simply connected planar 2-complex with edges labelled by letters of S , whose boundary cycle, read from a base vertex, is w , and in which the boundary cycle of every face, read from a suitable vertex in a suitable direction, is a word of R . We show, by induction on the number of edges, that every such diagram D with V vertices and A faces satisfies

$$E(\phi(w_D)) \leq ((2m-1)A + 2(V-1))e, \quad (16)$$

where w_D is its boundary word. If D is a single vertex, w_D is empty. Otherwise let ε be the first edge of the boundary cycle, from the base vertex v_0 to v_1 , with label $x^{\pm 1}$.

If ε lies on no face, the boundary cycle traverses it twice, and removing it leaves diagrams $D_0 \ni v_0$ and $D_1 \ni v_1$ with $w_D = x^{\pm 1} w_{D_1} x^{\mp 1} w_{D_0}$, the words read from v_1 and v_0 . By (W1) and (W4), $E(\phi(w_D)) \leq 2e + E(\phi(w_{D_1})) + E(\phi(w_{D_0}))$, and the vertices and faces of D_0 and D_1 add up to those of D .

Otherwise ε lies on the boundary of a face f . Let c be the boundary cycle of f read from v_0 and beginning with ε , so that $c = \varepsilon\pi$ with π a path in ∂f from v_1 to v_0 . Removing f and ε merges f with the outer region and leaves a van Kampen diagram D' with the same vertices and one face fewer, whose boundary word read from v_0 is $w_{D'} = \pi^{-1}\rho$, where $w_D = \varepsilon\rho$. Hence $w_D = cw_{D'}$ freely. The word c is a cyclic permutation of some $r^{\pm 1}$, $r \in R$, that is $c = y^{-1}r^{\pm 1}y$ freely with $|y| \leq m-1$, so $E(\phi(c)) \leq e + 2(m-1)e$ by (W4) and (W2), and $E(\phi(w_D)) \leq (2m-1)e + E(\phi(w_{D'}))$ by (W1).

In both cases (16) follows from the induction hypothesis. Finally $V-1$ is at most the number of edges of D , since D is connected, and counting the two sides of every edge gives twice the number of edges as at most $mA + |w|$, the sum of the lengths of the face boundaries and of the boundary cycle. So (16) gives $E(\phi(w)) \leq ((2m-1)A + mA + |w|)e$. $\square$

9.4 Exact S_3 rounding in place

An approximate copy of S_3 is rounded to an exact one by functional calculus on the same space. The state is untouched, so no entropy is gained or lost.

Lemma 9.3 (Exact rounding [20, Lemma B.3]). *Let X and Y be unitaries on $\mathbb{C}^M$, and suppose that the displacements of X^2 , Y^2 and $(XY)^3$ and the centrality defects of X and Y are at most δ . There are unitaries x, c on $\mathbb{C}^M$ with $x^2 = c^3 = I$ and $xcx = c^{-1}$, such that*

$$\|(x - X)\xi\|_F \leq \delta, \quad \chi(x) \leq 3\delta, \quad \|(c - XY)\xi\|_F \leq 29\delta, \quad \chi(c) \leq 56\delta.$$

Proof. Let $\text{sgn } t = 1$ for $t \geq 0$ and -1 for $t < 0$, and let $x = \text{sgn}(\text{Re } X)$ and $y = \text{sgn}(\text{Re } Y)$ by functional calculus; they are self-adjoint unitaries. On the unit circle $|z - \text{sgn } \text{Re } z| \leq |z^2 - 1|$: for $z = e^{i\vartheta}$ with $|\vartheta| \leq \pi/2$ this reads $2|\sin(\vartheta/2)| \leq 2|\sin \vartheta|$, which holds as $\cos(\vartheta/2) \geq \frac{1}{2}$, and the case $\text{Re } z < 0$ follows by symmetry. Hence $|x - X|^2 \leq |X^2 - I|^2$ as functions of the normal operator X , and $\|(x - X)\xi\|_F \leq E(X^2) \leq \delta$; by (W5) also $\|\xi(x - X)\|_F \leq \delta$, so $\chi(x) \leq \chi(X) + 2\delta \leq 3\delta$. The same holds for y .

Put $Z = XY$ and $w = xy$, so that $xwx = yx = w^*$. Then $\chi(Z) \leq 2\delta$, $\chi(w) \leq 6\delta$, and, by (W3), $\|(w - Z)\xi\|_F \leq \|x(y - Y)\xi\|_F + \|(x - X)Y\xi\|_F \leq \delta + 3\delta = 4\delta$. From $w^3 - Z^3 = w^2(w - Z) + w(w - Z)Z + (w - Z)Z^2$ and (W3),

$$E(w^3) \leq E(Z^3) + 4\delta + (4\delta + 2 \cdot 2\delta) + (4\delta + 2 \cdot 4\delta) \leq 25\delta.$$

Let f send each point of the unit circle to a nearest cube root of unity, with the ties $e^{\pm i\pi/3}$ and -1 sent to 1 . Then $f(\bar{z}) = \overline{f(z)}$, and $|z - f(z)| \leq |z^3 - 1|$: for $z = -1$ both sides equal 2 , and otherwise $z = f(z)e^{i\vartheta}$ with $|\vartheta| \leq \pi/3$, and the inequality reads $2|\sin(\vartheta/2)| \leq 2|\sin(3\vartheta/2)|$. Let $c = f(w)$. Then $c^3 = I$, and $xcx = f(xwx) = f(w^*) = c^* = c^{-1}$. As before $\|(c - w)\xi\|_F \leq E(w^3) \leq 25\delta$ and $\|\xi(c - w)\|_F \leq 25\delta$. Hence $\|(c - Z)\xi\|_F \leq 29\delta$ and $\chi(c) \leq \chi(w) + \|(c - w)\xi\|_F + \|\xi(c - w)\|_F \leq 56\delta$. $\square$

9.5 A weighted sector bound

Rounded copies of S_3 that nearly commute in the weighted sense force entropy. We work in the Hilbert space $\mathcal{H}$ of operators on $\mathbb{C}^M$ with the Hilbert-Schmidt inner product $\langle Y, Z \rangle = \text{Tr}(Y^*Z)$, identified with $\mathbb{C}^M \otimes \mathbb{C}^M$. Left multiplications act on the first factor, which we call the system, and right multiplications on the second, the reference; they commute. The unit vector ξ purifies ρ on the system, and its reference marginal has the same nonzero spectrum, so it has entropy $S(\rho)$.

Lemma 9.4 (Projection products [19, Lemma 4.1]). *Let $P_1, \dots, P_k$ be orthogonal projections on a Hilbert space. Every vector y satisfies $\|y\|^2 - \text{Re}\langle y, P_k \cdots P_1 y \rangle \leq 2 \sum_l \|(I - P_l)y\|^2$.*

Proof. By homogeneity let $\|y\| = 1$. Put $y_0 = y$, $y_l = P_l y_{l-1}$ and $s_l = y_{l-1} - y_l$, which lies in the range of $I - P_l$. Then $\sum_l \|s_l\|^2 = 1 - \|y_k\|^2 =: \beta$, and $\alpha := 1 - \text{Re}\langle y, y_k \rangle = \text{Re} \sum_l \langle (I - P_l)y, s_l \rangle \leq \sqrt{E\beta}$ with $E = \sum_l \|(I - P_l)y\|^2$. Since $0 \leq \|y - y_k\|^2 = 2\alpha - \beta$, we get $\alpha \leq \sqrt{2E\alpha}$, that is $\alpha \leq 2E$. $\square$

Let $c_1, \dots, c_N$ be unitaries on $\mathbb{C}^M$ with $c_i^3 = I$, and write $c_i = \sum_{k \in \mathbb{Z}/3} \omega^k P_i^k$ with $\omega = e^{2\pi i/3}$ and spectral projections P_i^k . For every $Y \in \mathcal{H}$,

$$\|(c_i - I)Y\|_F^2 = 3 \sum_{k \neq 0} \|P_i^k Y\|_F^2, \quad \|[c_i, Y]\|_F^2 = 3 \sum_{k \neq l} \|P_i^k Y P_i^l\|_F^2, \quad (17)$$

because $|\omega^k - \omega^l|^2 = 3$ for $k \neq l$ and the pieces $P_i^k Y P_i^l$ are orthogonal. For $\mathbf{a} = (a_1, \dots, a_{i-1}) \in (\mathbb{Z}/3)^{i-1}$ put $Q_{\mathbf{a}} = P_{i-1}^{a_{i-1}} \cdots P_1^{a_1}$. Summing over the coordinates one at a time gives $\sum_{\mathbf{a}} Q_{\mathbf{a}}^* Q_{\mathbf{a}} = \sum_{\mathbf{a}} Q_{\mathbf{a}} Q_{\mathbf{a}}^* = I$.

Lemma 9.5 (Transfer to the reference). *For every $Y \in \mathcal{H}$ and $1 \leq i \leq N$,*

$$\sum_{\mathbf{a}} \|Q_{\mathbf{a}} Y - Y Q_{\mathbf{a}}\|_F^2 \leq \frac{4}{3} \sum_{k < i} \|[c_k, Y]\|_F^2.$$

Proof. The pinchings $\mathcal{E}_k(Y) = \sum_l P_k^l Y P_k^l$ are orthogonal projections on $\mathcal{H}$, and by (17) $\|(I - \mathcal{E}_k)Y\|_F^2 = \sum_{l \neq l'} \|P_k^l Y P_k^{l'}\|_F^2 = \frac{1}{3} \|[c_k, Y]\|_F^2$. Both $\sum_{\mathbf{a}} \|Q_{\mathbf{a}} Y\|_F^2$ and $\sum_{\mathbf{a}} \|Y Q_{\mathbf{a}}\|_F^2$ equal $\|Y\|_F^2$, and summing over the coordinates one at a time, $\sum_{\mathbf{a}} \langle Y Q_{\mathbf{a}}, Q_{\mathbf{a}} Y \rangle = \sum_{\mathbf{a}} \text{Tr}(Q_{\mathbf{a}}^* Y^* Q_{\mathbf{a}} Y) = \langle \mathcal{E}_1 \cdots \mathcal{E}_{i-1} Y, Y \rangle$. So the left side equals $2\|Y\|_F^2 - 2 \text{Re}\langle Y, \mathcal{E}_1 \cdots \mathcal{E}_{i-1} Y \rangle$, and Lemma 9.4 bounds it by $4 \sum_{k < i} \|(I - \mathcal{E}_k)Y\|_F^2$. $\square$

A three-valued distribution that is nearly symmetric under $k \mapsto -k$ carries about one bit for each unit of weight off 0. Let h_2 be the binary entropy. With $t = (1 + y)/2$,

$$\ln 2 (1 - h_2(t)) = \sum_{k \geq 1} \frac{y^{2k}}{2k(2k-1)} \leq \sum_{k \geq 1} \binom{2k}{k} \frac{y^{2k}}{(2k-1)4^k} = 1 - \sqrt{1 - y^2} = 1 - 2\sqrt{t(1-t)},$$

comparing coefficients: $4^k \leq 2k \binom{2k}{k}$ holds for $k = 1$, and the right side grows by the factor $2(2k+1)/k \geq 4$ from k to $k+1$. For a distribution (r_0, r_1, r_2) with $s = r_1 + r_2 > 0$ this gives, with $t = r_1/s$,

$$H(r_0, r_1, r_2) \geq s h_2(r_1/s) \geq s - \frac{(\sqrt{r_1} - \sqrt{r_2})^2}{\ln 2}, \quad (18)$$

which also holds when $s = 0$.

Theorem 9.6 (Weighted sector bound). *Let c_i, x_i , $1 \leq i \leq N$, be unitaries on $\mathbb{C}^M$ with $c_i^3 = x_i^2 = I$ and $x_i c_i x_i = c_i^{-1}$, with no condition relating different indices. For a state ρ with $\xi = \sqrt{\rho}$ put $\Delta_i = E(c_i)$, $\theta_i = \chi(c_i)$, $\theta'_i = \chi(x_i)$, and*

$$T_i = \left(\sum_{k < i} \theta_k^2 \right)^{1/2}, \quad K_i = \left(\sum_{k < i} \|[c_i, c_k] \xi\|_F^2 \right)^{1/2}, \quad K'_i = \left(\sum_{k < i} \|[x_i, c_k] \xi\|_F^2 \right)^{1/2},$$

$$w_i = \frac{1}{3} \left(\Delta_i - \frac{4}{\sqrt{3}} T_i \right)_+^2, \quad \beta_i = \theta'_i + \frac{2}{\sqrt{3}} (2T_i + K'_i), \quad \pi_i = \frac{1}{3} \left(\theta_i + \frac{2}{\sqrt{3}} (2T_i + K_i) \right)^2.$$

If $\pi_i \leq 2/3$ for all i , then

$$S(\rho) \geq \sum_{i=1}^N \left[w_i - \frac{\beta_i^2}{2 \ln 2} - h_2(\pi_i) - \pi_i \right].$$

Proof. Sequential measurement. Measure $c_1, \dots, c_N$ in turn on the system, with the projections P_i^k , and let $A_i \in \mathbb{Z}/3$ be the outcomes. Then $\Pr(A_{<i} = \mathbf{a}, A_i = k) = \|P_i^k Q_{\mathbf{a}} \xi\|_F^2 =: p(\mathbf{a}, k)$. The measurements do not touch the reference, whose entropy is $S(\rho)$, so

$$S(\rho) \geq I(A_1 \cdots A_N : \text{Ref}) = \sum_{i=1}^N I(A_i : \text{Ref} \mid A_{<i}),$$

since the conditional entropy of the reference given the classical outcomes is nonnegative. Fix i .

The copy stays displaced. By (17), $\Pr(A_i \neq 0) = \frac{1}{3} \sum_{\mathbf{a}} \|(c_i - I) Q_{\mathbf{a}} \xi\|_F^2$. By Lemma 9.5 with $Y = \xi$, the vectors $Q_{\mathbf{a}} \xi$ and $\xi Q_{\mathbf{a}}$ differ by at most $\frac{2}{\sqrt{3}} T_i$ in ℓ^2 over $\mathbf{a}$, and $\sum_{\mathbf{a}} \|(c_i - I) \xi Q_{\mathbf{a}}\|_F^2 = \Delta_i^2$ because $\sum_{\mathbf{a}} Q_{\mathbf{a}} Q_{\mathbf{a}}^* = I$. Since $\|c_i - I\| \leq 2$, the triangle inequality in ℓ^2 gives $\Pr(A_i \neq 0) \geq w_i$.

The sign is a fair coin. Since $x_i P_i^k x_i = P_i^{-k}$ and right multiplication by x_i is unitary, $\sqrt{p(\mathbf{a}, -k)} = \|P_i^k x_i Q_{\mathbf{a}} \xi\|_F$ and $\sqrt{p(\mathbf{a}, k)} = \|P_i^k Q_{\mathbf{a}} \xi x_i\|_F$, so $\sum_{\mathbf{a}, k} (\sqrt{p(\mathbf{a}, k)} - \sqrt{p(\mathbf{a}, -k)})^2 \leq \sum_{\mathbf{a}} \|x_i Q_{\mathbf{a}} \xi - Q_{\mathbf{a}} \xi x_i\|_F^2$. Write

$$x_i Q_{\mathbf{a}} \xi - Q_{\mathbf{a}} \xi x_i = x_i (Q_{\mathbf{a}} \xi - \xi Q_{\mathbf{a}}) + ((x_i \xi) Q_{\mathbf{a}} - Q_{\mathbf{a}} (x_i \xi)) + Q_{\mathbf{a}} (x_i \xi - \xi x_i).$$

In ℓ^2 over $\mathbf{a}$ the first term is at most $\frac{2}{\sqrt{3}} T_i$ and the third is θ'_i . For the second apply Lemma 9.5 with $Y = x_i \xi$: since $[c_k, x_i \xi] = [c_k, x_i] \xi + x_i [c_k, \xi]$, it is at most $\frac{2}{\sqrt{3}} (K'_i + T_i)$. So the sum is at most β_i^2 , and since the terms $k = 1$ and $k = 2$ contribute equally, (18) gives

$$H(A_i \mid A_{<i}) \geq \Pr(A_i \neq 0) - \frac{1}{\ln 2} \sum_{\mathbf{a}} (\sqrt{p(\mathbf{a}, 1)} - \sqrt{p(\mathbf{a}, 2)})^2 \geq w_i - \frac{\beta_i^2}{2 \ln 2}.$$

The reference knows the outcome. Let the reference measure c_i from the right, with the projections $Y \mapsto Y P_i^l$, and call the outcome B_i . Left and right multiplications commute, so $\Pr(A_{<i} = \mathbf{a}, A_i = k, B_i = l) = \|P_i^k Q_{\mathbf{a}} \xi P_i^l\|_F^2$, and by (17) $\Pr(A_i \neq B_i) = \frac{1}{3} \sum_{\mathbf{a}} \|[c_i, Q_{\mathbf{a}} \xi]\|_F^2$. As above,

$$[c_i, Q_{\mathbf{a}} \xi] = c_i (Q_{\mathbf{a}} \xi - \xi Q_{\mathbf{a}}) + ((c_i \xi) Q_{\mathbf{a}} - Q_{\mathbf{a}} (c_i \xi)) + Q_{\mathbf{a}} (c_i \xi - \xi c_i),$$

and Lemma 9.5 with $Y = \xi$ and $Y = c_i \xi$ gives $\Pr(A_i \neq B_i) \leq \pi_i$. By data processing and Fano's inequality for three values,

$$I(A_i : \text{Ref} \mid A_{<i}) \geq I(A_i : B_i \mid A_{<i}) \geq H(A_i \mid A_{<i}) - h_2(\pi_i) - \pi_i,$$

where we used that $h_2(\pi) + \pi$ increases on $[0, 2/3]$. Summing over i proves the theorem. $\square$

Corollary 9.7. *In Theorem 9.6, suppose $\Delta_i \geq 1/2$ for all i , and that $\theta_i, \theta'_i, \|[c_i, c_k]\xi\|_F$ and $\|[x_i, c_k]\xi\|_F$ are at most b for all $i \neq k$, where $\sqrt{N}b \leq 10^{-3}$. Then $S(\rho) \geq N/16$.*

Proof. Now $T_i, K_i, K'_i \leq \sqrt{N}b \leq 10^{-3}$ and $\theta_i, \theta'_i \leq 10^{-3}$. So $w_i \geq \frac{1}{3}(\frac{1}{2} - \frac{4}{\sqrt{3}}10^{-3})^2 > 0.0825$, and β_i and $\sqrt{3\pi_i}$ are at most $10^{-3} + 2\sqrt{3} \cdot 10^{-3} < 4.5 \cdot 10^{-3}$. Hence $\beta_i^2/(2 \ln 2) < 1.5 \cdot 10^{-5}$ and $\pi_i < 6.8 \cdot 10^{-6}$, and with $h_2(\pi) \leq \pi \log(e/\pi)$ we get $h_2(\pi_i) < 1.3 \cdot 10^{-4}$. Each summand exceeds $0.0823 > 1/16$. $\square$

9.6 Completion

Proof of Theorem 8.2. Let ϕ be given by Lemma 9.1, with $e = 256m\sqrt{d\nu}$, and put

$$b_* = 100m(A + L + 1)e = 25600m^2(A + L + 1)\sqrt{d\nu}, \quad \delta_L = (2L + 1)e.$$

Condition (9) gives $\sqrt{N}b_* \leq 25600 \cdot 2^{-25} < 10^{-3}$.

Copies. Let $\psi_i = \phi(u_i)$, so $\chi(\psi_i) \leq Le$, and put $X_i = \psi_i \phi(a) \psi_i^*$, $Y_i = \psi_i \phi(b) \psi_i^*$ and $Z_i = X_i Y_i$. Since a^2, b^2 and $(ab)^3$ lie in R , (W4) and Lemma 9.1 give $E(X_i^2), E(Y_i^2), E(Z_i^3) \leq e + 2Le = \delta_L$, and (W2) gives $\chi(X_i), \chi(Y_i) \leq \delta_L$. Lemma 9.3 gives unitaries x_i, c_i with $x_i^2 = c_i^3 = I$, $x_i c_i x_i = c_i^{-1}$ and

$$\|(x_i - X_i)\xi\|_F \leq \delta_L, \quad \chi(x_i) \leq 3\delta_L, \quad \|(c_i - Z_i)\xi\|_F \leq 29\delta_L, \quad \chi(c_i) \leq 56\delta_L.$$

By (W4) and Lemma 9.1(iii), $E(Z_i) \geq E(\phi(a)\phi(b)) - 2Le \geq 1 - \delta_L$, so $\Delta_i = E(c_i) \geq 1 - 30\delta_L$.

Raw commutators. Let $i \neq k$ and $x, y \in \{a, b\}$, and let $P = \psi_i \phi(x) \psi_i^*$ and $P' = \psi_k \phi(y) \psi_k^*$. The group commutator $g = [u_i x u_i^{-1}, u_k y u_k^{-1}]$ is a word of length at most $8L + 4$ and area at most A , so Lemma 9.2 gives $E(\phi(g)) \leq (3mA + 8L + 4)e$. Since $PP' - P'P = (\phi(g) - I)P'P$ and $\chi(P'P) \leq 2\delta_L$, (W3) gives

$$\|(PP' - P'P)\xi\|_F \leq \gamma := (3mA + 8L + 4)e + 4\delta_L = (3mA + 16L + 8)e.$$

Expanding $[X_i, X_k Y_k] = [X_i, X_k]Y_k + X_k[X_i, Y_k]$ and $[X_i Y_i, Z_k] = X_i[Y_i, Z_k] + [X_i, Z_k]Y_i$ and using (W3), $\|[X_i, Z_k]\xi\|_F, \|[Y_i, Z_k]\xi\|_F \leq 2\gamma + 2\delta_L$ and $\|[Z_i, Z_k]\xi\|_F \leq 4\gamma + 6\delta_L$.

Rounded commutators. For unitaries U, V, U_0, V_0 ,

$$[U, V] - [U_0, V_0] = U(V - V_0) + (U - U_0)V_0 - V(U - U_0) - (V - V_0)U_0,$$

so by (W3) the weighted norms of the two commutators differ by at most $2\|(U - U_0)\xi\|_F + 2\|(V - V_0)\xi\|_F + 2\chi(U_0) + 2\chi(V_0)$. With $(U, U_0, V, V_0) = (x_i, X_i, c_k, Z_k)$ and (c_i, Z_i, c_k, Z_k) this gives

$$\begin{aligned} \|[x_i, c_k]\xi\|_F &\leq 2\gamma + 68\delta_L = (6mA + 168L + 84)e, \\ \|[c_i, c_k]\xi\|_F &\leq 4\gamma + 130\delta_L = (12mA + 324L + 162)e. \end{aligned}$$

Since $m \geq 6$, these and $\chi(c_i) \leq (112L + 56)e$ and $\chi(x_i) \leq (6L + 3)e$ are at most b_* , and $\Delta_i \geq 1 - (60L + 30)e \geq 1 - b_* > 1/2$. Corollary 9.7 gives $S(\rho) \geq N/16$. $\square$

The constant 2^{25} is not optimized; the proof uses only $25600 \cdot 2^{-25} < 10^{-3}$, and Corollary 9.7 has room to spare.

10 Local physics

Theorem B concerns an abstract channel. Here we record what it implies for physical systems that could produce the channel, and what autonomous dynamics can and cannot produce in the long run. We call a channel *cheap* if for all n and ϵ it has causal services for n rounds with error at most ϵ , $P = C = 0$ and memory polynomial in $\log(n/\epsilon)$, and *expensive* if for some $A, c > 0$ every service for n rounds with error at most $1/16$ and $P + C \leq A \log(n + 1)$ needs memory at least n^c for all large n . Neither notion constrains the exchange.

Local reservoirs. A *local reservoir* consists of a finite box of the lattice $\mathbb{Z}^\nu$, with a system of dimension at most q at each site, the probe $\mathbb{C}^d$ attached to the origin, and a Hamiltonian $H(t) = \sum_X h_X(t)$, possibly time dependent, whose terms are supported in sets X of diameter at most R_0 and satisfy $\|h_X(t)\| \leq J_0$. The reservoir starts in its tracial state, the probe is coupled to it for a time T , and $\Phi_T(X) = (\text{id} \otimes \tau)[W_T(X \otimes I)W_T^*]$ is the resulting channel on the probe, where W_T is the time-ordered evolution and τ the normalized trace of the reservoir. We call ν, q, R_0, J_0 and d the *parameters* of the reservoir; the size of the box is arbitrary.

Proposition 10.1 (Local reservoirs are cheap at every finite time). *There is K , depending only on the parameters, such that for every local reservoir, every $T \geq 0$, every $n \geq 1$ and every $0 < \epsilon \leq 1$, the channel Φ_T has a causal service for n rounds with error at most ϵ , $P = C = 0$, $J = (n - 1)Q$ and*

$$Q \leq K [1 + T + \log(n/\epsilon)]^\nu.$$

Proof. For $r \geq 0$ let $\Phi_T^{(r)}$ be the channel produced by the terms h_X with X inside the ball B_r of radius r about the origin. The Lieb–Robinson bound [41, 5, 45] gives constants $C_1, v, \mu > 0$, depending only on the parameters, such that the Heisenberg evolutions of an operator A at the origin under the full and the truncated dynamics differ in norm by at most $C_1 \|A\| e^{vT - \mu r}$. The bound does not depend on the dimension of the site at the origin, so it holds for operators on the probe tensored with a reference system that the dynamics does not touch. Hence the Heisenberg-picture maps of Φ_T and $\Phi_T^{(r)}$ differ in completely bounded norm by at most $C_1 e^{vT - \mu r}$, and by duality $d_\diamond(\Phi_T, \Phi_T^{(r)}) \leq C_1 e^{vT - \mu r}$. The truncated evolution acts only on the probe and the sites of B_r , and the other sites start in their tracial state, so $\Phi_T^{(r)}$ has a finite tracial factorization with bath dimension $m \leq q^{(2r+1)^\nu}$.

Choose $r = \lceil (vT + \ln(2C_1 n/\epsilon))/\mu \rceil$, so that $d_\diamond(\Phi_T, \Phi_T^{(r)}) \leq \epsilon/(2n)$, and let $Q = \lceil \log m + \log(2n/\epsilon) \rceil$. The exchange construction in the proof of Proposition 8.9, with U_f replaced by the unitary of this factorization and $\vartheta = 0$, gives a service of $\Phi_T^{(r)}$ in which every round is within $\epsilon/(2n)$ of $\Phi_T^{(r)}$, hence within ϵ/n of Φ_T ; a hybrid argument bounds the error by ϵ . Finally $Q \leq (2r + 1)^\nu \log q + \log(2n/\epsilon) + 1 \leq K[1 + T + \log(n/\epsilon)]^\nu$. $\square$

With Theorem B this says that the channel of the configuration lamps cannot be produced quickly by local physics.

Corollary 10.2 (Preparation time). *Let Φ be the channel of Theorem 8.5, and fix the parameters of a local reservoir with $d = 873$. There are $c, \delta_0 > 0$ such that, whenever such a reservoir produces at time T a channel within $\delta \leq \delta_0$ of Φ in diamond distance,*

$$T \geq c \delta^{-1/\nu} (\log(1/\delta))^{-(2\kappa+1)/\nu}, \quad 2\kappa + 1 < 26.49.$$

Proof. Let $n = \lfloor 1/(32\delta) \rfloor$. Proposition 10.1 gives a service of Φ_T with error at most $1/32$, zero purity and $Q \leq K[1 + T + \log(32n)]^\nu$. Replacing the n uses of Φ_T by uses of Φ one at a

time changes the observer's final state by at most $n\delta \leq 1/32$, so the same device serves Φ with error at most $1/16$ and zero purity. For δ_0 small, n is large, and Theorem 8.5(a) with $A_0 = 0$ gives $Q \geq c'n/(\log n)^{2\kappa+1}$. Hence $1 + T + \log(32n) \geq (c'n/(K(\log n)^{2\kappa+1}))^{1/\nu}$, and since $n \geq 1/(64\delta)$ this gives the claim for δ_0 small. $\square$

The maximally mixed start of the reservoir is essential: Φ has a Stinespring dilation with a pure environment of dimension r_* , so from a pure state a bounded number of sites produce Φ exactly in bounded time. The same argument applies to a circuit of depth T of gates of bounded range, whose light cone is exact, and so bounds the depth of any local circuit that prepares Φ . The corollary does not exclude that local dynamics produce Φ in the limit of long times; it says how slowly.

Autonomous dynamics. Let $\mathcal{B}$ be a von Neumann algebra with a faithful normal tracial state $\tau_{\mathcal{B}}$, put $\mathcal{M} = M_d \otimes \mathcal{B}$ with the trace $\tau = \text{tr}_d \otimes \tau_{\mathcal{B}}$, where tr_d is the normalized trace, and let $\iota(a) = a \otimes 1$ and $E = \text{id} \otimes \tau_{\mathcal{B}}$. An *autonomous dynamics* is a trace-preserving automorphism α of $\mathcal{M}$, and the channel after $t \in \mathbb{N}$ steps is $\Phi_t = E \circ \alpha^t \circ \iota$; for $\alpha = \text{Ad } U$ with $\mathcal{B} = M_m$ this is the tracial factorization $\Phi_t(X) = (\text{id} \otimes \tau_m)[U^t(X \otimes I)U^{-t}]$. Let $\mathcal{N}$ be the fixed-point algebra of α and $E_{\mathcal{N}}$ the τ -preserving conditional expectation onto it.

The next proposition combines the mean ergodic theorem with the observation of Junge, Le Merdy and Xu [35, Remark 10.3] and of Haagerup and Musat [25, Remark 5.2] that compressions $E \circ E_{\mathcal{N}} \circ \iota$ of trace-preserving conditional expectations are self-adjoint and positive; both of its clauses are standard.

Proposition 10.3 (Autonomous limits are self-adjoint; the mean ergodic theorem and [35, Remark 10.3], [25, Remark 5.2]). *The averages $\frac{1}{T} \sum_{t=0}^{T-1} \Phi_t$ converge in diamond norm to $\bar{\Phi} = E \circ E_{\mathcal{N}} \circ \iota$, and if Φ_t converges, its limit is $\bar{\Phi}$. The channel $\bar{\Phi}$ is self-adjoint and positive semidefinite for the Hilbert–Schmidt inner product on M_d . In particular, a channel Ψ with $\Psi^* \neq \Psi$ is not the long-time limit of any autonomous dynamics with a tracial bath, local or not.*

Proof. The automorphism extends to a unitary V on $L^2(\mathcal{M}, \tau)$. By the mean ergodic theorem [36], $\frac{1}{T} \sum_{t=0}^{T-1} V^t$ converges strongly to the orthogonal projection P onto the invariant vectors. For $x \in \mathcal{M}$ the averages of x lie in the ball of radius $\|x\|$ of $\mathcal{M}$, which is closed in L^2 , so $Px \in \mathcal{M}$, and Px is invariant; hence $P(\mathcal{M}) \subseteq \mathcal{N}$. Since $\mathcal{N}$ consists of invariant vectors and $\mathcal{M}$ is dense in $L^2(\mathcal{M})$, P is the orthogonal projection onto $L^2(\mathcal{N})$, whose restriction to $\mathcal{M}$ is $E_{\mathcal{N}}$. The map ι is an isometry from (M_d, tr_d) into $L^2(\mathcal{M}, \tau)$ with adjoint E , so $\Phi_t = \iota^* V^t \iota$, and the averages converge to $\iota^* P \iota = \bar{\Phi}$ on every $a \in M_d$, hence in diamond norm, since M_d is finite dimensional. An ordinary limit equals the limit of the averages. Finally $\text{tr}_d(b^* \bar{\Phi}(a)) = \langle P \iota(b), P \iota(a) \rangle$ is Hermitian in (a, b) and nonnegative for $a = b$. $\square$

Self-adjointness and positivity are not the only constraints. The limit $\bar{\Phi}$ is a compression of a conditional expectation, and this ties the decay of a coherence to the return probability of a state.

Proposition 10.4 (A return inequality). *Let $T = E \circ E_{\mathcal{N}} \circ \iota$ be as in Proposition 10.3, on M_D . If $T(e_{00}) = e_{00}$ and $T(e_{0j}) = a e_{0j}$ for some $j \neq 0$ and $0 \leq a \leq 1$, then*

$$\langle j | T(e_{jj}) | j \rangle \geq a^2 + \frac{(1-a)^2}{D-1}.$$

Proof. Write $P = E_{\mathcal{N}}$, the orthogonal projection onto $L^2(\mathcal{N})$, and identify $x \in M_D$ with $\iota(x)$; then $\text{tr}_D(x^* T(y)) = \langle x, Py \rangle$ with $\langle x, y \rangle = \tau(x^* y)$. Put $p_0 = e_{00}$, $p_j = e_{jj}$ and $x = e_{0j}$. Since $\|Pp_0\|_2^2 = \langle p_0, T(p_0) \rangle = \|p_0\|_2^2$, we have $Pp_0 = p_0$, so $p_0 \in \mathcal{N}$. Let $y = P(x)$,

$X = P(p_j)$ and $B = y^*y$. As P is $\mathcal{N}$ -bimodular, $p_0y = P(p_0x) = y$, $yp_0 = P(xp_0) = 0$ and $p_0X = P(p_0p_j) = 0$; so yy^* is supported in p_0 , and B and X in $1 - p_0$. The Schwarz inequality for conditional expectations gives $B \leq P(x^*x) = X$. Moreover $\tau(X) = \tau(p_j) = 1/D$ and $\tau(B) = \|Px\|_2^2 = \langle x, T(x) \rangle = a/D$, so $R = X - B \geq 0$ has $\tau(R) = (1 - a)/D$. By the Cauchy–Schwarz inequality, $\tau(B^2) = \tau((yy^*)^2) \geq \tau(yy^*)^2/\tau(p_0) = a^2/D$ and $\tau(R^2) \geq \tau(R)^2/\tau(1 - p_0) = (1 - a)^2/(D(D - 1))$, and $\tau(BR) \geq 0$. Hence

$$\frac{1}{D} \langle j|T(e_{jj})|j \rangle = \langle p_j, Pp_j \rangle = \tau(X^2) = \tau(B^2) + 2\tau(BR) + \tau(R^2) \geq \frac{a^2}{D} + \frac{(1 - a)^2}{D(D - 1)}. \quad \square$$

Remark 10.5 (Self-adjoint and positive is not enough). Haagerup and Musat give self-adjoint positive maps that are not such compressions because they are not factorizable [25, Theorem 5.4]. The following channel is an exact finite factor and is still excluded. Let $a = \sqrt{3}/2$ and $b = \frac{1}{2}$, and let Λ on M_3 have the Kraus operators $\text{diag}(1, a, a)$, $b e_{12}$ and $-b e_{21}$. With $\omega = e^{2\pi i/3}$ and $W_k = 1 \oplus \begin{pmatrix} a & b\omega^k \\ -b\omega^{-k} & a \end{pmatrix}$ one has $\Lambda = \frac{1}{3} \sum_{k=0}^2 \text{Ad } W_k$, so Λ has a finite tracial factorization with bath dimension three. It is self-adjoint, and positive for the Hilbert–Schmidt inner product, with eigenvalues between $\frac{1}{2}$ and 1. But $\Lambda(e_{00}) = e_{00}$, $\Lambda(e_{01}) = a e_{01}$ and $\langle 1|\Lambda(e_{11})|1 \rangle = a^2$, which violates Proposition 10.4. So Λ is not the long-time limit of any autonomous dynamics with a tracial bath.

Conversely, every channel of the form $\Psi\Psi^*$ with Ψ in the tracial closure is a compression of a conditional expectation (apply [25, Theorem 5.9] to Ψ^* , which is again factorizable), and in fact an autonomous long-time limit, of a dynamics that the construction does not make local; the construction adapts Kümmerer’s tensor dilations with a Bernoulli shift [38].

Proposition 10.6 (Autonomy without locality). *Let $\mathcal{N}_0$ be a von Neumann algebra with a faithful normal tracial state τ_0 , let U be a unitary in $M_d \otimes \mathcal{N}_0$, and put $\Psi(X) = (\text{id} \otimes \tau_0)[U(X \otimes 1)U^*]$. Let $\mathcal{B} = \overline{\bigotimes_{k \in \mathbb{Z}} (\mathcal{N}_0, \tau_0)}$, with U acting on the factor $k = 0$, and let β be the Bernoulli shift of $\mathcal{B}$. Then the autonomous dynamics $\alpha = \text{Ad } U \circ (\text{id} \otimes \beta) \circ \text{Ad } U^*$ has $\Phi_t \rightarrow \Psi\Psi^*$, where $\Psi^*(a) = (\text{id} \otimes \tau_0)[U^*(a \otimes 1)U]$ is the Hilbert–Schmidt adjoint of Ψ .*

Proof. We have $\alpha^t = \text{Ad } U \circ (\text{id} \otimes \beta^t) \circ \text{Ad } U^*$. The shift is mixing: $\tau_{\mathcal{B}}(w \beta^t(z)) \rightarrow \tau_{\mathcal{B}}(w)\tau_{\mathcal{B}}(z)$ for $w, z \in \mathcal{B}$, since equality holds as soon as w and $\beta^t(z)$ are elementary tensors with disjoint supports, and such tensors of norm at most $\|w\|$ and $\|z\|$ approximate w and z in 2-norm (Kaplansky’s density theorem). Let $a \in M_d$ and $y = U^*(a \otimes 1)U$, and write $y = \sum_{i,j} e_{ij} \otimes y_{ij}$. For every $x = \sum_{k,l} e_{kl} \otimes x_{kl}$ in $M_d \otimes \mathcal{B}$,

$$\tau(x (\text{id} \otimes \beta^t)(y)) = \frac{1}{d} \sum_{k,l} \tau_{\mathcal{B}}(x_{kl} \beta^t(y_{lk})) \longrightarrow \tau(x (E(y) \otimes 1)).$$

Taking $x = U^*(e_{ji} \otimes 1)U$ shows that every matrix entry of $\Phi_t(a) = E[U(\text{id} \otimes \beta^t)(y)U^*]$ converges to the corresponding entry of $E[U(E(y) \otimes 1)U^*] = \Psi(\Psi^*(a))$. Convergence on a basis of M_d gives convergence in diamond norm. $\square$

By a theorem of Haagerup and Musat [26, Theorem 3.6], every channel in $\mathcal{K}_d$ has the form Ψ of Proposition 10.6 for some $\mathcal{N}_0$, and the channel of Theorem B has it with $\mathcal{N}_0 = L(\Gamma)$.

Nothing in Proposition 10.6 makes the dynamics local: U is an arbitrary unitary of $M_d \otimes \mathcal{N}_0$, and we do not know whether a dynamics of this kind can be given by a local rule.

Remark 10.7 (Self-adjoint expensive channels). Self-adjointness is no obstacle to expense. Let z be the unitary of order three in $L(\mathbb{Z}_3)$, so that $\tau(z) = \tau(z^2) = 0$, let U be the gadget unitary of the channel Φ of Theorem B, and let $V = \begin{pmatrix} 0 & U^{\otimes z} \\ U^{*\otimes z*} & 0 \end{pmatrix}$, a self-adjoint unitary over $\mathbb{C}^2 \otimes \mathbb{C}^{873}$ and $L(\Gamma \times \mathbb{Z}_3)$. The channel it defines on M_{1746} is $\Theta\left(\begin{smallmatrix} x & y \\ u & w \end{smallmatrix}\right) = \text{diag}(\Phi(w), \Phi^*(x))$,

which is self-adjoint. It keeps the lower bounds of Theorem B. For part (a), a service of Θ becomes a service of Φ with one more memory qubit and one more bit of purity, at the same exchange, by feeding each input together with a retained tag qubit that starts in $|1\rangle$ and is flipped after every round, since Θ maps $|1\rangle\langle 1| \otimes \rho$ to $|0\rangle\langle 0| \otimes \Phi(\rho)$. For part (b), Θ has a flat probe of rank $2r_*$, with anchors in both blocks, so Proposition 8.7 gives $P + C < 3Q + 6$ at its own rank rate $J \leq \frac{n}{2} \log(2r_*)$; the tag transfer and part (a) then give $Q \geq c\sqrt{n}/(\log n)^\kappa$. But Θ maps $\text{diag}(I, -I)$ to $-\text{diag}(I, -I)$, so it is not positive and not an autonomous limit, while $\Theta^2 = \text{diag}(\Phi\Phi^*, \Phi^*\Phi)$ is one, by Proposition 10.6. The next proposition shows that Θ^2 is cheap.

Composing a gadget channel with its adjoint erases the multiplication of the group. In a triangle block the unitarity of the gadget forces the coefficient of the product label to be the product of the coefficients, and this is what the one-round theorem reads. In $\Phi\Phi^*$ a triangle only requires a permutation π_x of the labels that carries 1 to x and y to xy , and no relation between the permutations π_x for different x is needed.

Proposition 10.8 (Adjoint compositions of gadgets are exact finite factors). *Let Φ_G be a gadget channel with r_* labels. Then $\Phi_G\Phi_G^*$ and $\Phi_G^*\Phi_G$ have finite tracial factorizations with baths of at most $3r_* - 2$ qubits. Consequently, for every n , they have exact causal services with $Q \leq 3r_* - 2$, $P = C = 0$ and $J = (n - 1)Q$, and so does Θ^2 in Remark 10.7, with one more qubit.*

Proof. Let $\mathcal{F}$ be the set of labels, and write $A_g = \sum_{[w_a]=g} c_a E_a$, so that $\Phi_G\Phi_G^*(X) = \sum_{g,h} A_g A_h^* X A_h A_g^*$. A product $A_g A_h^*$ is a sum of terms $c_a \bar{c}_b E_a E_b^*$ over slots a, b in a common column, so it vanishes unless g and h label two slots of one column. In a triangle block with labels $1, y$ in the first row and x, z in the second, where $z = xy$, the columns carry the labels $\{1, x\}$ and $\{y, z\}$; so $A_g A_h^* \neq 0$ forces $hg^{-1} \in K = \{1\} \cup \{x^{\pm 1}\}$, the union over the triangles. For $k \in K$ let π_k be a permutation of $\mathcal{F}$ that extends the partial map $g \mapsto kg$ (defined when $g, kg \in \mathcal{F}$), with $\pi_1 = \text{id}$ and $\pi_{k^{-1}} = \pi_k^{-1}$: for $k \neq k^{-1}$ extend one of the two maps and invert it, and for an involution k the partial map is an involution on its domain, which we extend by the identity.

The bath is $r_* + |K| - 1 \leq 3r_* - 2$ qubits. On the first r_* , indexed by $\mathcal{F}$, let V_g be the Pauli matrix Z on the qubit g , so that $\tau(V_g V_h) = \delta_{gh}$, and let P_k permute these qubits by π_k , so that $P_k V_g P_k^* = V_{kg}$ whenever $g, kg \in \mathcal{F}$. On the other qubits choose diagonal unitaries χ_k with $\chi_1 = I$, $\chi_{k^{-1}} = \chi_k^*$ and $\tau(\chi_k^* \chi_l) = \delta_{kl}$: for an involution, Z on one qubit, and for a pair $k \neq k^{-1}$, $\text{diag}(1, i, -1, -i)$ and its adjoint on two qubits, different pairs on different qubits. Put $R_k = P_k \otimes \chi_k$, so that $R_{k^{-1}} = R_k^*$ and $R_k V_g R_k^* = V_{kg}$ for $g, kg \in \mathcal{F}$, and put $C_{g,h} = V_g R_{hg^{-1}}^*$. These are orthonormal for the normalized trace: if $hg^{-1} \neq h'g'^{-1}$ the factor χ has trace zero, and otherwise $\tau(C_{g,h}^* C_{g',h'}) = \tau(R_k V_g V_{g'} R_k^*) = \delta_{gg'}$, which then forces $h = h'$. In the same way $\tau(C_{g,h} C_{g',h'}^*) = \tau(V_g V_{g'}) = \delta_{gg'}$ when $hg^{-1} = h'g'^{-1}$, and it vanishes otherwise.

Let $W = \sum A_g A_h^* \otimes C_{g,h}$, summed over the pairs with $hg^{-1} \in K$; the other products $A_g A_h^*$ vanish. On the block of a symbol with label g , including the first block, where $g = 1$, it is V_g . On a triangle block, with H the Hadamard matrix and using $R_x V_1 R_x^* = V_x$ and $R_x V_y R_x^* = V_z$,

$$W = \frac{1}{2} \begin{pmatrix} V_1 + V_y & (V_1 - V_y)R_x^* \\ R_x(V_1 - V_y) & V_x + V_z \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & R_x \end{pmatrix} (H \otimes I) \begin{pmatrix} V_1 & 0 \\ 0 & V_y \end{pmatrix} (H \otimes I) \begin{pmatrix} I & 0 \\ 0 & R_x^* \end{pmatrix},$$

a self-adjoint unitary; triangles with coinciding labels are covered by the same computation. So W is unitary, and by orthonormality of the $C_{g,h}$, $(\text{id} \otimes \tau)[W(X \otimes I)W^*] = \sum_{g,h} A_g A_h^* X A_h A_g^* = \Phi_G\Phi_G^*(X)$. For $\Phi_G^*\Phi_G$ use right translations $g \mapsto gk$ with $k \in \{1\} \cup \{y^{\pm 1}\}$, the operators

$C_{g,h} = V_g R_{g^{-1}h}^*$, and rows instead of columns; on a triangle block the rows carry $\{1, y\}$ and $\{x, z\}$, and the same computation applies with x and y exchanged. For Θ^2 , one further maximally mixed qubit dephases the two blocks, and controlled on the block the two factorizations act on a common bath. Refreshing the whole bath in every round gives the services. $\square$

These services refresh the whole bath in every round, so their exchange is far above the rank rate; whether $\Phi\Phi^*$ has services with $P = C = 0$ and memory polynomial in $\log(n/\epsilon)$ at its own rank rate we do not know.

So a self-running system with a tracial bath cannot reach expense through $\Phi\Phi^*$. It can through a lazy version of the doubling.

A lazy doubling. The doubled channel Θ of Remark 10.7 is expensive but not positive, and its square is cheap. Averaging Θ with the identity restores positivity and keeps the expense. Put

$$\Theta_+ = \frac{1}{2}(\text{id} + \Theta), \quad \Theta_+ \begin{pmatrix} x & y \\ u & w \end{pmatrix} = \frac{1}{2} \begin{pmatrix} x + \Phi(w) & y \\ u & w + \Phi^*(x) \end{pmatrix} \quad \text{on } M_{1746}.$$

In Remark 10.7 only $zz^* = 1$ and $\tau(z^2) = 0$ are used, so z may also be the unitary $\text{diag}(1, i)$ of a qubit with its normalized trace. Since Θ is a unital channel that is self-adjoint for the Hilbert–Schmidt inner product, its spectrum lies in $[-1, 1]$ (unital channels are contractions for the Hilbert–Schmidt norm), and Θ_+ is self-adjoint and positive semidefinite, as Proposition 10.3 requires of an autonomous limit.

Proposition 10.9 (An expensive autonomous limit). *Let Φ , V and Θ be as in Remark 10.7, and $\kappa = 6 + \log 107$.*

- (a) *Autonomy. Let Z be the coordinate of $L^\infty[-1, 1]$ with the normalized Lebesgue measure, put $\mathcal{B} = L(\Gamma \times \mathbb{Z}_3) \otimes L^\infty[-1, 1]$ with the product trace, and $W = \exp(i\frac{\pi}{2}V \otimes Z) = \cos(\frac{\pi}{2}Z) + i\sin(\frac{\pi}{2}Z)V \in M_{1746} \otimes \mathcal{B}$. The autonomous dynamics $\alpha = \text{Ad } W$ has $\Phi_t = \Theta_+$ for every $t \geq 1$. More generally, $\exp(isV \otimes Z)$ gives the channel $\Xi_s = \frac{1+\text{sinc } 2s}{2} \text{id} + \frac{1-\text{sinc } 2s}{2} \Theta$, with $\text{sinc } x = \sin x/x$, and $d_\diamond(\Xi_s, \Theta_+) = |\sin 2s|/(4s)$.*
- (b) *Memory against purity, lower bound. There are $c, c_0 > 0$ and n_0 such that for $n \geq n_0$ and $\log n \leq B \leq c_0 n$ every causal service of Θ_+ for n rounds with error at most $1/16$ and purity $P + C \leq B$ has, at any exchange,*

$$Q \geq c \frac{n}{B(\log n)^{2\kappa}}.$$

In particular, for every $A_0 \geq 0$ there is $c' > 0$ such that services with $P + C \leq A_0 \log n$ have $Q \geq c'n/(\log n)^{2\kappa+1}$ for all large n .

- (c) *Upper bound. There is K such that for every $n \geq 1$ and $0 < B \leq n$, Θ_+ has an exact causal service with purity $P + C < B$, $J = (n - 1)Q$ and $Q \leq K(n/B) \log(2n/B)$. Moreover $\Theta_+ \in \mathcal{K}_{1746}$.*

So Θ_+ is a long-time limit, indeed an exact finite-time value, of an autonomous dynamics with a tracial bath; it is expensive in the sense defined at the beginning of this section, and it trades memory and purity at the same fixed rate as Φ (Theorem 8.5(a)). Nothing in the construction is local. Part (b) loses nothing against Theorem 8.5(a) because the round extracted from a service of Θ_+ is converted into a round of Φ with a *linear* loss in leakage and in the commutator $\|[U, 1 \otimes \sqrt{\rho}]\|_F^2$, which replaces entropy production in the one-round theorem (Lemmas 10.10 and 10.12), on the same bath.

Proof of (a). Since $V^2 = 1$ and Z commutes with V , $\exp(isV \otimes Z) = \cos(sZ) + i \sin(sZ)V$, so that $W^t = \exp(it\frac{\pi}{2}V \otimes Z)$ and

$$W^t(X \otimes 1)W^{-t} = \cos^2(\frac{t\pi}{2}Z)X + \sin^2(\frac{t\pi}{2}Z)V(X \otimes 1)V + \frac{i}{2}\sin(t\pi Z)(V(X \otimes 1) - (X \otimes 1)V).$$

The trace of $\mathcal{B}$ is the product of the trace of $L(\Gamma \times \mathbb{Z}_3)$ and the integral over Z . The last term integrates to 0 because $\sin(t\pi Z)$ is odd, $\int \cos^2(\frac{t\pi}{2}Z) = \frac{1}{2}(1 + \text{sinc } t\pi) = \frac{1}{2}$ for $t \geq 1$, and $(\text{id} \otimes \tau)[V(X \otimes 1)V] = \Theta(X)$. So $\Phi_t = \frac{1}{2}(\text{id} + \Theta)$. The same computation with s in place of $\frac{t\pi}{2}$ gives Ξ_s . Finally $\Xi_s - \Theta_+ = \frac{1}{2}\text{sinc}(2s)(\text{id} - \Theta)$, and $\|\text{id} - \Theta\|_\diamond = 2$: the input $|0\rangle\langle 0| \otimes \sigma$ is mapped by Θ into the block $|1\rangle\langle 1|$. The automorphism $\text{Ad } W$ is inner, hence trace preserving. $\square$

The same channel is reached, for every $t \geq 1$, by a Bernoulli-shift construction like that of Proposition 10.6: with $\mathcal{B} = L(\Gamma \times \mathbb{Z}_3) \overline{\otimes} \bigotimes_{k \in \mathbb{Z}} (M_2, \tau_2)$, β the shift of the chain and p_0 the projection $|1\rangle\langle 1|$ of the qubit at site 0, the automorphism $\text{Ad}((1 - p_0) + p_0 V) \circ (\text{id} \otimes \beta)$ raises V to the parity of t fair bits.

We now work with a general unital channel Φ on M_d , its doubling $\Theta(\begin{smallmatrix} x & y \\ u & w \end{smallmatrix}) = \text{diag}(\Phi(w), \Phi^*(x))$ and $\Theta_+ = \frac{1}{2}(\text{id} + \Theta)$ on M_{2d} . Let $\mathcal{S}_\Phi \subseteq M_d$ be the span of the Kraus operators of Φ , so that the Choi support of Φ is $\{|A\rangle\rangle : A \in \mathcal{S}_\Phi\}$, and let Π_Φ also denote the Hilbert–Schmidt projection of M_d onto $\mathcal{S}_\Phi$. Since $\Theta(X) = \sum_g (e_{01} \otimes A_g)X(e_{01} \otimes A_g)^* + (e_{10} \otimes A_g^*)X(e_{10} \otimes A_g^*)^*$ for Kraus operators A_g of Φ , the Choi support of Θ_+ corresponds to the span of the three pairwise orthogonal subspaces

$$\mathbb{C}1_{2d}, \quad e_{01} \otimes \mathcal{S}_\Phi, \quad e_{10} \otimes \mathcal{S}_\Phi^* \quad (19)$$

of $M_{2d} = M_2 \otimes M_d$. Throughout, $\|\cdot\|_F$ is the Hilbert–Schmidt norm and $\|\cdot\|$ the operator norm, as in Section 9; an operator X on the bath $\mathbb{C}^M$ is identified with $1 \otimes X$ when it acts on $\mathbb{C}^d \otimes \mathbb{C}^M$, so that $\|1 \otimes X\|_F = \sqrt{d}\|X\|_F$.

Lemma 10.10 (Commutator form of the one-round theorem). *Lemma 9.1, Theorem 8.2 and Corollary 8.4 remain true if, in the definition $\nu = \ell + \Delta S$, the entropy production ΔS of the round $\Psi(X) = \text{Tr}_M U(X \otimes \rho)U^*$ is replaced by any $\tilde{s} \geq 0$ with $\|[U, 1 \otimes \sqrt{\rho}]\|_F^2 \leq 4d\tilde{s} \ln 2$, and if the hypothesis $d_\diamond(\Psi, \Phi_G) \leq 1/8$ is replaced by the weaker hypothesis*

$$|\langle z|\Psi(|z\rangle\langle 0|)|0\rangle| \leq \frac{3}{4}, \quad (20)$$

where 0 and z are the anchor positions of the empty word and of j ; in Lemma 9.1(iii) the bound then reads $E(\phi(a)\phi(b)) \geq 0.66 - e$. Conversely, every round can be rewritten, without changing its channel or its leakage, as a round (U', ρ) with $U' = (1 \otimes W)U$ for which $\tilde{s} = \Delta S$ is admissible.

Proof. The rewriting is the normalization step of the proof of Lemma 9.1, after which (14) is the inequality $\|[U', 1 \otimes \sqrt{\rho}]\|_F^2 \leq 4d\Delta S \ln 2$; its proof uses only that U' is a unitary on $\mathbb{C}^d \otimes \mathbb{C}^M$. In the proof of Lemma 9.1, ΔS enters only through (14): afterwards only the bounds $\epsilon_S \leq 2\sqrt{\ln 2}\eta$, $\epsilon_\ell + \epsilon_S \leq \sqrt{1 + 4\ln 2}\eta$ and $2\epsilon_\ell^2 + 2\epsilon_S^2 \leq 8\ln 2\eta^2$ with $\eta = \sqrt{d\nu}$ are used, together with the unitarity of U , its leakage and $d_\diamond(\Psi, \Phi_G)$. These bounds hold with $\epsilon_S^2 \leq 4d\tilde{s} \ln 2$ and $\nu = \ell + \tilde{s}$. The diamond hypothesis enters only in the displacement step, through $|\text{Tr}(\rho B_\varnothing^* B_j)| = |\langle z|\Psi(|z\rangle\langle 0|)|0\rangle| \leq 2u \leq \frac{1}{4}$. Under (20) instead, $E(\phi(j))^2 \geq 2 - \frac{3}{2} - 4\epsilon_\ell \geq \frac{1}{2} - 4\eta$, so $E(\phi(a)\phi(b)) \geq \sqrt{1/2 - 4\eta} - e \geq 0.66 - e$ if $\eta \leq \frac{1}{64}$, while $e > 1$ otherwise. In the proof of Theorem 8.2 this gives $E(Z_i) \geq 0.66 - \delta_L$ and $\Delta_i \geq 0.66 - 30\delta_L \geq 0.66 - b_* > \frac{1}{2}$ in place of $\Delta_i \geq 1 - b_*$, and $\Delta_i \geq \frac{1}{2}$ is all that Corollary 9.7 needs. Theorem 8.2 and Corollary 8.4 use the round only through Lemma 9.1. $\square$

Lemma 10.11 (Weighted intertwining). *Let B be an operator on $\mathbb{C}^d \otimes \mathbb{C}^M$ with $\|B\| \leq 1$, let R, R' be self-adjoint operators on $\mathbb{C}^M$, ξ an operator on $\mathbb{C}^M$, and $\Sigma = BR - R'B$.*

(i) *If h is Lipschitz with constant L_h on an interval containing the spectrum of R , then $\|[h(R), \xi]\|_F \leq L_h \|[R, \xi]\|_F$.*

(ii) *If $g(x) = \int \hat{g}(t)e^{itx}dt$ with $M_1 = \int |t\hat{g}(t)|dt$ and $M_2 = \int t^2|\hat{g}(t)|dt$ finite, then*

$$\|(Bg(R) - g(R')B)\xi\|_F \leq M_1 \|\Sigma\xi\|_F + \frac{1}{2}M_2 \|\Sigma\| \sqrt{d} \|[R, \xi]\|_F.$$

Proof. (i) In an eigenbasis of R with eigenvalues r_k , the entries of $[h(R), \xi]$ and $[R, \xi]$ are $(h(r_k) - h(r_l))\xi_{kl}$ and $(r_k - r_l)\xi_{kl}$. (ii) Differentiating $\theta \mapsto e^{i\theta t R'} B e^{i(1-\theta)t R}$ gives $B e^{itR} - e^{itR'} B = it \int_0^1 e^{i\theta t R'} \Sigma e^{i(1-\theta)t R} d\theta$. Write $e^{i(1-\theta)t R} \xi = \xi e^{i(1-\theta)t R} + [e^{i(1-\theta)t R}, \xi]$ and use (i) with $h(x) = e^{i(1-\theta)tx}$; unitaries do not change Hilbert–Schmidt norms, so $\|(B e^{itR} - e^{itR'} B)\xi\|_F \leq |t| \|\Sigma\xi\|_F + \frac{t^2}{2} \|\Sigma\| \sqrt{d} \|[R, \xi]\|_F$. Integrate against $|\hat{g}(t)| dt$. $\square$

We fix the cutoff once and for all. Let $S : \mathbb{R} \rightarrow [0, 1]$ be smooth with $S = 0$ on $(-\infty, 0]$ and $S = 1$ on $[1, \infty)$, let $0 < s_0 \leq \frac{1}{4}$, and put

$$g(x) = \sin\left(\frac{\pi}{2}S(2x/s_0 - 1)\right) S(3 - 2x), \quad g_2(x) = \cos\left(\frac{\pi}{2}S(2x/s_0 - 1)\right), \quad f(x) = \sqrt{x} g(x).$$

Then g is smooth with compact support, so $M_g := \int (|t| + t^2)|\hat{g}(t)| dt < \infty$; on $[0, 1]$ we have $0 \leq g \leq 1$, $g = 0$ on $[0, s_0/2]$, $g = 1$ and $g_2 = 0$ on $[s_0, 1]$, and $g^2 + g_2^2 = 1$. Let L_g be a common Lipschitz constant of g and g_2 on $[0, 1]$, put $\sigma = \sqrt{2/s_0}$ and

$$K_a = 11M_g + 1 + 1.5L_g + 2\sigma(4 + 1.5L_g).$$

These depend only on s_0 .

Lemma 10.12 (Block extraction). *Let Φ be a unital channel on M_d , and let $\Psi(X) = \text{Tr}_M U(X \otimes \rho)U^*$ be a round of Θ_+ , with U unitary on $\mathbb{C}^2 \otimes \mathbb{C}^d \otimes \mathbb{C}^M$, $u = d_\diamond(\Psi, \Theta_+)$ and leakage ℓ relative to Θ_+ . Put $\epsilon_\ell = \sqrt{2d\ell}$, $\epsilon_S = \|[U, 1 \otimes \sqrt{\rho}]\|_F$, $\gamma = \epsilon_\ell + \epsilon_S$ and $\mathcal{E} = 2u + s_0 + 4K_a\sqrt{d}\gamma$. If $\mathcal{E} < \frac{1}{2}$, there are a unitary U' on $\mathbb{C}^d \otimes \mathbb{C}^M$ and a state ρ' on $\mathbb{C}^M$ such that the round $\Psi'(X) = \text{Tr}_M U'(X \otimes \rho')U'^*$ of Φ , with leakage $\ell_\Phi(\Psi')$ relative to Φ , satisfies*

$$d_\diamond(\Psi', \Phi) \leq \frac{\mathcal{E}}{\frac{1}{2} - \mathcal{E}}, \quad \ell_\Phi(\Psi') \leq \frac{4K_a^2\gamma^2}{d(\frac{1}{2} - \mathcal{E})}, \quad \|[U', 1 \otimes \sqrt{\rho'}]\|_F^2 \leq \frac{8K_a^2\gamma^2}{\frac{1}{2} - \mathcal{E}},$$

and $|\langle x|\Psi'(|x\rangle\langle y|)|y\rangle| \leq \mathcal{E}/(\frac{1}{2} - \mathcal{E})$ for all unit vectors x, y with $\langle x|\Phi(|x\rangle\langle y|)|y\rangle = 0$.

Proof. Put $\xi = \sqrt{\rho}$ and write $U = \sum_{a,b \in \{0,1\}} e_{ab} \otimes U_{ab}$ with $D_0 = U_{00}$, $B = U_{01}$, $D_1 = U_{11}$, operators on $\mathbb{C}^d \otimes \mathbb{C}^M$.

Residuals. As in the proof of Lemma 9.1, $\mathcal{J}(\Psi)$ is the marginal of the unit vector $(2d)^{-1/2}U\xi \in M_{2d} \otimes M_M$, so $\ell = (2d)^{-1}\|(\Pi^\perp \otimes \text{id})(U\xi)\|_F^2$, with Π the projection of M_{2d} onto the span of (19). Put $\alpha = (2d)^{-1}(\text{Tr} \otimes \text{id})(U)$, the average of the $2d$ diagonal entries of U in M_M , and $E_a = D_a - \alpha$, $F = (\Pi_\Phi^\perp \otimes \text{id})(B)$. By (19), $(\Pi^\perp \otimes \text{id})(U)$ has the diagonal blocks E_0, E_1 and the upper right block F ; since $\Pi^\perp \otimes \text{id}$ commutes with right multiplication by ξ , the numbers $e_a = \|E_a\xi\|_F$ and $e_F = \|F\xi\|_F$ are at most ϵ_ℓ . Also $\|\alpha\| \leq 1$ and $\|E_a\| \leq 2$.

Commutators. Put $k_X = \|[X, \xi]\|_F$ for $X \in \{D_0, B, D_1\}$; these are at most ϵ_S . The diagonal entries of U are entries of D_0 and D_1 , so by Cauchy–Schwarz $k_\alpha := \sqrt{d}\|[\alpha, \xi]\|_F \leq (k_{D_0}^2 + k_{D_1}^2)^{1/2}/\sqrt{2} \leq \epsilon_S/\sqrt{2}$. With $R = 1 - \alpha^*\alpha$ and $R' = 1 - \alpha\alpha^*$, both between 0 and 1, $k_R := \sqrt{d}\|[R, \xi]\|_F \leq 2k_\alpha \leq \sqrt{2}\epsilon_S$, and likewise $k_{R'} \leq \sqrt{2}\epsilon_S$.

Defects. Unitarity gives $B^*B = 1 - D_1^*D_1$ and $BB^* = 1 - D_0D_0^*$, hence

$$\begin{aligned}\Delta &:= R - B^*B = \alpha^*E_1 + E_1^*D_1, & \Delta' &:= R' - BB^* = \alpha E_0^* + E_0D_0^*, \\ \Sigma &:= BR - R'B = B\Delta - \Delta'B,\end{aligned}$$

with $\|\Delta\|, \|\Delta'\| \leq 1$ and $\|\Sigma\| \leq 2$. From $\xi E_1 = E_1\xi - [D_1, \xi] + [\alpha, \xi]$ and $D_1\xi = \xi D_1 + [D_1, \xi]$, $\|\Delta\xi\|_F \leq e_1 + \|\xi E_1\|_F + 2k_{D_1} \leq 2e_1 + 3k_{D_1} + k_\alpha \leq 4\gamma$; symmetrically $\|\Delta'\xi\|_F \leq 2e_0 + 3k_{D_0} + k_\alpha \leq 4\gamma$; and $\Sigma\xi = B\Delta\xi - \Delta'\xi B - \Delta'[B, \xi]$ gives $\|\Sigma\xi\|_F \leq 9\gamma$.

Two factorizations. Let P and P' be the spectral projections of R and R' for $(s_0/2, \infty)$. Since g vanishes on $[0, s_0/2]$, $g(R) = R^{-1/2}Pf(R)$ with $\|R^{-1/2}P\| \leq \sigma$, and likewise for R' . The operators $B_1 = BR^{-1/2}P$ and $B'_1 = R'^{-1/2}P'B$ satisfy

$$B_1^*B_1 = P - \tilde{\Delta}, \quad \tilde{\Delta} = PR^{-1/2}\Delta R^{-1/2}P, \quad B'_1B_1^* = P' - \tilde{\Delta}', \quad \tilde{\Delta}' = P'R'^{-1/2}\Delta'R'^{-1/2}P'.$$

Take polar decompositions $B_1 = V_1|B_1|$ and $B'_1 = |B'_1|V'_1$ with V_1, V'_1 unitary. On the range of $1 \otimes P$ we have $|B_1| = 1 + \varphi(\tilde{\Delta})$ with $\varphi(x) = \sqrt{1-x} - 1$ and $\tilde{\Delta} \leq 1$, and $\varphi(x)^2 \leq x^2$; so $\|(|B_1| - P)Y\|_F \leq \|\tilde{\Delta}Y\|_F$ for every Y with range in that subspace, and similarly for B'_1 . Put $Z_0 = Bg(R)\xi = B_1f(R)\xi$. Then $Z_0 - V_1f(R)\xi = V_1(|B_1| - P)f(R)\xi$, and $\tilde{\Delta}f(R)\xi = PR^{-1/2}\Delta g(R)\xi$, while $\|\Delta g(R)\xi\|_F \leq \|\Delta\xi\|_F + \sqrt{d}\|[g(R), \xi]\|_F \leq \|\Delta\xi\|_F + L_g k_R$ by Lemma 10.11(i). Hence

$$a_1 := \|Z_0 - V_1f(R)\xi\|_F \leq \sigma(4 + 1.5L_g)\gamma.$$

On the other side, $Z_0 = (Bg(R) - g(R')B)\xi + g(R')[B, \xi] + [g(R'), \xi]B + \xi f(R')B'_1$ and $\xi f(R')B'_1 = \xi f(R')V'_1 + \xi f(R')(|B'_1| - P')V'_1$, where $\|\xi f(R')(|B'_1| - P')\|_F \leq \|\tilde{\Delta}'f(R')\xi\|_F \leq \sigma(\|\Delta'\xi\|_F + L_g k_{R'})$. By Lemma 10.11(ii), $\|(Bg(R) - g(R')B)\xi\|_F \leq M_g(\|\Sigma\xi\|_F + k_R) \leq 11M_g\gamma$. Hence

$$a_2 := \|Z_0 - \xi f(R')V'_1\|_F \leq 11M_g\gamma + (1 + 1.5L_g)\gamma + \sigma(4 + 1.5L_g)\gamma, \quad a_1 + a_2 \leq K_a\gamma.$$

Polar decomposition of Z_0 . Put $\zeta_1 = |f(R)\xi| = (\xi f(R)^2\xi)^{1/2}$ and $\zeta_2 = |(\xi f(R'))^*| = (\xi f(R')^2\xi)^{1/2}$. Then $|V_1f(R)\xi| = \zeta_1$ and $|(\xi f(R')V'_1)^*| = \zeta_2$, as operators $1 \otimes \zeta_i$. By the Araki–Yamagami inequality $\| |X| - |Y| \|_F \leq \sqrt{2}\|X - Y\|_F$ [2], applied to Z_0 and to Z_0^* , $\| |Z_0| - \zeta_1 \|_F \leq \sqrt{2}a_1$ and $\| |Z_0^*| - \zeta_2 \|_F \leq \sqrt{2}a_2$. Let $Z_0 = O|Z_0| = |Z_0^*|O$ with O unitary. Then

$$\|O\zeta_1 - \zeta_2O\|_F \leq \|O(\zeta_1 - |Z_0|)\|_F + \|(|Z_0^*| - \zeta_2)O\|_F \leq \sqrt{2}(a_1 + a_2) =: \delta.$$

The round. The Hermitian operators $O(1 \otimes \zeta_1)O^*$ and $1 \otimes \zeta_2$ are within δ in $\|\cdot\|_F$, and their spectra are those of ζ_1 and ζ_2 , each repeated d times. By the Hoffman–Wielandt inequality [30], the decreasing eigenvalue lists (x_k) of ζ_1 and (y_k) of ζ_2 satisfy $d\sum_k (x_k - y_k)^2 \leq \delta^2$; so the unitary W_h of $\mathbb{C}^M$ that maps an eigenbasis of ζ_1 to one of ζ_2 in the same order has $\|1 \otimes (W_h\zeta_1W_h^* - \zeta_2)\|_F \leq \delta$. Put $U' = (1 \otimes W_h^*)O$, $c_1 = \text{Tr } \zeta_1^2 = \text{Tr } \rho f(R)^2$ and $\rho' = \zeta_1^2/c_1$. Then $\|[U', 1 \otimes \zeta_1]\|_F \leq \|O\zeta_1 - \zeta_2O\|_F + \|(\zeta_2 - W_h\zeta_1W_h^*)O\|_F \leq 2\delta$, which is the third bound once we know $c_1 \geq \frac{1}{2} - \mathcal{E}$. The channel of (U', ρ') is $\Psi' = \Phi_{O\zeta_1}/c_1$, where $\Phi_X(w) = \text{Tr}_M[X(w \otimes 1)X^*]$. Its leakage is $(dc_1)^{-1}\|(\Pi_\Phi^\perp \otimes \text{id})(O\zeta_1)\|_F^2$, and

$$\|(\Pi_\Phi^\perp \otimes \text{id})(O\zeta_1)\|_F \leq \|(\Pi_\Phi^\perp \otimes \text{id})(Z_0)\|_F + \sqrt{2}a_1 \leq e_F + L_g k_R + \sqrt{2}a_1 \leq 2K_a\gamma,$$

since $(\Pi_\Phi^\perp \otimes \text{id})(Bg(R)\xi) = F\xi g(R) + (\Pi_\Phi^\perp \otimes \text{id})(B[g(R), \xi])$. This is the second bound.

Closeness. For a completely positive map Φ_X as above, $\|\Phi_X\|_\diamond = \|\text{Tr}_M(X^*X)\|$, and $\|\Phi_X - \Phi_Y\|_\diamond \leq \|X - Y\|_F(\|\text{Tr}_M X^*X\|^{1/2} + \|\text{Tr}_M Y^*Y\|^{1/2})$, where Tr_M traces out the input bath. Let $Y = B\xi$. The upper left block of $\Psi(e_{11} \otimes w)$ is $\Phi_Y(w)$ and that of

$\Theta_+(e_{11} \otimes w)$ is $\frac{1}{2}\Phi(w)$; compressing to the block, $\|\Phi_Y - \frac{1}{2}\Phi\|_\diamond \leq 2u$. Since $g^2 + g_2^2 = 1$ on $[0, 1]$, $\Phi_Y - \Phi_{Z_0} - \Phi_{B_{g_2(R)}\xi}$ is $w \mapsto \text{Tr}_M[B(w \otimes X)B^*]$ with $X = \sum_{h \in \{g, g_2\}} h(R)[h(R), \rho]$, and $\|X\|_1 \leq 4L_g\|[R, \xi]\|_F$, so its diamond norm is at most $4L_g k_R/\sqrt{d}$ (split X into positive and negative parts and use $B^*B \leq 1$). Next, $\|\Phi_{B_{g_2(R)}\xi}\|_\diamond \leq \text{Tr} \rho g_2(R)Rg_2(R) + \|\xi g_2(R)\Delta g_2(R)\xi\|_1 \leq s_0 + \sqrt{d}(\|\Delta\xi\|_F + L_g k_R)$, because $xg_2(x)^2 \leq s_0$ on $[0, 1]$. Finally $\text{Tr}_M \zeta_1^2 = c_1 \leq 1$ and $\text{Tr}_M Z_0^* Z_0 \leq 1$ give $\|\Phi_{O_{\zeta_1}} - \Phi_{Z_0}\|_\diamond \leq 2\sqrt{2}a_1$. Adding up,

$$\|\Phi_{O_{\zeta_1}} - \frac{1}{2}\Phi\|_\diamond \leq 2u + s_0 + 2\sqrt{2}a_1 + \sqrt{d}(4 + 1.5L_g)\gamma + 6L_g\gamma/\sqrt{d} \leq \mathcal{E}.$$

Evaluating at τ_d gives $|c_1 - \frac{1}{2}| \leq \mathcal{E}$, so $c_1 \geq \frac{1}{2} - \mathcal{E}$, and $\|\Psi' - \Phi\|_\diamond \leq \mathcal{E}/c_1 + |1/(2c_1) - 1| \leq 2\mathcal{E}/c_1$. If $\langle x|\Phi(|x\rangle\langle y|)|y\rangle = 0$, then $|\langle x|\Phi_{O_{\zeta_1}}(|x\rangle\langle y|)|y\rangle| \leq \|\Phi_{O_{\zeta_1}} - \frac{1}{2}\Phi\|_\diamond \leq \mathcal{E}$, and dividing by c_1 gives the last bound. $\square$

Proof of (b). Let a service of Θ_+ for n rounds have error $\epsilon \leq \frac{1}{16}$ and $P + C \leq B$ with $\log n \leq B$. Theorem 8.1, applied to Θ_+ on M_{1746} , gives a round with bath $\mathbb{C}^M$, $M = 2^Q$, $u = d_\diamond(\Psi_t, \Theta_+) \leq \frac{1}{15}$ and $\nu_t = \ell_t + \Delta S_t \leq t_n \leq \frac{16}{15} \cdot \frac{B+1+\log n}{n} \leq \frac{3.2B}{n}$, as in the proof of Theorem 8.5. The normalization step of the proof of Lemma 9.1, which uses only that U_t is a unitary on $\mathbb{C}^{2d} \otimes \mathbb{C}^M$, replaces U_t by $(1 \otimes W)U_t$ without changing Ψ_t or its leakage, after which $\epsilon_S^2 = \|[U_t, 1 \otimes \sqrt{\rho_t}]\|_F^2 \leq 4(2d)\Delta S_t \ln 2$; so $\gamma^2 \leq 2(2d\ell_t + 8d \ln 2 \Delta S_t) \leq 12d\nu_t$. Apply Lemma 10.12 with $s_0 = \frac{1}{100}$ and $d = 873$: $\mathcal{E} \leq \frac{2}{15} + \frac{1}{100} + 14K_a d \sqrt{\nu_t} \leq \frac{3}{14}$ as soon as $14K_a d \sqrt{3.2B/n} \leq 0.07$. Then $c_1 \geq \frac{1}{2} - \mathcal{E} \geq \frac{2}{7}$, and the anchors 0 and $z = z(j)$ satisfy $\langle z|\Phi(|z\rangle\langle 0|)|0\rangle = \tau(\lambda([j])) = 0$, so the round Ψ' satisfies (20). By Lemma 10.10 it may be used in Corollary 8.4 with

$$\nu' = \ell_\Phi(\Psi') + \frac{\|[U', 1 \otimes \sqrt{\rho'}]\|_F^2}{4d \ln 2} \leq \left(4 + \frac{2}{\ln 2}\right) \frac{K_a^2 \gamma^2}{d c_1} \leq 290K_a^2 \nu_t \leq 928K_a^2 \frac{B}{n}.$$

Let C_R and y_0 be the constants of Corollary 8.4(a) for R_Φ and the transport of Section 8.3, and put $y_n = n/(928K_a^2 C_R B)$, so that $C_R \nu' y_n \leq 1$ and $y_n \leq n$. With $c_0 = \min\{1/(928K_a^2 C_R y_0), (0.07/(14K_a d))^2/3.2\}$ and $B \leq c_0 n$, the corollary gives

$$Q = \log M \geq S(\rho') \geq \frac{y_n}{32(\log y_n)^{2\kappa}} \geq \frac{1}{29696 K_a^2 C_R} \cdot \frac{n}{B(\log n)^{2\kappa}}.$$

For the second statement take $B = \max(1, A_0) \log n$. $\square$

Proof of (c). Let E , K_G and a $(1 - r^{-1})$ -expansive r^{-1} -morphism $f : E \rightarrow \text{Sym}(\Omega)$ be as in Proposition 8.9, with its unitary $U_f = \sum_a c_a E_a \otimes P_{\pi_a}$ and its set Ω_0 of good points, at which $\text{Tr}_\Omega U_f(w \otimes |\omega\rangle\langle\omega|)U_f^* = \Phi(w)$. Let Ω_1 be the set of ω with $(f([x])f([y]))^{-1}(\omega) = f([z])^{-1}(\omega)$ for every triangle (x, y, z) and $f(g)^{-1}(\omega) \neq f(h)^{-1}(\omega)$ for distinct labels g, h . As $U_f^*(\psi \otimes |\omega\rangle) = \sum_a \bar{c}_a E_a^* \psi \otimes |\pi_a^{-1}(\omega)\rangle$, the computation of Proposition 8.9 gives $\text{Tr}_\Omega U_f^*(x \otimes |\omega\rangle\langle\omega|)U_f = \Phi^*(x)$ for $\omega \in \Omega_1$. A point outside Ω_1 is the image under the bijection $f([z])$ of a point where $f([x])f([y])$ and $f([z])$ differ, or under $f(g)$ of a point where $f(g)$ and $f(h)$ agree; so $|\Omega \setminus \Omega_1| \leq K_G M/r$, and $\Omega_* = \Omega_0 \cap \Omega_1$ has $|\Omega_*| \geq (1 - 2K_G/r)M$.

Let the bath be a coin qubit, a phase qubit carrying $z = \text{diag}(1, i)$, and $\mathbb{C}^\Omega$, and put $W_f = |0\rangle\langle 0| \otimes 1 + |1\rangle\langle 1| \otimes V_f$ with $V_f = \begin{pmatrix} 0 & U_f \otimes z \\ U_f^* \otimes z^* & 0 \end{pmatrix}$. For every state β that is a mixture of the $|\omega\rangle\langle\omega|$, $\omega \in \Omega_*$, the bath state $\tau_2 \otimes \tau_2 \otimes \beta$ makes W_f apply exactly Θ_+ : the coin splits the channel into $\frac{1}{2}\text{id}$ and $\frac{1}{2}$ the channel of V_f , the phase qubit removes the off-diagonal blocks since $\tau(z^2) = 0$, and the diagonal blocks are $\Phi(w)$ and $\Phi^*(x)$ at the points of Ω_* . Now repeat the exact service of Proposition 8.9(c) with K_G replaced by $2K_G$ and two more qubits, which are imported maximally mixed: with $\eta = B/(4n)$ and $r = 2K_G/\eta$ the imported register is maximally mixed on $\text{span}\{|\omega\rangle : \omega \in \Omega_*\} \otimes \mathbb{C}^{k_0}$ (and on the two qubits), its deficit is below B/n ,

every round applies exactly Θ_+ , and $Q = \lceil \log \sigma_E(8K_G n/B) + \log(4n/B) \rceil + 2$. The bound $\log \sigma_E(R) \leq C_E R \log R$ of Theorem 7.5(b) gives $Q \leq K(n/B) \log(2n/B)$. With β maximally mixed on all of Ω , the same unitary gives a finite tracial factorization of $(1 - \vartheta)\Theta_+ + \vartheta \Upsilon$ for some channel Υ and some $\vartheta \leq 2K_G/r$, for every r ; hence $\Theta_+ \in \mathcal{K}_{1746}$. $\square$

Remark 10.13 (Exact finite factorizations of Θ_+). Θ_+ is an exact autonomous value but not an exact finite factor. Indeed, if $\Theta_+(X) = (\text{id} \otimes \tau_m)[W(X \otimes 1)W^*]$ for a unitary W on $\mathbb{C}^{2d} \otimes \mathbb{C}^m$, the proof of Lemma 10.12 with $\rho = \tau_m$ has $\ell = 0$ and $\epsilon_S = 0$: the diagonal blocks of W are both $1 \otimes \alpha$, its upper right block B lies in $\mathcal{S}_\Phi \otimes M_m$, $B^*B = 1 \otimes R$ and $BB^* = 1 \otimes R'$. Then $B(1 \otimes R) = (1 \otimes R')B$, so for every eigenvalue $\mu > 0$ of R the operator $\mu^{-1/2}B$ maps $\mathbb{C}^d \otimes \ker(R - \mu)$ unitarily onto $\mathbb{C}^d \otimes \ker(R' - \mu)$, and the upper left block of $\Theta_+(e_{11} \otimes w)$ writes Φ as the convex combination $\sum_\mu \frac{2\mu m_\mu}{m} \Phi_\mu$ of exact factors with baths of dimension $m_\mu = \dim \ker(R - \mu) \leq m$. For the channel of Theorem 8.5 this is impossible: such a combination is a single round with $\ell = \Delta S = 0$ (a direct sum of flat baths), and Theorem 8.2 with $\nu = 0$ would give it entropy at least $N/16$ for every N .

Remark 10.14 (A lower exchange). At an anchor z of the gadget, $\Phi(|z\rangle\langle z|) = |z\rangle\langle z|$, so Θ_+ maps $|1\rangle\langle 1| \otimes |z\rangle\langle z|$ to the flat state $\frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) \otimes |z\rangle\langle z|$ of rank two. Proposition 8.7 with $r = 2$ therefore gives $P + C < 3Q + 6$ for services of Θ_+ with error at most $1/16$ and exchange $J \leq n/2$, and Proposition 10.9(b), used as in the proof of Theorem 8.5(b) with $B = \max(3Q + 6, \log n)$ (if $B > c_0 n$, then already $Q \geq (c_0 n - 6)/3$), gives $Q \geq c\sqrt{n}/(\log n)^\kappa$ for such services, whatever their purity.

So autonomous dynamics with a tracial bath can produce expensive channels. The reduced dynamics of the probe matters: if the channels Φ_t themselves form a semigroup of unital channels, they are mixed unitary for all large t [3], hence cheap, whereas $\Theta_+^2 \neq \Theta_+$. Locality constrains how Θ_+ can be approached: Proposition 10.1 and Proposition 10.9(b), combined as in the proof of Corollary 10.2, show that a local reservoir with $d = 1746$ that produces a channel within $\delta \leq \delta_0$ of Θ_+ at time T has $T \geq c\delta^{-1/\nu}(\log(1/\delta))^{-(2\kappa+1)/\nu}$, for constants $c, \delta_0 > 0$ that depend only on the parameters of the reservoir. So no local reservoir produces Θ_+ exactly at a finite time, or approaches it exponentially fast. Whether a local dynamics can produce an expensive channel at all is open (Section 11).

11 Open problems

- (1) *The exact profile.* Theorem A gives $\exp(r^{1+o(1)})$, with polylogarithmic factors in the exponent between $cr/(\log r)^\kappa$, $\kappa = 6 + \log 107$, and $C r \log r$. The exponent κ comes from the Dehn function of H_3 , known to be polynomial of degree at most $\log 107 + 4$ and at least 2 [19]; a quadratic Dehn function would give $\kappa = 4$. What is the correct polylogarithmic factor; is $\log \sigma_E(r) = \Theta(r \log r)$?
- (2) *Beyond the counting ceiling.* Does some amenable group have a chunk with profile $\exp(cr)$? Does any sofic group have a chunk with profile $\exp(r^{1+c})$ for some $c > 0$? The counting criterion cannot give either; a different lower-bound mechanism would be needed. For the second question, grids of fixed dimension should not help: the argument sketched in Remark 7.8 would bound the profile of their chunks by $\exp(O(r \log r))$. By Remark 8.12, a gadget channel of a sofic group can need superlinear memory at zero purity only if its chunk profile exceeds $\exp(O(r))$. Every fixed chunk of a sofic group has a computable profile [33, Lemma 1] (search the degrees $1, 2, \dots$ and the finitely many maps at each), so no chunk outgrows every computable function; how fast a computable profile can grow is open.

- (3) *The optimal exchange rate in general.* For Φ the least memory at the rank rate and the least memory with purity at most the memory are both $n^{1/2+o(1)}$ (Remark 8.13). For every channel with a flat probe the second is at most three times the first, plus six. Is the converse true up to polylogarithmic factors for every channel in $\mathcal{K}_d$ with a flat probe? Theorem 8.10 bounds the rank-rate memory by $\log k + n \log(k'/k) + O(\log n)$, so the converse holds whenever some exact rectangular dilation makes this at most a polylogarithmic factor times the least memory at purity at most memory.
- (4) *The dyadic group.* Its profile is finite and at least $\exp(cr^{1/3})$. In $\mathcal{G}$ the words $w_0 = 1$, $w_{2n} = uw_n$ and $w_{2n+1} = tuw_n$ reach the integer n with length $O(\log n)$, and commutators of copies at points of equal parity reduce freely to smaller indices. What is the area of the commutators at points of opposite parity?
- (5) *Thompson's groups.* Is $\mathcal{L}$ sofic whenever F is? The hyperlinearity transfer of Proposition 3.4 does not obviously give permutations, and not every action of a sofic group is sofic: some coset actions of residually finite Kazhdan groups are not [39]. We do not know whether the action of F on $\mathbb{D}$ is sofic when F is. Does Γ embed in V or in $2V$? The embedding of Proposition 4.3 uses one coordinate for each ray.
- (6) *Local autonomous physics.* Proposition 10.9 gives an expensive long-time limit of an autonomous dynamics with a tracial bath, but the dynamics is not local. Can an expensive channel be the long-time limit of a fixed translation-invariant local rule, such as a quantum cellular automaton or a time-independent finite-range Hamiltonian, with the bath in its tracial state? By Proposition 10.1 such a limit would be approached slowly; for Θ_+ the distance at time t is at least of order $t^{-\nu}$ up to logarithmic factors (Section 10). By [3] the reduced dynamics of the probe could not be a semigroup.

Use of AI

This work was carried out with extensive use of generative AI systems. They proposed constructions and proof strategies, wrote and checked proofs, carried out computations, searched the literature, wrote the verification code, and drafted and edited the text. The authors directed the research, assessed and selected the results, and take full responsibility for the content. The proofs are given in full in the text; repeated adversarial checking and the accompanying programs support them but do not replace them. The verification notes state which statements are checked by machine.

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